ELACESTRANT

(PROTOCOL STML-ELA-0222)

A Phase 1b/2, Open-Label Umbrella Study to Evaluate Safety and Efficacy of Elacestrant in Various Combinations in Patients with Metastatic Breast Cancer (ELEVATE)

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The protocol has been approved by Stemline Therapeutics, Inc.

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Date

INVESTIGATOR PROTOCOL AGREEMENT

STML-ELA-0222: A Phase 1b/2, Open-label Umbrella Study to Evaluate Safety and Efficacy of **El**a**c**estrant in **Va**rious Combinations in Pa**t**ients with Metastatic Br**e**ast Cancer (ELEVATE)

I hereby agree to:

- Conduct the study as outlined in this protocol with reference to national/local regulations and current ICH/GCP guidelines.
- Maintain the confidentiality of all information received or developed in connection with this protocol.
- Fully cooperate with monitoring and auditing and allow access to all documentation by authorized individuals representing Stemline Therapeutics, Inc., or Health authorities.
- I will, and I will cause any of my collaborators to, perform any activity in accordance with any applicable international and local anti-corruption legislations, such as OECD Convention on Combating Bribery of Foreign Public Officials in International Business Transactions, UK Bribery Act and US Foreign Corrupt Practices Act, including Italian Legislative Decree 231/2001. In particular, during the performance of the study, I will not, and I will cause any of my collaborators not to, directly or indirectly offer, pay, give, or promise to pay or give or receive any payment or gift of any money or thing of value to or from any government officer to influence any acts or decisions or to induce such officer to use its influence to effect or influence the decision of the relevant government body or any other decision maker. I accept to promptly inform the Sponsor in writing in case of violations of or deviations from any of the above prescriptions in the conduct of the study and I acknowledge and accept Sponsor's rights to conduct audits in order to verify compliance with the above during or in connection with the performance of the study. I agree and accept that a violation of any of the above prescriptions may result in the termination of the research activities of the site I work in and/or the entire study. Any episode of non-compliance will be documented.

Protocol Version (Date): Version 4.0 22 December 2023

Approved and signed by Investigator:

Print Name		
Signature		**Date**
Institution		

PROCEDURES IN CASE OF EMERGENCY

Emergency contact information can be found on the subject identification card.

1. CLINICAL PROTOCOL SYNOPSIS

<table>
<tr><td colspan="2">Name of Sponsor/Company: Stemline Therapeutics, Inc.</td></tr>
<tr><td>Protocol No.: STML-ELA-0222</td><td>EU CT number: 2022-503114-23
EudraCT No.: 2022-002609-10</td></tr>
<tr><td colspan="2">Title of Study: A Phase 1b/2, Open-label Umbrella Study to Evaluate Safety and Efficacy of Elacestrant in Various Combinations in Patients with Metastatic Breast Cancer (ELEVATE)</td></tr>
<tr><td colspan="2">Rationale: Combining medications for mBC may overcome resistance seen after early treatment. One such strategy is to combine medications. This trial will find combination doses for further study and determine if combining elacestrant with drugs approved for advanced or metastatic breast cancer in ER+/HER2– subtypes improves efficacy.</td></tr>
<tr><td colspan="2">Objective
Main Objectives
• Phase 1b: Determine the RP2D of elacestrant combined with other study drugs.
• Phase 2: Evaluate the efficacy of elacestrant combined with each other drug for PFS.
Secondary Objectives
• Describe the PK of elacestrant, combination drugs, major metabolites, and potential DDIs.
• Evaluate efficacy of elacestrant in each combination for ORR, DoR, CBR, PFS, and OS.</td></tr>
<tr><td colspan="2">Main Trial Endpoints
• Phase 1b: Number of DLTs observed during the first cycle
• Phase 2: Estimation of PFS rate at 6 months for each of the combination arms in patients who received prior ET and CDK4/6i in the metastatic setting; estimation of PFS rate at 12 months for all the combinations with palbociclib, abemaciclib, and ribociclib as second-/third-line treatment in the metastatic setting in patients who received no prior CDK4/6i in the metastatic setting (Arm D)</td></tr>
<tr><td colspan="2">Secondary Trial Endpoints
Phase 1b
• AEs and SAEs, as well as changes in clinical laboratory values, vital signs measurements, and ECGs
• Standard PK parameters including $AUC_{0\text{-}tau}$, C_{max}, T_{max}, and C_{trough}
• ORR: proportion of patients who achieve a BOR of confirmed PR or CR
• DoR: time from the date of the first documented CR/PR until the first radiological documentation of disease progression or death, whichever comes first
• CBR: proportion of patients who achieve a BOR of confirmed CR/PR or durable stable disease (duration at least 24 weeks from the date of first dose)
• PFS: time from the date of the first dose to the date of the first radiological documentation of disease progression or death, whichever occurs first
• OS: time from the date of the first dose to the date of death from any cause
Phase 2
• ORR, DoR, CBR, PFS, and OS
• AEs and SAEs, as well as changes in clinical laboratory values, vital signs, and ECGs</td></tr>
<tr><td colspan="2">Trial Design: This is an open-label, multicenter, Phase 1b/2 study in patients with ER+/HER2– advanced/metastatic breast cancer. Phase 1b will determine the RP2Ds of elacestrant in combination with alpelisib, everolimus, palbociclib, ribociclib, and capivasertib. In Phase 2, the efficacy and safety of the various combinations will be evaluated. In both phases, study drugs will be administered in 28-day cycles.</td></tr>
</table>

Treatment duration is estimated to be 6 months in all cohorts except Arm D, in which the duration is estimated to be 9 months.
Trial Population: This study will evaluate the safety and efficacy of elacestrant combined with alpelisib, everolimus, palbociclib, abemaciclib, ribociclib, and capivasertib in women and men ≥18 years: • **Medical Condition:** Histopathologically or cytologically confirmed ER+/HER2–, advanced or metastatic breast cancer • **Principal Inclusion Criteria:** At least 1 measurable lesion per RECIST 1.1; adequate bone marrow and organ function • **Principal Exclusion Criteria:** Active central nervous system metastases or advanced, symptomatic visceral spread that are at risk of life-threatening complications in the short term; prior chemotherapy or elacestrant in the advanced/metastatic setting
Interventions: Elacestrant and the combination drugs to be evaluated are considered experimental in this patient population. Elacestrant will be administered orally QD as 100, 200, 300, and 400 mg elacestrant dihydrochloride tablets with alpelisib (200 to 250 mg), everolimus (5 to 10 mg), palbociclib (100 to 125 mg), ribociclib (400 to 600 mg), and capivasertib (320 to 400 mg BID). The RP2D of elacestrant and abemaciclib will be determined in the ELECTRA Study (ClinicalTrials.gov Identifier: NCT 05386108) before Phase 2 begins in this combination. Luteinizing hormone-releasing hormone agonists are required for pre- and perimenopausal women and men as per standard of care in breast cancer. Treatment will continue until disease progression, death, development of unacceptable toxicity, withdrawal of consent, etc.
Ethical Considerations: The Principal Investigator(s) at each site will ensure that the patient is given full and adequate oral and written information about the nature, purpose, possible risk, and benefit of the study. The Dose Escalation Committee in consultation with the Study Steering Committee will monitor treatment-emergent data for subject safety. The anticipated favorable clinical benefits of elacestrant in other indications and combinations are projected to outweigh the risks based on previous experience.

2. TABLE OF CONTENTS, LIST OF TABLES, AND LIST OF FIGURES

TABLE OF CONTENTS

1. CLINICAL PROTOCOL SYNOPSIS 5
2. TABLE OF CONTENTS, LIST OF TABLES, AND LIST OF FIGURES 7
LIST OF TABLES 12
LIST OF FIGURES 13
LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS 14
3. INVESTIGATOR(S) AND STUDY ADMINISTRATIVE STRUCTURE 19
4. INTRODUCTION 20
4.1. Epidemiology 20
4.2. Current Treatments and Unmet Medical Need 20
4.3. Combination Rationale 22
4.4. In Vitro Studies 22
4.5. In Vivo Studies 24
4.6. Elacestrant Background 25
4.7. Overview of Elacestrant in Combination with Other Drugs 27
4.7.1. Efficacy Assessments 27
4.7.2. Safety 28
4.8. Rationale 31
4.9. Overall Benefit-Risk Assessment 31
5. TRIAL OBJECTIVES AND ENDPOINTS 33
6. INVESTIGATIONAL PLAN 36
6.1. Overall Study Design 36
6.2. Phase 1b 36
6.2.1. Dose-Limiting Toxicity 38
6.2.2. Determination of the Recommended Phase 2 Dose 39
6.3. Phase 2 40
6.3.1. Arm A 40
6.3.2. Arm B 40
6.3.3. Arm C 40
6.3.4. Arm D 40

6.3.5. Arm E....41
6.4. Treatment Assignment....41
6.5. Intra-patient Dose Escalation....41
6.6. Dose Adjustment Criteria....41
6.6.1. Combination Doses and Rationale....41
6.6.2. Combination with Alpelisib....42
6.6.3. Combination with Everolimus....47
6.6.4. Combination with Palbociclib....50
6.6.5. Combination with Abemaciclib....52
6.6.6. Combination with Ribociclib....55
6.6.7. Combination with Capivasertib....58
6.7. Study Steering Committee and Dose Escalation Committee....61
6.7.1. Membership....61
6.7.2. Responsibilities....61
6.8. Participant Diversity....62
6.9. Criteria for Study Termination....62
6.9.1. End of Clinical Trial and Archiving....62
6.10. Schematic Study Treatments and Visit Schedule....64
7. SELECTION AND WITHDRAWAL OF PATIENTS....73
7.1. Patient Inclusion Criteria Common to Phase 1b and 2 for All Treatment Arms....73
7.2. Patient Exclusion Criteria Common to Phase 1b and 2 for All Treatment Arms....74
7.3. Additional Eligibility Criteria by Combination....76
7.3.1. Alpelisib Combination (Phase 1b and Arm A)....76
7.3.2. Everolimus Combination (Phase 1b and Arm B)....76
7.3.3. Abemaciclib Combination (Arm C)....76
7.3.4. Ribociclib Combination (Phase 1b and Arm C)....77
7.3.5. Palbociclib Combination (Phase 1b)....77
7.3.6. Palbociclib Combination (Arm D)....77
7.3.7. Abemaciclib Combination (Arm D)....78
7.3.8. Ribociclib Combination (Arm D)....78
7.4. Patient Withdrawal Criteria....79

7.4.1. Treatment Discontinuation 79
7.5. Duration of Treatment 80
7.6. Duration of Study 80
7.7. Study Stopping Rules 80
7.7.1. End of Study 80
8. TREATMENT OF PATIENTS 82
8.1. Description of Study Drug 82
8.2. Study Drug Storage 85
8.3. Administration 85
8.4. Study Drug Accountability 85
8.5. Study Drug Handling and Disposal 86
8.6. Treatment Compliance 86
8.7. Concomitant Medications 86
8.7.1. Permitted Concomitant Medications 87
8.7.2. Medications to be Administered with Caution 87
8.7.3. Prohibited Concomitant Medications 88
9. STUDY ASSESSMENTS 90
9.1. Assessment of Efficacy 90
9.2. Assessment of Safety 91
9.2.1. Demographic/Medical History 91
9.2.2. Physical Examination and Vital Signs 91
9.2.3. Electrocardiogram 92
9.2.4. Laboratory Assessments 92
9.2.5. Hematology 92
9.2.6. Comprehensive Metabolic Panel 92
9.2.7. Virus Serology 92
9.2.8. Pregnancy Screen 92
9.3. Pharmacokinetic Assessments 93
9.3.1. Receipt of Pharmacokinetic Samples 93
9.3.2. Pharmacokinetic Samples Bioanalysis 93
9.3.3. Retention of Pharmacokinetic Samples 94
9.4. Pharmacodynamic and Other Biomarkers Assessments 94
9.5. Adverse and Serious Adverse Events 94

9.5.1. Adverse Event Reporting Period 94
9.5.2. Definition of Adverse Events 94
9.5.2.1. Adverse Event 94
9.5.2.2. Serious Adverse Event/Serious Adverse Reaction 95
9.5.2.3. Treatment-Emergent Adverse Events 97
9.5.2.4. Adverse Event/Adverse Drug Reaction Intensity (Severity) 97
9.5.2.5. Adverse Events of Special Interest 97
9.6. Relationship to Study Drug 97
9.7. Recording Adverse Events 98
9.8. Reporting Adverse Events 99
9.9. Suspected Unexpected Serious Adverse Reaction 99
9.10. Management of Serious Adverse Events 99
9.10.1. Reporting Duties of the Investigator 99
9.10.2. Reporting Duties of the Sponsor 101
9.11. Management of Pregnancy Exposure Cases 101
9.12. Management of Misuse and Overdose Cases 102
9.13. Follow-up of Subjects After Adverse Events 102
9.13.1. Sponsor Follow-up 102
9.13.2. Investigator Follow-up 102
9.13.3. Post Study Follow-up 103
9.14. Periodic Safety Reporting 103
9.14.1. Annual Safety Reporting 103
9.14.2. Periodic Line-listings Safety Reporting 103
9.15. Remote Visits 103
10. STATISTICS 104
10.1. Definitions of Analysis Sets 104
10.2. Data Analysis 104
10.3. Subject Disposition 104
10.4. Demographic and Other Baseline Characteristics 105
10.5. Prior and Concomitant Therapy 105
10.6. Sample Size Phase 1b 105
10.7. Sample Size Phase 2 105
10.7.1. Arm A 105

10.7.2. Arm B 106
10.7.3. Arm C 106
10.7.4. Arm D 106
10.7.5. Arm E....107
10.8. Extent of Exposure 107
10.9. Efficacy Analysis....107
10.10. Safety Analysis 108
10.11. Pharmacokinetics Analysis....108
10.12. Pharmacodynamics Analysis....108
10.12.1. Receipt of the Samples 109
10.12.2. Sample Analysis 109
10.12.3. Retention of Samples....110
10.13. Interim Analyses....110
11. DIRECT ACCESS TO SOURCE DATA/DOCUMENTS....110
11.1. Study Monitoring....110
11.2. Audits and Inspections....111
12. DATA QUALITY MANAGEMENT....112
12.1. Data Collection 112
12.1.1. Electronic Case Report Form....112
12.1.2. Interactive Response Technology....112
12.1.3. Patient Diary 113
12.1.4. Central Laboratory....113
12.1.5. Data Capture Systems Versions and Validation Documentation 113
12.1.6. Clinical Data Management 113
12.2. Source Data....113
13. QUALITY CONTROL AND QUALITY ASSURANCE 114
14. ETHICS 114
14.1. Ethics Review 114
14.2. Ethical Conduct of the Study and General Aspects....115
14.3. Written Informed Consent 115
14.4. Patient Insurance....115
15. DATA HANDLING AND RECORDKEEPING 116
15.1. Inspection of Records 116

15.2. Retention of Records 116
15.3. General Principles on Personal Data Privacy Compliance 116
15.3.1. Acknowledgment 116
15.3.2. Data Controllers and Data Processors 117
15.3.3. Duties of The Parties Involved in The Performance of The Study 117
15.3.4. Archiving of The Clinical Trial Master File and Code Pairing List 119
15.3.5. Data Breach 120
15.3.6. Information Notice on Personal Data Protection and Pseudonymization 121
15.3.7. Genetic Data 122
15.3.8. Transfer of Patients' Data Outside the European Union 123
15.3.9. Exercise of Patients' Data Privacy Rights 123
15.3.10. Future Research 124
16. PUBLICATION POLICY 124
17. LIST OF REFERENCES 125
18. APPENDICES 130
APPENDIX A. ECOG PERFORMANCE STATUS CRITERIA 131
APPENDIX B. HBV/HCV TESTING CRITERIA 132
APPENDIX C. RESPONSE EVALUATION CRITERIA IN SOLID TUMORS (RECIST) VERSION 1.1 133
APPENDIX D. COCKROFT-GAULT FORMULA 134
APPENDIX E. DRUGS KNOWN TO PROLONG THE QT INTERVAL 135
APPENDIX F. CONTRACEPTION GUIDANCE 136

LIST OF TABLES

Table 1 Abbreviations 14
Table 2: Epidemiology of ER+/HER2– Breast Cancer (× 1,000) 20
Table 3: Adverse Drug Reactions Observed in ≥15% in Any of the Pivotal Studies of Combination Drugs 29
Table 4: Objectives and Endpoints, Phase 1b and Phase 2 33
Table 5 Dose Level of Elacestrant and Alpelisib to be Assessed in Phase 1b 37
Table 6: Dose Level of Elacestrant and Everolimus to be Assessed in Phase 1b 37
Table 7: Dose Level of Elacestrant and Palbociclib to be Assessed in Phase 1b 37

Table 8: Definition of Dose-limiting Toxicity 38
Table 9: Dose Levels and Reduction Guidelines for the Combination of Elacestrant and Alpelisib 42
Table 10: Alpelisib Combination - Recommendation for Dose Modifications 43
Table 11: Dose Levels and Reduction Guidelines for the Combination of Elacestrant and Everolimus 47
Table 12: Everolimus Combination - Recommendation for Dose Modifications 48
Table 13: Dose Levels and Reduction Guidelines for the Combination of Elacestrant and Palbociclib 51
Table 14: Palbociclib Combination - Recommendation for Dose Modifications 51
Table 15: Abemaciclib Combination - Recommendation for Dose Modifications 53
Table 16: Dose Levels and Reduction Guidelines for the Combination of Elacestrant and Ribociclib 55
Table 17: Ribociclib Combination - Recommendation for Dose Modifications 56
Table 18 Dose Levels and Reduction Guidelines for the Combination of Elacestrant and Capivasertib 58
Table 19 Capivasertib Combination - Recommendation for Dose Modifications 58
Table 20: Schedule of Assessments, Combinations with Alpelisib, Everolimus, Palbociclib, Ribociclib, and Abemaciclib 67
Table 21 Schedule of Assessments, Combination With Capivasertib 70
Table 22: Investigational Product 82

LIST OF FIGURES

Figure 1: Progression-free Survival of Elacestrant and Standard of Care - All Patients 26
Figure 2: Progression-free Survival of Elacestrant and Standard-of-Care Patients with *ESR1* Mutations 27
Figure 3: Phase 1b Study Design 64
Figure 4: Phase 2 Study Design 65

LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

The following abbreviations and specialist terms are used in this study protocol.

Table 1 Abbreviations

Abbreviation	Explanation
λz	terminal elimination rate constant
ADR	adverse drug reaction
AE	adverse event
AESIs	adverse events of special interest
AI	aromatase inhibitor
AKT	enzymes involved in processes for cell growth and survival
ALP	alkaline phosphatase
ALT	alanine aminotransferase
A.MRBS	A. Menarini Research & Business Service
ANC	absolute neutrophil count
ASCO	American Society of Clinical Oncology
AST	aspartate aminotransferase
ATC	Anatomical Therapeutic Chemical
AUC	area under the plasma concentration-time curve
AUC_{0-24}	area under the plasma concentration-time curve from time 0 to 24 hours
AUC_{0-tau}	area under the plasma concentration-time curve over the dosing interval
BCRP	breast cancer resistant protein
BOR	best overall response
BP	blood pressure
BSA	body surface area
BUN	blood urea nitrogen
C1/D1	Cycle 1, Day 1
C_{ave}	average concentration
CA	competent authority
CAP	College of American Pathologists
CBC	complete blood count
CBR	clinical benefit rate
CDK4/6i	cyclin-dependent kinase targeting enzymes CDK4 and CDK6 inhibitor
cfNA	cell-free nucleic acid
CI	confidence interval
CL/F	apparent total body clearance following oral administration

Abbreviation	Explanation
C_{min}	minimum observed plasma concentration
C_{max}	maximum observed plasma concentration
CMP	comprehensive metabolic panel
CNS	central nervous system
CR	complete response
Cr	creatinine
CrCl	creatinine clearance
CRF	chronic renal failure
CRO	contract research organization
CT	computed tomography
CTCAE	Common Terminology Criteria for Adverse Events
C_{trough}	concentration at the end of the dose interval
CV%	coefficient of variation
DDI	drug-drug interaction
DEC	Dose Escalation Committee
DHEA	dehydroepiandrosterone
DLP	data lock point
DLT	dose-limiting toxicity
DoR	duration of response
DPO	Data Protection Officer
DSUR	Development Safety Update Report
EC	Ethics Committee
ECG	electrocardiogram
ECOG	Eastern Cooperative Oncology Group
eCRF	electronic case report form
EDC	electronic data capture
eGFR	estimated glomerular filtration rate
EOT	end of treatment
ER	estrogen receptor
ER+	estrogen receptor positive
ER+/HER2–	estrogen receptor positive/human epidermal growth factor receptor 2 negative
ESR1	estrogen receptor gene alpha
ET	endocrine therapy
eTMF	electronic Trial Master File
EU	European Union

Abbreviation	Explanation
FPG	fasting plasma glucose
FSH	follicle-stimulating hormone
GCP	Good Clinical Practice
gDNA	genomic DNA
GDPR	General Data Protection Regulation
GLP	Good Laboratory Practice
Hgb	hemoglobin
HbA1c	glycosylated hemoglobin
HBV	hepatitis B virus
HCC1428-LTED	palbociclib-sensitive
HCC1428-LTED-palboR	palbociclib-resistant
Hct	hematocrit
HCV	hepatitis C virus
HDL	high-density lipoprotein
HEENT	Head, Eyes, Ears, Nose, and Throat
HER2–	human epidermal growth factor receptor 2 negative
HIPAA	Health Insurance Portability and Accountability Act
HIV	human immunodeficiency virus
HR	hormone receptor
IB	Investigators' Brochure
IC50	half-maximal inhibitory concentration
ICF	informed consent form
ICH	International Council for Harmonisation
IEC	Independent Ethics Committee
ILD	interstitial lung disease
IM	intramuscular
IMP	Investigational Medicinal Product
IMPD	Investigational Medicinal Product Dossier
IND	Investigational New Drug
INR	international normalized ratio
IRB	Institutional Review Board
IRT	interactive response technology
ITT	intent-to-treat
IV	intravenous

Abbreviation	Explanation
JCO	Journal of Clinical Oncology
Ki	score indicating the percentage of positively stained tumor cells among the total number of malignant cells assessed
LC-MS/MS	liquid chromatography tandem mass spectrometry
LDL	low-density lipoprotein
LHRH	luteinizing hormone-releasing hormone
LTED	long-term estrogen deprived
MAH	Marketing Authorisation Holder
mBC	metastatic breast cancer
MedDRA	Medical Dictionary for Regulatory Activities
mTOR	mammalian target of rapamycin
MRI	magnetic resonance imaging
NCI	National Cancer Institute
ORR	overall response rate
OS	overall survival
PBMC	peripheral blood mononuclear cell
PD	pharmacodynamic
PDX	patient-derived xenograft
PFS	progression-free survival
PGR	progesterone
P-gp	P-glycoprotein
PI	Principal Investigator The Investigator who leads the study conduct at an individual study site. Every study site has a PI.
PIK3CA	phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit alpha
PK	pharmacokinetics
PPIs	proton-pump inhibitors
PR	partial response
PS	performance status
PT	prothrombin time
QD	once daily
QTcB	QT interval corrected by Bazett formula
QTcF	QT interval corrected by Fridericia formula
Rb	retinoblastoma gene
RECIST	Response Evaluation Criteria in Solid Tumors

Abbreviation	Explanation
RES	Response-Evaluable Set
RMP	Risk Monitoring Plan
RP2D	recommended Phase 2 dose
RR	response rate
SAE	serious adverse event
SAP	Statistical Analysis Plan
SAS	Safety Analysis Set
SD	standard deviation
SDSU	Study Drug Safety Unit
SEER	Surveillance, Epidemiology, and End Results
SERD	selective estrogen receptor degrader
SMP	Safety Management Plan
SmPC	Summary of Product Characteristics
SOC	system-organ-class; alternatively standard of care
SOP	standard operating procedure
SSC	Study Steering Committee
SUSAR	suspected unexpected serious adverse reactions
$t_{1/2}$	terminal elimination half-life
$t_{1/2,z}$	apparent plasma terminal elimination half-life
TEAE	treatment-emergent adverse event
T_{max}	time of the maximum observed plasma concentration
ULN	upper limit of normal
USPI	US Prescribing Information
V/F	apparent volume of distribution
WBC	white blood cell
WHO DD	World Health Organization Drug Dictionary
WT	wild-type

3. INVESTIGATOR(S) AND STUDY ADMINISTRATIVE STRUCTURE

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4. INTRODUCTION

4.1. Epidemiology

Breast cancer is the leading cause of cancer and cancer deaths in women (Bray et al, 2018). The incidence and prevalence of patients with invasive breast cancer, as well as estimates for the prevalence of subjects with ER+/HER2– breast cancer, are presented in Table 2.

Table 2: Epidemiology of ER+/HER2– Breast Cancer (× 1,000)

	US	EU[a]	Japan[a]	Global[a]
Incidence: New yearly cases of invasive breast cancer	253[a], 282[b]	531	92	2,261
Prevalence of invasive breast cancer	1,071[a]	2,138	328	7,791
Prevalence of ER+, HER2– breast cancer (approximately 70% of invasive breast cancer)	750	1,497	230	5,454

Abbreviations: EU=European Union; SEER=Surveillance, Epidemiology, and End Results; US=United States.
a. International Agency for Research on Cancer and World Health Organization 2021
b. National Cancer Institute and Surveillance, Epidemiology, and End Results (SEER) Program 2021

In the United States, an estimated 43,600 women died from breast cancer in 2021 (Adamo et al, 2020). Mariotto and colleagues estimated that by January 2017, there were 154,794 women living with mBC in the United States, with 3 in 4 initially diagnosed with stage I to III breast cancer that later progressed to mBC (Mariotto et al, 2017). In Europe, Ferlay and colleagues estimated that 157,100 women died from breast cancer in 2020 (Ferlay et al, 2021).

Male breast cancer is a rare disease representing 1% of all breast carcinomas diagnosed in the United States. In 2019, an estimated 2,670 new cases of breast cancer will be diagnosed among men in the United States, and about 500 men will die of breast cancer (Hassett et al, 2020). The lifetime risk of breast cancer is about 1 in 1,000 for a man, and the average age of a new breast cancer diagnosis for men (67 years) is 5 years older than for women (62 years). Risk factors for male breast cancer include a family history of breast cancer, black race, exposure to radiation, or carrying a predisposition germline genetic mutation (eg, BRCA2, BRCA1, CHEK2, PALB2).

4.2. Current Treatments and Unmet Medical Need

The primary target for treatment of ER+ breast cancer is to prevent estrogen from acting as the fuel in malignant cells to promote tumor growth. There are many ways to target estrogen signaling that are the backbone for current practice paradigms. These include, but are not limited to, preventing the conversion of other hormones into estrogen in the body (AIs), blocking estrogen from binding directly to its receptor (tamoxifen) and through selective degradation of the estrogen receptor (fulvestrant and elacestrant). Prior to menopause, the ovaries act as a primary source for estrogen production, and ovarian suppression and/or ablation is often a part of treatment for ER+ pre/perimenopausal women through chemical suppression with LHRH agonists (also known as gonadotropin-releasing hormone analogues, overall ATC code L02AE, and specifically L02AE01, L02AE02, L02AE03, L02AE04 and L02AE05 per local practice and regulations) or oophorectomy. In men with ER+ breast cancer, LHRH agonists are also vital for

decreasing androgen production, which are converted to estrogens that can drive cancer growth, through pituitary regulation of the testes (DeSimone et al, 2008).

Currently, recommended first-line standard of care for locally advanced or metastatic ER+/HER2– breast cancer is endocrine therapy, with either AIs or fulvestrant, plus a CDK4/6i (NCCN, 2023; Bentzon et al, 2008; Burstein et al, 2021; Gennari et al, 2021). Once the disease progresses, there are limited therapeutic options. Treatment guidelines recommend the use of sequential endocrine therapy in the absence of visceral crisis until all endocrine therapy options have been exhausted (NCCN, 2023; Gennari et al, 2021). Endocrine therapy includes endocrine monotherapy, such as fulvestrant, if the first-line therapy contained an AI-based therapy, or AIs if the first-line therapy applied a fulvestrant-based therapy (NCCN, 2023; Burstein et al, 2021; Gennari et al, 2021).

Until the US FDA approval of elacestrant in January 2023, fulvestrant was the only approved SERD for the treatment of subjects with ER+/HER2– mBC (Niikura et al, 2014). Fulvestrant effectively degrades ER and has demonstrated clinical benefit in ER+/HER2– mBC. A 500 mg monthly dose of fulvestrant, after a biweekly dose during the first month, in subjects with ER+/HER2–mBC who failed previous endocrine therapy, was associated with a median PFS of 6.5 months (Di Leo et al, 2010). However, these data were generated prior to the approval of CDK4/6 inhibitors. Recently, data on fulvestrant monotherapy post-CDK4/6 inhibitor treatment are starting to emerge.

- In a recent Phase 2 trial of second/third line venetoclax + fulvestrant versus fulvestrant alone in ER+/HER2– mBC patients who experienced disease recurrence/progression during/after CDK4/6 inhibitor therapy (VERONICA study), treatment with fulvestrant as a single agent was associated with a median PFS of 1.94 months with a clinical benefit rate of 13.7% (Lindeman et al, 2021).
- In the MATCH Phase 2 clinical trial, high-dose fulvestrant was associated with a median PFS of 2.2 months and a clinical benefit rate of 16% among subjects with a detectable *ESR1* mutation, in which few subjects received prior CDK4/6 inhibitors (Turner et al, 2020).

In addition to the need for effective treatment options for patients with ER+/HER2– mBC after progression on CDK4/6 inhibitors, the IM route of administration of fulvestrant underscores the need for novel oral ER antagonists in this setting. Although a long-acting IM formulation of fulvestrant was approved that can be administered monthly after 3 biweekly doses, the total volume of the injections per dose is 10 mL, a volume that is difficult to tolerate by some patients (Wardell et al, 2015; Bihani et al, 2017).

Available second-line combination therapy options for patients with HR+/HER2– metastatic breast cancer include everolimus + exemestane, everolimus + fulvestrant, and the newly FDA-approved capivasertib + fulvestrant for adult patients with 1 or more *PIK3CA/AKT1/PTEN*-alterations, as detected by an FDA-approved test. Additionally, for subjects with *PIK3CA*-mutant breast cancer, the combination of fulvestrant and alpelisib is another option. These everolimus/alpelisib combinations are associated with approximately 20% to 25% treatment discontinuation rate because of AEs in clinical trials (everolimus USPI; alpelisib USPI).

Due to the infrequent incidence of male breast cancer, there are no specific clinical trials that have addressed this patient population, and to date treatment regimens mirror what is

recommended for pre/perimenopausal women with breast cancer. The current JCO recommendation for male breast cancer in the advanced state is as follows: Men with advanced or metastatic, HR+/HER2– breast cancer should be offered endocrine therapy as first-line therapy except in cases of visceral crisis or rapidly progressive disease. Options include tamoxifen, an AI with a GnRH agent, and fulvestrant. CDK4/6 inhibitors can be used in men as they are used in women (type: formal consensus; evidence quality: low; strength of recommendation: strong).

In addition, the USPIs for both ribociclib (Kisquali®) and palbociclib (Ibrance®) indicate pre/perimenopausal women and men should be treated with LHRH agonists according to current clinical practice standards. In this protocol, we ask that any men considered for enrollment also be treated with LHRH agonists as indicated by local practice guidelines.

4.3. Combination Rationale

ER+/HER2– breast cancer is the most prevalent breast cancer subtype. Endocrine therapy is the mainstay of treatment; however, due to the heterogeneous nature of the disease, development of resistance to this therapeutic approach is very common in the metastatic setting.

After progression on first-line treatment, both hormonal monotherapy and combination therapy are available. Currently available combination therapies are everolimus + exemestane and, for patients with activating mutations of PIK3CA gene, alpelisib + fulvestrant. On November 16, 2023, the Food and Drug Administration approved capivasertib with fulvestrant for adult patients with hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative locally advanced or metastatic breast cancer with one or more *PIK3CA/AKT1/PTEN*-alterations, as detected by an FDA-approved test, following progression on at least one endocrine-based regimen in the metastatic setting or recurrence on or within 12 months of completing adjuvant therapy (fda.gov). For patients who did not receive prior treatment with a CDK4/6i, this combination of fulvestrant with a CDK4/6i is among the recommended treatment options for this patient population (NCCN, 2023).

The deep crosstalk between the 3 pathways (ER, PI3K/AKT/mTOR and CDK) provide a rationale for the combination of elacestrant and inhibitors of the PI3K pathway, such as alpelisib, capivasertib and everolimus, as well as the combination of elacestrant and CDK4/6i (palbociclib, ribociclib, and abemaciclib) to overcome endocrine resistance.

The elacestrant Phase 3 EMERALD trial demonstrated that elacestrant monotherapy was significantly superior to standard-of-care hormonal monotherapy (AIs and fulvestrant) in terms of PFS and was associated with a manageable safety profile (see Section 4.6).

Since elacestrant showed superior activity versus fulvestrant or AI in the EMERALD study, the replacement of the endocrine backbone by elacestrant in these combinations can potentially enhance patient outcomes. This will also enable an oral–oral combination treatment to replace the undesirable injectable SERD with the first oral SERD (elacestrant), which had demonstrated improvement in statistical significance over fulvestrant in a post hoc analysis in EMERALD.

4.4. In Vitro Studies

The in vitro antiproliferative activity of elacestrant was evaluated in MCF-7 and CDK4/6iR cell lines. Treatment of MCF-7 cells with elacestrant at 10, 100, or 1000 nM decreased basal rates of cell proliferation, with an IC50 of 5.6 nM. Cotreatment of cells with both elacestrant and E2

resulted in a dose-dependent decrease in E2-induced proliferation, with IC50 values for elacestrant of 4.2 and 27 nM, in the presence of 0.01 and 0.1 nM E2, respectively.

These results demonstrate that elacestrant does not stimulate proliferation of MCF-7 cells, such as 17β-estradiol, but suppresses their proliferation, both basal and E2-stimulated (Report RAD-002).

The in vitro antiproliferative effects of elacestrant were also evaluated in models that mimic post-AI and post-AI-CDK4/6 inhibitor settings.

Elacestrant inhibited the growth of HCC1428-LTED and HCC1428-LTED-palboR cell lines. The mean IC50 of growth inhibition mediated by elacestrant in palbociclib-sensitive and palbociclib-resistant cells was 1.77 and 1.69 nM, respectively. In summary, elacestrant inhibited the growth of HCC1428-LTED and HCC1428-LTED-palboR cell lines with similar potency and to a similar extent. The data suggest that elacestrant has the potential to inhibit the proliferation of tumor cells in a post-CDK4/6 inhibitor setting (Report 18RAD2023).

In addition, elacestrant antiproliferative effects were evaluated in *ESR1* mutant and WT cell lines mimicking AI (LTED cell lines) and CDK4/6 resistance.

In the post-AI setting, exposure to elacestrant inhibited the growth of HCC1428-LTED, MCF7-Y537S, and MCF7-D538G cells, which proliferated in the absence of estrogen. Palbociclib, ribociclib, and abemaciclib did not inhibit colony formation of their respective CDK4/6iR cell lines. However, elacestrant (300 nM) inhibited colony formation of both CDK4/6iS and CDK4/6iR breast cancer cells when compared to vehicle control-treated cells. The growth inhibitory effect observed was independent of *ESR1* status (WT or mutant) and CDK4/6 inhibitor used to develop resistance.

Capivasertib, a pan-AKT inhibitor, demonstrated inhibition of all 3 isoforms of AKT with an IC50 < 10 nmol/L and also exerted activity against 15 kinases, most of which were members of the cAMP-dependent, cGMP-dependent and protein kinase C (AGC) kinase family (Andrikopoulou et al, 2022). In vitro testing of capivasertib monotherapy was conducted in 182 cell lines derived from solid and hematologic tumors, revealing persistent susceptibility in HER2+ and ER+ breast cancer cell lines, while lung and colorectal cancer cell lines exhibited a lower frequency of response (Davies et al, 2012). Additionally, capivasertib induced dose-dependent inhibition of growth and survival in invasive lobular carcinoma (ILC) human and mouse breast cancer cell lines (Teo et al, 2018). Inhibition of AKT led to a reduction in ER-mediated transcription by attenuating the recruitment of ER and CREB-binding protein (CBP) to estrogen response elements (Ribas et al, 2015). Furthermore, the combination of capivasertib and endocrine treatment with fulvestrant, anastrozole, or tamoxifen was found to be superior to each respective monotherapy in ER+ endocrine-resistant breast cancer cell lines, particularly in cells harboring mutations of *PIK3CA* or loss of *PTEN* function (Ribas et al, 2015).

In summary, elacestrant exhibited activity in the post-AI and post CDK4/6 setting by inhibiting the growth of cell line models harboring either WT or mutant ER. Elacestrant also inhibited the colony formation of CDK4/6iS and CDK4/6iR cells harboring either WT or mutant *ESR1* (19RAD2034). The in vitro data of capivasertib provides a preclinical basis for combination of AKT inhibitor and SERDs in endocrine-resistant ER-positive breast cancer.

4.5. In Vivo Studies

Simultaneous inhibition of the ER and PI3K pathways or CDK4/6i was well-characterized by in vivo models of ER+/HER2– breast cancer. The effect of fulvestrant combined with the PI3K inhibitor alpelisib and/or CDK4/6i in several CDK4/6i-resistant xenograft models both PIK3CA mutant or WT was evaluated. Combination treatments significantly suppressed tumor growth compared with single-agent treatments and were supported by PD biomarkers of inhibition (Bosch et al, 2015).

Elacestrant was tested in combination with CDK4/6i, everolimus, and PI3Ki, alpelisib, in multiple ER+/HER2– cell line- and patient-derived xenograft models, characterized by different genetic alterations, including those conferring resistant to CDK4/6i. Elacestrant showed significant antitumor activity in combination with CDK4/6i, alpelisib, and everolimus in these resistant models, regardless of any alterations in cell cycle regulators (such as Rb loss, CDK6 over-expression, upregulated cyclin E1 and E2F1, both in *ESR1* WT and mutant) (Bihani et al, 2017). Resulting data strongly support the combination of a PI3K/mTOR pathway or CDK4/6i with anti-estrogen therapy. Antitumor efficacy of elacestrant was assessed in MCF7 xenograft model in combination with palbociclib and everolimus. In this study both the combination of elacestrant (60 mg/kg QD) plus the CDK4/6 inhibitor palbocilcib (45 mg/kg QD) and the combination of elacestrant (60 mg/kg) plus the mTOR inhibitor everolimus (2.5 mg/kg QD) resulted in improved antitumor activity compared with the single-agent treatments (Bihani et al, 2017).

The antitumor efficacy of elacestrant was assessed in combination with the CDK4/6 inhibitor palbociclib and ribociclib in several PDX models.

Elacestrant in combination with palbocilcib demonstrated superior antitumor activity compared with the single-agent treatments in the HBCx-3 *ESR1* WT model, the ST-941 *ESR1* Y537S model, and palbociclib-resistant model. In addition, in the ST-941 PDX model, elacestrant showed higher antitumor activity in combination with ribociclib compared with ribociclib and elacestrant single-agent treatments (elacestrant IB).

The antitumor efficacy of elacestrant combined with the PI3K inhibitor alpelisib was evaluated in 2 PDX models: the ST3932 PDX model, which harboured WT ER and mutant PIK3CA (Patel et al, 2019), and the ST941 PDX model, which harboured the *ESR1* Y537S mutation and a PIK3CA mutation (elacestrant IB).

Elacestrant administered orally, at different doses, demonstrated improved antitumor activity versus the vehicle control and the single-agent treatments in both models tested.

In the CTG-2432 PIK3CA mutated model and palbociclib-resistant PDX model, elacestrant in combination with everolimus demonstrated improved antitumor activity compared with single-agent treatments (Patel et al, 2019).

There are currently no in vivo studies specifically testing the combination of elacestrant and capivasertib. Nonetheless, combined treatment with capivasertib and fulvestrant showed synergy in an ER+ patient-derived xenograft and delayed tumor progression after cessation of therapy (Ribas et al, 2015). Precisely, combination of capivasertib and fulvestrant resulted in almost complete inhibition of tumor growth (80% inhibition $P < 1E-05$) compared with vehicle control. After 90 days of treatment, the therapies were withdrawn and tumor volume assessed to establish the efficacy of the combination in delaying tumor progression. Removal of capivasertib showed

a dramatic rise in tumor volume as early as 10 days after withdrawal, while the combination of capivasertib plus fulvestrant showed sustained antitumor effect even after 50 days after cessation of treatment (Ribas et al, 2015). Overall, AKT being a central mediator within this PI3K/AKT/mTOR pathway poses a rational therapeutic target in endocrine resistant breast cancer. In addition to the recent US FDA approval of capivasertib (16 November 2023), numerous studies already supported mechanistic evidence for the combination of AKT inhibitor and endocrine therapy to delay the onset of resistance, as well as re-sensitize endocrine-resistant tumors to the antiproliferative effects of endocrine therapy (Campbell et al, 2001; Miller et al, 2010; Ribas et al, 2015).

4.6. Elacestrant Background

Elacestrant is a tetrahydronaphthalene compound that acts as a SERD; it is being developed for treatment of ER+ advanced/mBC.

Elacestrant has been evaluated in vitro models and in patient-derived xenografts that represent acquired and de novo CDK4/6i resistance (Patel et al, 2019). Elacestrant demonstrated growth inhibition in cells resistant to palbociclib, abemaciclib, and ribociclib in both *ESR1* WT and mutant backgrounds. The authors also reported elacestrant downregulates several key cell cycle players and halts cell cycle progression in vitro and in vivo. They concluded breast cancer tumor cells continue to rely on ER signaling to drive tumor growth despite exposure to CDK4/6i. Elacestrant can inhibit this ER-dependent growth despite previously reported mechanisms of CDK4/6i resistance observed, such as Rb loss, CDK6 overexpression, upregulated cyclin E1, and E2F1, among others (Patel et al, 2019).

A recent study evaluated the safety and efficacy of elacestrant in a Phase 1 study of heavily pretreated postmenopausal women with ER+/HER2– mBC, including those with the *ESR1* mutation. Among the 57 postmenopausal women enrolled, 50 patients received the RP2D of 400 mg QD. The median age was 63 years, and there was a median of three prior anticancer therapies including CDK4/6i (in 52% of the population), SERD (52%), and *ESR1* mutation (circulating tumor DNA 50%). There were no DLTs and the most common adverse events at the RP2D (400 mg tablet, n=24) were nausea (33.3%), increased blood triglycerides, and decreased blood phosphorus (25.0% each). The objective response rate (ORR) was 19.4% (n=31 evaluable patients receiving the RP2D), 15.0% in patients with prior SERD, 16.7% in patients with prior CDK4/6i, and 33.3% in patients with *ESR1* mutation (n=5; 5/15). The clinical benefit rate (24 weeks) was 42.6% overall (n=47 patients receiving the RP2D), 56.5% (n=23, *ESR1* mutation), and 30.4% (n=23, prior CDK4/6i). Elacestrant clinical benefit was associated with a decline in *ESR1* mutant allele fraction (Bardia et al, 2021).

Oral administration of elacestrant had an absolute bioavailability of 10% and a mean $t_{1/2}$ ranging from 27 to 47 hours and reached steady state after 5 to 6 days. Mean occupancy of the ER in the uterus after 7 daily doses was 83% among those who received 200 mg and 92% among those who received 500 mg QD (Conlan et al, 2020).

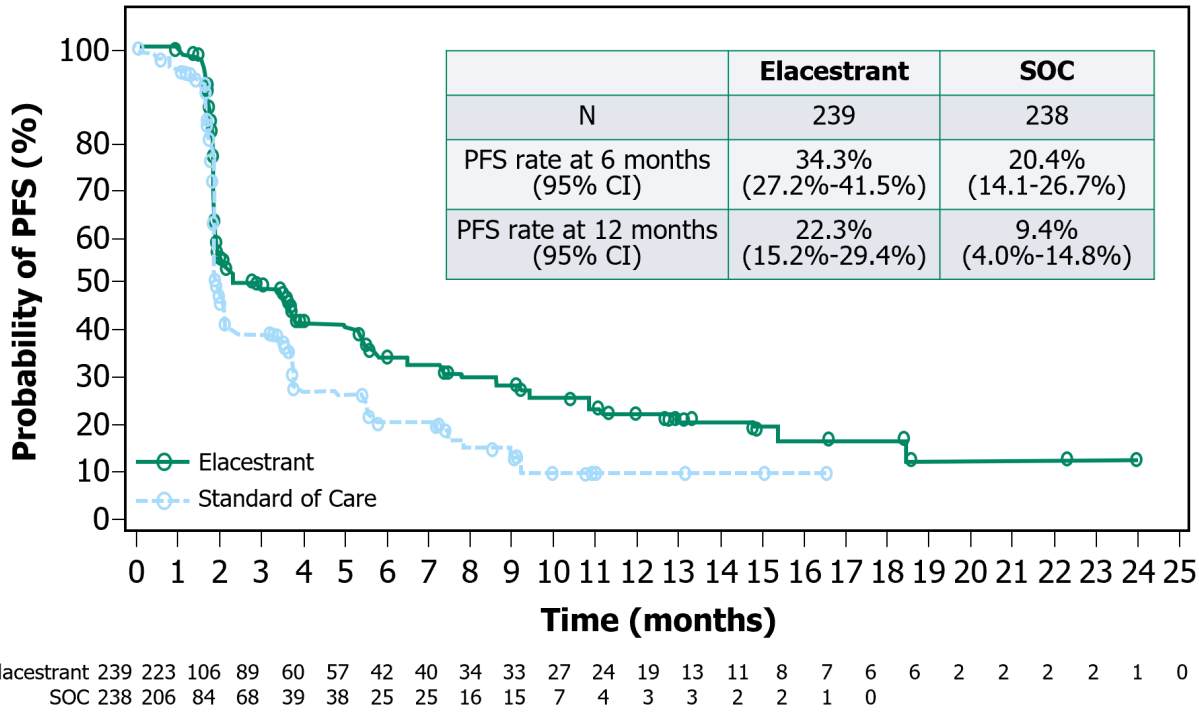
Results from the Phase 3 EMERALD study compared elacestrant monotherapy 400 mg QD (239 patients) with the Investigator's choice of endocrine monotherapy SOC using fulvestrant, anastrozole, letrozole, or exemestane in 238 patients who had ER+/HER2– advanced/mBC following CDK4/6i therapy. The study had 2 primary endpoints: PFS in all patients and PFS in

patients with tumors harboring *ESR1* mutations. Baseline patient and tumor characteristics were similar in both treatment arms (Bidard et al, 2022).

In all patients, elacestrant was associated with a 30% reduction in hazard of progression or death, with a median PFS of 2.79 months versus 1.91 months on elacestrant and SOC, respectively (*P*=0.0018).

The 6- and 12-month PFS estimates in all patients were 34.3% and 22.3%, respectively, in the elacestrant arm and 20.4% and 9.4%, respectively, in the SOC arm (Figure 1).

Figure 1: Progression-free Survival of Elacestrant and Standard of Care - All Patients

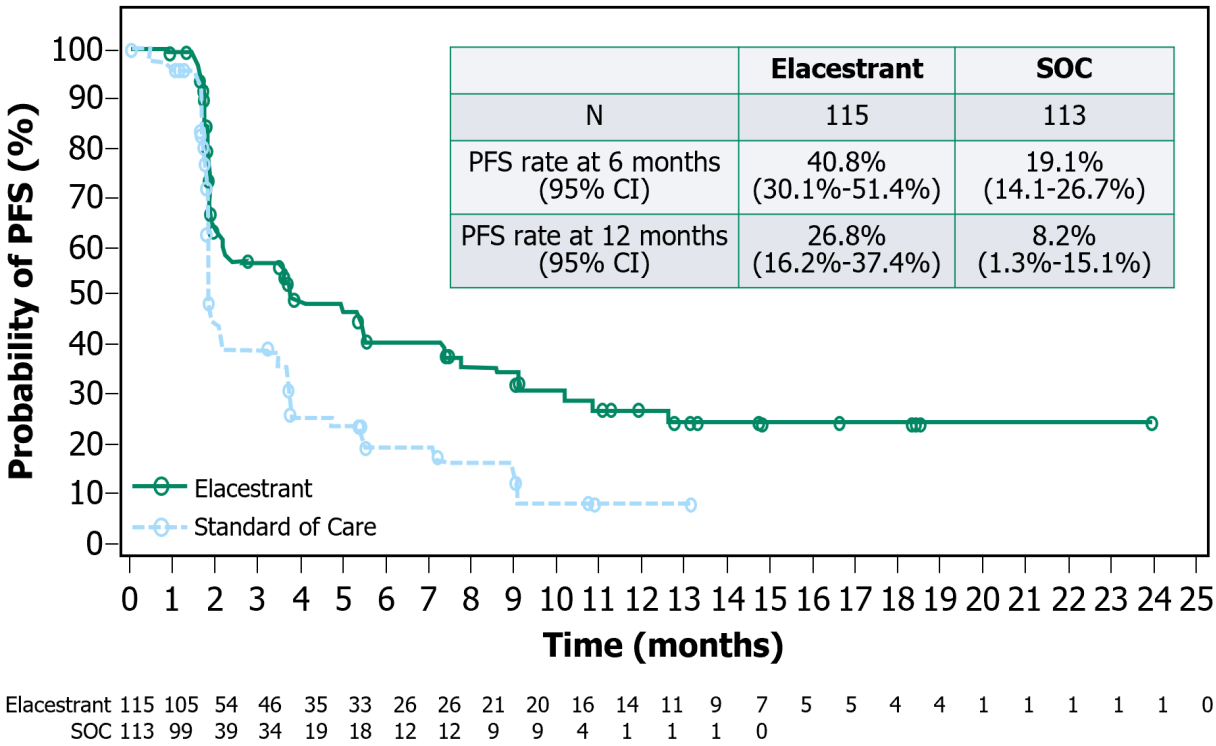


	Elacestrant	SOC
N	239	238
PFS rate at 6 months (95% CI)	34.3% (27.2%-41.5%)	20.4% (14.1-26.7%)
PFS rate at 12 months (95% CI)	22.3% (15.2%-29.4%)	9.4% (4.0%-14.8%)

CI=confidence interval; PFS=progression-free survival; SOC=standard of care.

Among patients with *ESR1* mutations, elacestrant was associated with 45% reduction in hazard of progression or death, with a median PFS of 3.78 versus 1.87 months on elacestrant and SOC, respectively (*P*=.0005). Similarly, the 6- and 12-month PFS estimates were 40.8% and 26.8%, respectively, in the elacestrant arm and 19.1% and 8.2%, respectively, in the SOC arm. The PFS results were consistent in many prespecified subpopulations (Figure 2).

Figure 2: Progression-free Survival of Elacestrant and Standard-of-Care Patients with *ESR1* Mutations



	Elacestrant	SOC
N	115	113
PFS rate at 6 months (95% CI)	40.8% (30.1%-51.4%)	19.1% (14.1-26.7%)
PFS rate at 12 months (95% CI)	26.8% (16.2%-37.4%)	8.2% (1.3%-15.1%)

Elacestrant	115	105	54	46	35	33	26	26	21	20	16	14	11	9	7	5	5	4	4	1	1	1	1	1	0
SOC	113	99	39	34	19	18	12	12	9	9	4	1	1	1	0										

CI=confidence interval; PFS=progression-free survival; SOC=standard of care.

Elacestrant was well tolerated with a manageable safety profile consistent with other hormonal therapies (Table 3). No fatal TEAEs were reported in either of the 2 treatment arms, and only 3 SAEs were considered related to elacestrant (elacestrant IB).

It was determined that elacestrant has the potential to become a new treatment option in the studied patient population (Bidard et al, 2022).

4.7. Overview of Elacestrant in Combination with Other Drugs

4.7.1. Efficacy Assessments

Each of the following drugs used in combination with elacestrant has been approved for treatment of breast cancer. Additional information concerning the efficacy of each is available in USPI and SmPC documents for each drug.

Alpelisib (ATC code L01EM03) is an alpha-specific phosphatidylinositol 3-kinase inhibitor indicated in combination with fulvestrant for treatment of males and postmenopausal females who have HR+/HER2–, PIK3CA-mutated, advanced or mBC, as detected by an FDA-approved test following progression on or after an endocrine-based drug regimen (alpelisib USPI, 2022).

Everolimus (ATC code L01EG02) is an mTOR indicated for treatment of postmenopausal women who have advanced HR+/HER2– breast cancer in combination with exemestane after failure of treatment with letrozole or anastrozole; treatment with everolimus and AIs enhances the antitumor activity (everolimus USPI, 2022).

Palbociclib (ATC code L01EF01) is an inhibitor of CDK4/6 indicated for treatment of adult patients with HR+/HER2– advanced or mBC in combination with:

- An AI as initial endocrine-based therapy in postmenopausal women or in men; or
- Fulvestrant in patients with disease progression following endocrine therapy (palbociclib USPI, 2019).

Abemaciclib (ATC code L01EF03) is a CDK4/6 inhibitor indicated:

- In combination with an AI as initial endocrine-based therapy for treatment of postmenopausal women, and men who have HR+/HER2– advanced or mBC;
- In combination with fulvestrant for treatment of adult patients who have HR+/HER2– advanced or mBC with disease progression following endocrine therapy;
- As monotherapy for treatment of adult patients with HR+/HER2– advanced, or mBC with disease progression following endocrine therapy and prior chemotherapy in the metastatic setting (abemaciclib USPI, 2023);
- In combination with endocrine therapy (tamoxifen or an AI) for adjuvant treatment of adult patients who have HR+/HER2–, node-positive, early breast cancer at high risk of recurrence and a Ki-67 score ≥20%, as determined by an FDA-approved test

Ribociclib (ATC code L01EF02) is a CDK4/6 inhibitor indicated in combination with:

- An AI for treatment of pre/perimenopausal or postmenopausal women with HR+/HER2– advanced or mBC as initial endocrine-based therapy; or
- Fulvestrant for treatment of postmenopausal women with HR+/HER2– advanced, or mBC as initial endocrine-based therapy or following disease progression on endocrine therapy (ribociclib USPI, 2022).

Capivasertib is a kinase inhibitor indicated, in combination with fulvestrant for the treatment of adult patients with HR+/HER2– locally advanced or metastatic breast cancer with one or more PIK3CA/AKT1/PTEN- alterations as detected by an FDA-approved test following progression on at least one endocrine-based regimen in the metastatic setting or recurrence on or within 12 months of completing adjuvant therapy (capivasertib USPI, 2023).

4.7.2. Safety

Review of available elacestrant safety information shows minimal overlap of common ADRs, except nausea, defined here as any adverse reaction reported in ≥15% of patients in any of the pivotal studies, which include alpelisib (SOLAR-1 [Andre et al, 2019]), everolimus (BOLERO-2 [Baselga et al, 2012]), palbociclib (PALOMA-2 [Finn et al, 2016]), abemaciclib (MONARCH 2 [Sledge et al, 2017] and MONARCH 3 [Goetz et al, 2017]), ribociclib (MONALEESA-2 [Hortobagyi et al, 2016]), capivasertib (CAPItello-291 [Turner et al, 2023]) and elacestrant (EMERALD [Bidard F-C et al, 2022]). Details are available in US prescribing information for each of the combination drugs and in the elacestrant IB and Table 3.

Table 3: Adverse Drug Reactions Observed in ≥15% in Any of the Pivotal Studies of Combination Drugs

Adverse Reaction	Alpelisib + fulvestrant (N=284, SOLAR 1)	Everolimus + exemestane (N=482, BOLERO 2)	Palbociclib + letrozole (N=444, PALOMA-2)	Abemaciclib + AI (N=327, MONARCH 3) / + fulvestrant (N=441, MONARCH 2)	Ribociclib + letrozole (N=334, MONALEESA-2)	Capivasertib+ fulvestrant (N = 355, CAPItello-291)	Elacestrant (N=237, EMERALD
Infections	NR	50	60	39 / 43	NR	NR	1
Pneumonitis	NR	19	NR	NR	NR	NR	0
Neutropenia	NR	NR	80	41 / 46	75	NR	1
Leukopenia	NR	NR	39	21 / 28	33	NR	<1
Anemia	NR	NR	24	28 / 29	18	10	3
Thrombocytopenia	NR	NR	16	10 / 16	NR	NR	<1
Diarrhea	58	33	26	81 / 86	35	72	8
Constipation	NR	NR	NR	16 / NR	25	NR	5
Nausea	45	29	35	39 / 45	52	35	25
Stomatitis	30	67	30	NR / 26	12	15	1
Vomiting	27	17	16	28 / 26	29	NR	11
Abdominal pain	17	NR	NR	29 / 35	11	21	2
Fatigue	42	36	37	40 / 46	37	21	11
Back pain	NR	NR	NR	NR	20	NR	<1
Asthenia	NR	NR	17	NR		13	5
Mucosal inflammation	19	NR	NR	NR / NR	NR	NR	0
Edema peripheral	15	19	NR	NR / NR	12	NR	<1
Pyrexia	14	15	12	NR	13	NR	0
Weight decreased	27	25	NR	NR / NR	NR	NR	<1
Decreased appetite	36	30	NR	24 / 27	19	17	8
Hyperglycemia	NR	14	NR	NR / NR	NR	16	0
Dysgeusia	18	22	10	NR / 18	NR	NR	<1

Adverse Reaction	Alpelisib + fulvestrant (N=284, SOLAR 1)	Everolimus + exemestane (N=482, BOLERO 2)	Palbociclib + letrozole (N=444, PALOMA-2)	Abemaciclib + AI (N=327, MONARCH 3) / + fulvestrant (N=441, MONARCH 2)	Ribociclib + letrozole (N=334, MONALEESA-2)	Capivasertib+ fulvestrant (N = 355, CAPItello-291)	Elacestrant (N=237, EMERALD
Headache	18	21	NR	NR / 20	22	17	4
Cough	NR	24	NR	13 / 13	NR	NR	1
Blood creatinine increased	NR	NR	NR	19 / 12	NR	NR	0
Pruritis	18	13	NR	NR / NR	14	12	1
Dry skin	18	NR	12	NR / NR	NR	NR	1
Hot flashes	NR	6	NR	NR / NR	NR	NR	10
Dyspepsia	11	NR	NR	NR / NR	NR	NR	6
ALT increase	NR	NR	NR	16 / 13	18	NR	2
AST increase	NR	NR	NR	15 / 12	18	NR	3
Arthralgia	NR	20	NR	NR / NR	NR	NR	4
Epistaxis	NR	17	NR	NR / NR	NR	NR	1
Rash*	36	36	18	NR/11	17	38	NR

ALT=alanine aminotransferase; AST=aspartate aminotransferase; NR=not reported. * Some reports include combined terms for rash.
Sources: Andre et al, 2019; Bardia et al, 2021; Baselga et al, 2012; Bidard F-C et al, 2022; Finn 2016; Goetz et al, 2017; Hortobagyi et al, 2016; Sledge et al, 2017;Turner et al, 2023.

4.8. Rationale

The data presented in this section underscore the potential clinical utility for this clinical trial considering elacestrant as an endocrine backbone for combination with agents targeting resistant or alternative signaling pathways in both CDK4/6i-pretreated or CDK4/6i-naïve patients.

Based on the growing evidence of preclinical elacestrant activity together with the emerging superior efficacy data of elacestrant monotherapy in mBC patients after receiving prior endocrine therapy alone or in combination, there is support for the continued development of elacestrant in combination with inhibitors of the PI3K pathway, such as alpelisib, capivasertib and everolimus, as well as the combination of elacestrant and the currently available CDK4/6i (e.g., palbociclib, ribociclib, and abemaciclib), to overcome emerging resistance.

Also, it is reasonable to expect that replacing the hormonal component with elacestrant in combination treatments, should not only improve clinical outcomes in this patient population, but should also provide a much needed, desirable, and convenient alternative of an oral–oral combination for patients.

4.9. Overall Benefit-Risk Assessment

The safety profile of single agent elacestrant is considered tolerable and manageable as demonstrated by the low frequency of dose modifications and incidence of TEAEs leading to permanent discontinuation, and no important identified or potential risks have been established for elacestrant in the target population to date. Of note, elacestrant was not associated with cardiac safety issues, and hematological AEs were rare.

In addition, the oral route of administration of elacestrant is more convenient and tolerable relative to the IM administration of fulvestrant. Fulvestrant requires monthly IM injections. Injection site–related events including sciatica, neuralgia, neuropathic pain, and peripheral neuropathy have been reported with fulvestrant injection.

- According to an exposure-response analysis for safety based on 32 response evaluable- patients with advanced ER+/HER2– breast cancer from studies RAD1901-005 and RAD1901-106, a positive and statistically significant relationship was found between elacestrant exposure (AUC_{0-24}, C_{ave}, C_{min}, and C_{max}) and probability of AEs (MENA-PMX-RAD1901-3863). This analysis indicates that elacestrant exposures associated with doses above 400 mg are expected to lead to increased risk of AEs of Grade 3 or more, AEs leading to treatment discontinuation, and SAEs. These findings are consistent with the trend observed in Study RAD1901-005, in which patients in the 400 mg elacestrant dihydrochloride QD dose group experienced fewer gastrointestinal TEAEs than patients in the 600 mg elacestrant hydrochloride QD dose group.
- Preclinical studies also showed a favorable effect of elacestrant versus fulvestrant.
- Available clinical data in patients (EMERALD Clinical Trial data; 06 September 2021 data cut-off; elacestrant IB) is consistent with preclinical results.

- In addition to preclinical activity and clinical efficacy, the manageable safety profile, and the oral convenience, indicate that elacestrant can be an important treatment option for patients with ER+/HER2– breast cancer.

Given the positive benefit demonstrated of elacestrant in patients with mBC from Study RAD1901-308, the anticipated favorable clinical benefits of elacestrant in other indications and combinations are projected to outweigh the risks. As noted in Section 4.7.2, the AEs associated with elacestrant and the combination drugs in this study show minimal overlap. Given the manageable AE profiles demonstrated with fulvestrant with alpelisib (SOLAR-1), fulvestrant and everolimus (PrE0102 study [Kornblum et al, 2018]), fulvestrant and palbociclib (PALOMA-3 [Goetz et al, 2017]), fulvestrant and abemaciclib (MONARCH-2 [Sledge et al, 2017]), and fulvestrant with ribociclib (MONALEESA-3 [Slamom et al, 2020]), and fulvestrant with capivasertib (CAPItello-291 [Turner et al, 2023]) the benefit-risk profile of elacestrant remains positive and supports further development in these specific combinations.

Based on the emerging safety data of the elacestrant plus alpelisib combination (see Section 6.2.), the Clinical recommendation for patients receiving this combination includes prophylaxis with loratadine for up to one month as tolerated in the first cycle of treatment with alpelisib and elacestrant. Additionally, short-course systemic steroids can be initiated if Grade 3 rash occurs, as per investigator's clinical judgment.

Additionally, based on the SmPC for everolimus, the use of prophylactic steroid mouthwash is strongly encouraged to help for the prevention of mucositis in patients receiving the combination of elacestrant plus everolimus.

5. TRIAL OBJECTIVES AND ENDPOINTS

The objectives of Phase 1b and Phase 2 and corresponding endpoints are provided in Table 4.

Table 4: Objectives and Endpoints, Phase 1b and Phase 2

Objectives	Endpoints
Phase 1b	
Primary • Determine the RP2D of elacestrant in combination with each of the other study drugs	• Number of DLTs observed during the first cycle
Secondary • Characterize the safety of elacestrant in combination with each of the other study drugs • Describe the plasma (blood for everolimus) PK of elacestrant and each of the combination drugs and their major metabolites and explore any potential DDIs • Evaluate the efficacy of elacestrant in combination with each of the other study drugs for ORR as per RECIST v. 1.1, DoR, CBR, PFS, and OS	• AEs and SAEs, as well as changes in clinical laboratory values, vital signs measurements and ECGs • Standard plasma (blood for everolimus) PK parameters including, but not limited to, the area under the $AUC_{0\text{-}tau}$, C_{max}, T_{max}, and C_{trough} • ORR: proportion of patients who achieve a BOR of confirmed PR or CR • DoR: time from the date of the first documented CR/PR until the first radiological documentation of disease progression or death, whichever comes first • CBR: proportion of patients who achieve a BOR of confirmed CR/PR or durable stable disease (duration at least 24 weeks from the date of first dose) • PFS: time from the date of the first dose to the date of the first radiological documentation of disease progression or death, whichever occurs first • OS: time from the date of the first dose to the date of death from any cause

Exploratory • Assess the PD of elacestrant in combination with each of the other drugs • Evaluate relationship between efficacy and PD • Evaluate the efficacy of each combination according to *ESR1* mutation status at baseline • Evaluate the exposure-response for efficacy and safety of elacestrant and the combination drugs.	• Changes in biomarkers including, but not limited to, allele mutation frequencies in cfNA in blood • Relationship between efficacy endpoints and allele mutation frequencies (at baseline and post baseline) • ORR, DoR, CBR, PFS, and OS • Relationship between elacestrant and combination drugs exposure and response (efficacy and safety).
Phase 2	
Primary • Evaluate the efficacy of elacestrant in combination with each of the other drugs for PFS	• Estimation of PFS rate at 6 months for each of the combination arms in patients who received prior ET and a CDK4/6i in the metastatic setting • Estimation of PFS rate at 12 months for all the combinations with palbociclib, abemaciclib, and ribociclib as second/third-line treatment in the metastatic setting in patients who received no prior CDK4/6i in the metastatic setting (Arm D)
Secondary • Evaluate the efficacy of elacestrant in combination with each of the other drugs for additional efficacy endpoints • Further characterize the safety of elacestrant in combination with each of the other study drugs	• ORR, DoR, CBR, PFS, and OS • AEs and SAEs, as well as changes in clinical laboratory values, vital signs, and ECG
Exploratory • Assess the PD of each combination • Evaluate the relationship between efficacy and PD • Evaluate the efficacy of each combination according to *ESR1* mutation status at baseline • Evaluate the exposure-response for efficacy and safety of elacestrant and the combination drugs.	• Changes in biomarkers including, but not limited to, allele mutation frequencies in cfNA in blood • Relationship between efficacy endpoints and allele mutation frequencies (at baseline and post baseline) • ORR, DoR, CBR, PFS, and OS • Relationship between elacestrant and combination drugs exposure and response (efficacy and safety).

AE=adverse event; AI=aromatase inhibitor; $AUC_{0\text{-}tau}$=area under the plasma concentration-time curve over the dosing interval; BOR=best overall response; CBR=clinical benefit rate; cfNA=cell-free nucleic acid; C_{max}=maximum observed plasma concentration; CR/PR=complete response/partial response; C_{trough}=predose trough concentration; DDI=drug-drug interaction; DLT=dose-limiting toxicity; DoR=duration of response;

ECG=electrocardiogram; *ESR1*= estrogen receptor gene alpha; SAE=serious adverse event; ORR=overall response rate; PFS=progression-free survival, and OS=overall survival, PD=pharmacodynamic; PK=pharmacokinetic; RECIST=Response Evaluation Criteria in Solid Tumors; RP2D=recommended Phase 2 dose; T_{max}=time of the maximum observed plasma concentration.

6. INVESTIGATIONAL PLAN

6.1. Overall Study Design

This is a multicenter, Phase 1b/2 trial. Phase 1b aims to determine the RP2D of elacestrant when administered in combination with alpelisib, everolimus, palbociclib, ribociclib, and capivasertib. Phase 2 will evaluate the efficacy and safety of the various combinations in patients with ER+/HER2– advanced/metastatic breast cancer.

6.2. Phase 1b

The Phase 1b portion of the trial will enroll participants in each combination at starting doses that are lower than their respective recommended dosage instructions (Table 5, Table 6, Section 6.6.1 to Section 6.6.6). The lower starting doses are used in the Phase 1b portion to ensure safety in combination with elacestrant through the DLT evaluation period (Section 6.2). Once patients in the lower doses have completed their first cycle without DLTs or safety concerns as determined by the Dose Escalation Committee (Section 6.7), dose levels will be increased in the stepwise manner as described (Table 5 and Table 6) to get to the highest possible combination doses without incurring unwanted safety issues. For each combination, guidance is provided to the Investigator for stepwise dose reductions and possible scenarios that may require dose modifications (Table 9 to Table 17). Once the RP2D has been determined for each combination based on the totality of safety, tolerability, and PK data, the Phase 2 arm may be opened for enrollment. Of note, the Sponsor in consultation with the SSC, may close a specific combination without determining a RP2D if the combination becomes outdated (i.e., change in treatment paradigms or landscape) or if the combination does not seem to offer a clear benefit compared to risk.

For each combination, this phase is expected to have between 1 and 3 cohorts of up to 6 DLT-evaluable patients each, however, additional patients, additional cohorts and/or intermediate doses may be enrolled as needed to determine the RP2D if recommended by the study steering committee.

Table 5 and Table 6 present the dose levels for elacestrant and each of the combination drugs that will be evaluated in Phase 1b based on the number of DLT in each cohort. The doses of 100, 200, 300, and 400 mg of elacestrant dihydrochloride correspond to the doses of 86, 172, 258, and 345 mg of elacestrant, respectively.

As of 20 December 2023, the status of each combination is as follows:

In the **elacestrant plus alpelisib** treatment arm, 2 DLTs of Grade 3 rash attributed to alpelisib were observed in the first 3 patients treated in Cohort 1 (300 mg QD elacestrant, 250 mg QD alpelisib), meeting the pre-specified DLT criteria as per protocol v1.0. An ad hoc DEC meeting was held to discuss the 3 patients in this combination, and it was decided to move to an alpelisib dose level -1 (200 mg QD) and to treat an additional 6 patients in this combination (300 mg elacestrant + 200 mg alpelisib). Depending on how many DLTs are observed in the Cohort alpelisib dose level-1 per Section 6.2.2, if moving forward with this combination, the next cohort for the alpelisib and elacestrant may be Cohort 0 (≤1 DLT observed) or Cohort -2 (≥ 2 DLTs observed) (Table 5).

Table 5 Dose Level of Elacestrant and Alpelisib to be Assessed in Phase 1b

Dose Level	Elacestrant dihydrochloride (mg)	Alpelisib (mg)
Cohort 1- closed with 2 DLTs	300 QD	250 QD
Cohort alpelisib dose level-1[a]	300 QD	200 QD
Cohort 0	400 QD	200 QD
Cohort -2	300 QD	150 QD

[a] Please note: Based on programming constraints, in the database and IRT, all 9 patients treated (N=3 for 250 mg alpelisib and N=6 for 200 mg alpelisib) are grouped together as "Cohort 1" patients. For reporting purposes, the different starting doses of these 9 patients will be presented and clarified whenever Cohort 1 is mentioned.

In the **elacestrant plus everolimus** treatment arm, Cohort 1 (300 mg QD elacestrant + 5.0 mg QD everolimus) and Cohort 2 (400 mg QD elacestrant + 5.0 mg QD everolimus) have enrolled 6 patients each with no DLTs observed in the first cycle of treatment on study as reviewed and discussed at their respective DEC meetings. After review of the Cohort 1 and Cohort 2 data, the Study Steering Committee agreed to proceed with Cohort 3 treating patients with 400 mg QD elacestrant and 10 mg QD everolimus.

If 2 or more DLTs are identified in Cohort 3, the SSC may request to investigate a fourth cohort (400 mg QD elacestrant and 7.5 mg QD of everolimus); however, enrollment of Cohort 4 is not required if Cohort 3 becomes the RP2D dose.

Table 6: Dose Level of Elacestrant and Everolimus to be Assessed in Phase 1b

Dose Level	Elacestrant dihydrochloride (mg)	Everolimus (mg)[a]
Cohort 1 – completed, no DLTs	300 QD	5.0 QD
Cohort 2 – completed, no DLTs	400 QD	5.0 QD
Cohort 3	400 QD	10 QD
Cohort 4 (if needed)	400 QD	7.5 QD

[a] Based on data from Cohorts 1 and 2, everolimus 10 mg may be used in Cohort 3 after discussion with the Study Steering Committee.

In the **elacestrant plus palbociclib** treatment arm, Cohort 1 (300 mg elacestrant + 100 mg palbociclib) has enrolled 6 patients with no DLTs observed in the first cycle of treatment on study as reviewed and discussed at the DEC meeting. Cohort 2 is currently enrolling patients.

Table 7: Dose Level of Elacestrant and Palbociclib to be Assessed in Phase 1b

Dose Level	Elacestrant dihydrochloride (mg)	Palbociclib (mg)
Cohort 1- completed, no DLTs	300 QD	100 QD **21 days on, 7 days off**
Cohort 2	400 QD	100 QD **21 days on, 7 days off**
Cohort 3	400 QD	125 QD **21 days on, 7 days off**

In the **elacestrant plus ribociclib** treatment arm, Cohort 1 (100 mg elacestrant + 400 mg ribociclib) and Cohort 2 (200 mg elacestrant + 400 mg ribociclib) have enrolled 6 patients each, with no DLTs observed in the first cycle of treatment on study, as reviewed and discussed at the respective DEC meetings. After review of the Cohort 1 and Cohort 2 data, the Dose Escalation Committee agreed to proceed with Cohort 3, treating 6 patients with 300 mg QD elacestrant and 400 mg QD ribociclib taken 21 days with 7 days off drug in each cycle. Cohort 3 is currently enrolling patients.

Table 8: Dose Level of Elacestrant and Ribociclib to be Assessed in Phase 1b

Dose Level	Elacestrant dihydrochloride (mg)	Ribociclib (mg)[a]
Cohort 1- completed, no DLTs	100 QD	400 QD **21 days on, 7 days off**
Cohort 2- completed, no DLTs	200 QD	400 QD **21 days on, 7 days off**
Cohort 3	300 QD	400 QD **21 days on, 7 days off**
Cohort 4 (if needed)	100, 200, 300 or 400 QD based on PK	400, 600 QD **21 days on, 7 days off**

[a] Based on data from Cohort 1, Cohort 2 and Cohort 3, elacestrant 400 mg and/or ribociclib 600 mg may be used in Cohort 4 after discussion with the Study Steering Committee.

The combination of **elacestrant and capivasertib** was added to the protocol with the approval of capivasertib by the US FDA on 16 November 2023. The proposed Cohorts 1-3 (+/- Cohort –1) doses are provided in Table 9, in which elacestrant is given QD and capivasertib is taken BID every 4 days with 3 days off drug.

Table 9: Dose Level of Elacestrant and Capivasertib to be Assessed in Phase 1b

Dose Level	Elacestrant dihydrochloride (mg)	Capivasertib (mg)
Cohort –1[a]	300 QD	200 BID 4 days on/3 days off
Cohort 1	300 QD	320 BID 4 days on/3 days off
Cohort 2	400 QD	320 BID 4 days on/3 days off
Cohort 3	400 QD	400 BID 4 days on/ 3 days off

[a] Based on data from previous cohorts, Cohort –1 may be conducted at the indicated dose levels if needed.

6.2.1. Dose-Limiting Toxicity

Phase 1b aims at selecting the RP2D dose, defined as a dose associated with less than 33% of patients experiencing a DLT, that is, ≤1 patient experiencing a DLT out of 6 DLT-evaluable patients. DLT is based on the NCI CTCAE v5.0. DLT refers to toxicities experienced during Cycle 1 by patients enrolled in Phase 1b.

Patients who do not receive approximately 70 percent of the scheduled therapy of either drug during Cycle 1, for reasons other than toxicity, will not be considered evaluable for DLT assessment and will be replaced if recommended by the steering committee. Disease progression and AEs deemed related to disease progression will not be considered DLTs.

For each combination, Phase 1b may have between 6 and 18 DLT-evaluable patients, however, additional patients, additional cohorts and/or intermediate doses may be enrolled as needed to determine the RP2D if agreed to by the DEC and/or study steering committee.

Any of the toxicities listed in Table 8 that are not clearly due to breast cancer or extraneous causes will be considered a DLT.

Table 8: Definition of Dose-limiting Toxicity

Toxicity	Criteria
Death	—
Hematology	○ Febrile neutropenia ○ Grade 4 neutropenia that does not resolve to Grade ≤3 within 8 days ○ Grade 4 thrombocytopenia

Toxicity	Criteria
	○ Grade 3 thrombocytopenia lasting >8 days ○ Grade 3 thrombocytopenia with bleeding
Fatigue	○ Grade 3 lasting for ≥7 days
Nausea	○ Grade ≥3 lasting >72 hours despite optimal antiemetic treatment and other supportive treatment
Vomiting	○ Grade 4 ○ Grade 3 lasting >72 hours despite optimal antiemetic treatment
Diarrhoea	○ Grade 4 diarrhoea ○ Grade 3 lasting >72 hours despite optimal antidiarrhoeal treatment
Electrolytes	○ Grade 3 or 4 associated with clinical signs and symptoms, regardless of duration
Renal	○ Grade 3 or 4 serum creatinine increase
Hepatic	○ Grade 3 or 4 bilirubin increase ○ ALT or AST ≥10 × ULN
Hy's law	○ ALT or AST >3 × ULN with bilirubin > 2 × ULN in the absence of viral hepatitis A, B or C; preexisting or acute liver disease, or another drug causing the observed injury
Biochemistry	○ Grade 4 ○ Grade 3 lasting >8 days, unless considered not clinically significant as per the discretion of the Investigator For the elacestrant plus alpelisib combination, ○ Note: Grade 3 hyperglycemia[a] is a DLT if thought to be related to elacestrant or alpelisib if fasting blood glucose values >250-500 mg/dl or >13.9-27.8 mmol/l with no other possible explanation
Other non-hematological	○ Grade 3 or 4 For the elacestrant plus alpelisib combination, ○ Onset of DLT of Grade 3 rash despite prophylaxis with antihistamines will qualify as a DLT; if prophylaxis not given, clinical analysis of each case at dose escalation committee meeting will determine if Grade 3 rash qualifies as DLT.

ALT=alanine aminotransferase; AST=aspartate aminotransferase; ULN=upper limit of normal.
[a] per CTCAE v4.03

6.2.2. Determination of the Recommended Phase 2 Dose

Cohort 1:

- Cohort 1: ≤1 patient experiences a DLT during the first cycle (28 days), enrollment to Cohort 2 will be initiated.

- If ≥2 patients experience DLTs in the first cohort, the combination at the doses evaluated at Cohort 1 will be considered not feasible and the Sponsor and the DEC and/or SSC may decide to test lower doses based on all available data including safety, PK, and PD. At this time, Cohort -1 or Cohort 0 (reduced dose of one or more drugs) may be opened.

Cohort 2:

- If ≤1 patient experiences a DLT during the first cycle, enrollment to Cohort 3 will be initiated.
- If ≥2 patients experience DLTs in the second cohort, the doses evaluated in Cohort 1 will be considered the RP2D of this combination.

Cohort 3:

- If ≤1 patient experiences a DLT during the first cycle, the doses evaluated in this cohort will be considered the RP2D for this combination, unless the SSC and/or DEC determine more patients should be treated at this or an intermediate dose level to determine the RP2D
- If ≥2 patients experience DLTs, the doses evaluated in Cohort 2 will be considered the RP2D of this combination, unless the SSC and/or DEC determine an intermediate dose level is needed to determine the RP2D

6.3. Phase 2

Eligible patients will receive elacestrant and combination treatment at the RP2D.

6.3.1. Arm A

Fifty (50) eligible patients (after prior therapy with 1 or 2 prior hormonal therapies in combination with a CDK4/6i in the metastatic setting) will be enrolled in Arm A and will receive the combination of elacestrant and alpelisib.

6.3.2. Arm B

Fifty (50) eligible patients (after prior therapy with 1 or 2 prior hormonal therapies in combination with a CDK4/6i in the metastatic setting) will be enrolled in Arm B and will receive the combination of elacestrant and everolimus.

6.3.3. Arm C

Sixty (60) eligible patients (after prior therapy with 1 or 2 prior hormonal therapies in combination with CDK4/6i in the metastatic setting) will be enrolled in Arm C and will receive the combination of elacestrant and abemaciclib or ribociclib (30 patients each).

6.3.4. Arm D

Ninety (90) eligible patients (after prior therapy with 1 or 2 prior hormonal therapies and no prior CDK4/6i in the metastatic setting) will be enrolled in Arm D will receive the combination of elacestrant and palbociclib, abemaciclib, or ribociclib (30 patients each).

6.3.5. Arm E

Sixty (60) eligible patients (after prior therapy with 1 or up to 2 prior hormonal therapies, CDKi are allowed but not required) with tumors that harbor one or more PIK3CA/AKT1/PTEN alterations will be enrolled in Arm E will receive the combination of elacestrant and capivasertib.

6.4. Treatment Assignment

Patient enrollment, assignment of patients to specific cohorts/treatment arms, and selection of certain combinations of drugs in certain arms of Phase 2 will be selected by the Investigator, based on the inclusion/exclusion criteria specific for each arm.

A patient list will be produced by the IRT provider using a validated system that automates the assignment of patient numbers to a treatment arm based on the Investigator's choice. Enrollment within the cohorts/treatment arm or a combination of drugs within an arm will be closed when the target number of patients is achieved.

6.5. Intra-patient Dose Escalation

For patients enrolled in the Phase 1b part of the trial, once a patient has completed Cycle 3 at the assigned dose level, intrapatient dose escalation will be allowed. Only patients who have not progressed on treatment and did not have a dose reduction because of AEs, may escalate to the highest dose level that has completed DLT assessment and has been shown to be safe (no more than 1 DLT in 6 patients). Patients who follow the intrapatient dose escalation will not be included in the DLT analysis at the higher dose level.

6.6. Dose Adjustment Criteria

Dose delay, interruption, or reduction and initiation of supportive care are allowed as clinically indicated by the treating investigator. Subjects who require a treatment interruption due to drug-related toxicity for more than 4 weeks should discontinue study treatment after consultation with the sponsor.

Monitor CBC prior to the start of trial therapy, and at the beginning of each cycle, and as clinically indicated. Laboratory values that meet the criteria listed in this section should be repeated and confirmed within 24 hours of the initial results being reported.

Recommendations for dose modification (delay, interruption, reduction, or discontinuation) for any of the study drugs in each combination are summarized in Table 10, Table 12, Table 14, Table 15, and Table 17. Additional dose reductions and/or interruptions are also allowed based on local practice and clinical judgment of the treating investigator.

6.6.1. Combination Doses and Rationale

Elacestrant was recently approved for treatment of postmenopausal women or adult men, with ER+/HER2–, *ESR1*-mutated advanced or metastatic breast cancer with disease progression following at least 1 line of endocrine therapy. The recommended dosage of elacestrant is 345 mg (1 × 400 mg tablet of elacestrant dihydrochloride) taken orally with food QD until disease progression or unacceptable toxicity occurs (elacestrant USPI, 2023). The daily 400 mg dose of elacestrant dihydrochloride was also the recommended Phase 2 dose established in early studies of elacestrant in advanced/metastatic breast cancer patients (Bardia et al, 2021).

In combination, the selection of elacestrant doses to be studied in Phase 1b was based on the risk of potential PK DDIs between elacestrant and the combination drug for each arm. Overall, when a potential PK DDI was expected with elacestrant either as a perpetrator or as a victim, the starting dose of elacestrant was decreased with respect to the approved dose of 345 mg to either 258 mg (i.e., 300 mg elacestrant dihydrochloride) when the DDI risk at the highest dose was deemed low (<2) or to 86 mg (i.e., 100 mg elacestrant dihydrochloride) when the DDI risk was considered high (>5).

The primary outcome of Phase 1b is to determine the RP2D of elacestrant in combination other each of the other study drugs, the starting dose of each drug is lower than their recommended dosages, as described in the sections below. The starting doses in Phase 1b are described below. Individual dosing recommendations for the combination drugs can be found in the IMPDs.

6.6.2. Combination with Alpelisib

The recommended dose of alpelisib is 300 mg (2 × 150 mg tablets) taken orally QD with food. There is a potential for PK DDI between elacestrant and alpelisib mediated by BCRP, as elacestrant is an inhibitor of BCRP and alpelisib is a BCRP substrate, leading to increase in alpelisib exposure. The increase in alpelisib exposure due to BCRP inhibition by elacestrant is expected to be low (an increase of 45% in C_{max} and 23% in AUC was observed for rosuvastatin [BCRP substrate] when coadministered with 345 mg elacestrant in a DDI clinical study). Consequently, the starting dose of alpelisib was selected to be 250 mg, or 50 mg/day lower than the recommended dose, while the starting dose of elacestrant was decreased to 258 mg (300 mg elacestrant dihydrochloride).

Alpelisib will be administered QD for 28 consecutive days (1 cycle).

Based on the occurrence of Grade 3 rash in Cohort 1 on study, the Clinical Recommendation for alpelisib + elacestrant patients includes: Prophylaxis with loratadine for up to one month as tolerated in the first cycle of treatment with alpelisib and elacestrant. Additionally, short-course systemic steroids can be initiated if Grade 3 rash occurs, as per investigator's clinical judgment.

The dose levels that will be evaluated during Phase 1b and reduction guidelines are presented in Table 9.

Table 9: Dose Levels and Reduction Guidelines for the Combination of Elacestrant and Alpelisib

Drug in Combination	Starting Dose	Dose -1	Dose -2
Alpelisib (mg)	250 QD	200 QD	150 QD
	200 QD	150 QD	100 QD
	150 QD	100 QD	Discontinue alpelisib
Elacestrant (mg)	300 QD	200 QD	100 QD
	400 QD	300 QD	200 QD

QD=once daily.

Recommendations for dose modification are presented in Table 10.

Table 10: Alpelisib Combination - Recommendation for Dose Modifications

CTCAE Grade	Alpelisib	Elacestrant
Hematologic Toxicity		
Grade 1 or 2	No dose adjustment is required.	No dose adjustment is required.
Grade 3 or 4	Withhold drug until recovery to Grade ≤2; resume at the same dose.	No dose adjustment is required.
Febrile neutropenia Grade 3 (ANC <1.0 × 10^9/L with single temperature of >38.3°C [101°F] or a sustained temperature of ≥38°C [100.4°F] for more than 1 hour)	Withhold drug until recovery to Grade ≤2; resume at the next lower dose.	No dose adjustment is required.
Febrile neutropenia (Grade 4: Life-threatening consequences; urgent intervention indicated)	Permanently discontinue.	Interrupt until improvement to Grade ≤ 2, and no fever. Resume at same dose level.
Hyperglycemia[a]		
FPG >ULN – 160 mg/dL or > ULN – 8.9 mmol/L (CTCAE v4.03 Grade 1)	No dose adjustment required. Initiate or intensify antidiabetic treatment.	No dose adjustment required.
FPG >160 – 250 mg/dL or >8.9 – 13.9 mmol/L (CTCAE v4.03 Grade 2)	No dose adjustment required. Further intensify antidiabetic treatment. If FPG does not decrease to ≤160 mg/dL or 8.9 mmol/L, within 21 days under appropriate antidiabetic treatment, reduce alpelisib dose by one dose level and follow FPG value-specific recommendations.	No dose adjustment required.
FPG >250 –500 mg/dL or >13.9 – 27.8 mmol/L (CTCAE v4.03 Grade 3)	Interrupt alpelisib. Initiate or intensify oral antidiabetic treatment[b] and consider additional antidiabetic medications[c] for 1 to 2 days until hyperglycemia improves. Administer intravenous hydration and consider appropriate treatment (e.g., intervention for electrolyte/ketoacidosis /hyperosmolar disturbances). If FPG decreases to ≤160 mg/dL or 8.9 mmol/L within 3 to 5 days under appropriate antidiabetic treatment, resume alpelisib at 1 lower dose level.	No dose adjustment required.

CTCAE Grade	Alpelisib	Elacestrant
	If FPG does not decrease to ≤160 mg/dL or 8.9 mmol/L within 3 to 5 days under appropriate antidiabetic treatment, consultation with a physician with expertise in the treatment of hyperglycemia is recommended. If FPG does not decrease to ≤ 160 mg/dL or 8.9 mmol/L within 21 days following appropriate antidiabetic treatment[b], permanently discontinue alpelisib treatment.	
FPG >500 mg/dL or >27.8 mmol/L (CTCAE v4.03 Grade 4)	Interrupt alpelisib. Initiate or intensify appropriate antidiabetic treatment[b] (administer intravenous hydration andconsider appropriate treatment [e.g., intervention for electrolyte/ ketoacidosis/ hyperosmolar disturbances]), recheck FPG within 24 hours and as clinically indicated. If FPG decreases to ≤500 mg/dL or 27.8 mmol/L, follow FPG value-specific recommendations for Grade 3. If FPG is confirmed at >500 mg/dL or 27.8 mmol/L, permanently discontinue alpelisib treatment.	No dose adjustment required.
Rash and SCARs		
Grade 1 (<10% BSA with active skin toxicity)	No dose adjustment required. Initiate topical corticosteroid treatment. Consider adding oral antihistamine to manage symptoms. If active rash is not improved within 28 days of appropriate treatment, add a low -dose systemic corticosteroid. If the etiology is SCAR, permanently discontinue alpelisib.	No dose adjustment required.
Grade 2 (10-30% BSA with active skin toxicity)	No alpelisib dose adjustment required. Initiate or intensify topical corticosteroid and oral	No dose adjustment required.

CTCAE Grade	Alpelisib	Elacestrant
	antihistamine treatment. Consider low-dose systemic corticosteroid treatment. If rash improves to Grade ≤1 within 10 days, systemic corticosteroid may be discontinued. If the etiology is SCAR, permanently discontinue alpelisib.	
Grade 3 (e.g., severe rash not responsive to medical management) (> 30% BSA with active skin toxicity)	Interrupt alpelisib. Initiate or intensify topical/systemic corticosteroid and oral antihistamine treatment. If the etiology is SCAR, permanently discontinue alpelisib. If the etiology is not a SCAR, interrupt dose until improvement to Grade ≤1, then resume alpelisib at next lower dose level.	No dose adjustment required.
Grade 4 (e.g., severe bullous, blistering or exfoliating skin conditions) (any % BSA associated with extensive superinfection, with IV antibiotics indicated; life-threatening consequences)	Permanently discontinue alpelisib.	No dose adjustment required.
Diarrhoea or Colitis		
Grade 1	No alpelisib dose adjustment is required. Initiate appropriate medical therapy and monitor as clinically indicated.	No dose adjustment required.
Grade 2	o Initiate or intensify appropriate medical therapy and monitor as clinically indicated. o Interrupt alpelisib dose until recovery to Grade ≤1, then resume alpelisib at same dose level. o For recurrent diarrhoea or colitis Grade ≥2, interrupt alpelisib dose until improvement to Grade ≤1, then resume alpelisib at the next lower dose level.	Interrupt elacestrant dose until recovery to Grade ≤1, then resume at same dose level.
Grade 3	Initiate or intensify appropriate medical therapy and monitor as	Interrupt elacestrant until recovery to Grade ≤1,

CTCAE Grade	Alpelisib	Elacestrant
	clinically indicated. Interrupt alpelisib dose until recovery to Grade ≤1, then resume alpelisib at the next lower dose level.	resume at the next lower dose level.
Grade 4	Permanently discontinue alpelisib.	Interrupt elacestrant until recovery to Grade ≤1, resume at the next lower dose level.
Other Toxicity		
Grade 1 or 2	No alpelisib dose adjustment is required. Initiate appropriate medical therapy and monitor as clinically indicated[b,c].	Grade 1: No dose adjustment is required. Grade 2: If toxicity becomes intolerable, withhold until improvement to Grade ≤1. Resume at same dose.
Grade 3	Interrupt alpelisib dose until recovery to Grade ≤1, then resume alpelisib at the next lower dose level.	Withhold if clinically indicated. When toxicity resolves to Grade 1 or baseline, resume at a lower dose level. If the Grade 3 toxicity recurs, withhold treatment. When toxicity resolves to Grade 1 or baseline, resume at another lower dose level.
Grade 4	Permanently discontinue alpelisib.	Withhold treatment. When toxicity resolves to Grade 1 or baseline, resume at a lower dose level. If the Grade 4 toxicity recurs, permanently discontinue.

ANC=absolute neutrophil count; BSA=body surface area; CTCAE=Common Terminology Criteria for Adverse Events, version 5.0; FPG=fasting plasma glucose; SCAR=severe cutaneous adverse reaction; ULN=upper limit of normal.
Note: Laboratory values that meet criteria listed in this table should be repeated and confirmed within 24 hours of the initial results being reported.

a. Hyperglycemia grading follows the CTCAE criteria version 4.03

b. Initiate applicable anti-diabetic medications, including metformin and insulin sensitizers (such as thiazolidinediones or dipeptidyl peptidase-4 inhibitors), and review respective prescribing information for dosing and dose titration recommendations, including local diabetic treatment guidelines. Metformin was recommended in the SOLAR-1 trial with the following guidance: Initiate metformin 500 mg QD. Based on tolerability, metformin dose may be increased to 500 mg twice daily, followed by 500 mg with breakfast, and 1,000 mg with dinner, followed by further increase to 1,000 mg twice daily if needed.

c. As recommended in the SOLAR-1 trial, insulin may be used for 1 to 2 days until hyperglycemia resolves. However, this may not be necessary in the majority of alpelisib-induced hyperglycemia, given the short half-life of alpelisib and the expectation of glucose levels normalizing after interruption of alpelisib.

Grading according to CTCAE Version 5.0. For Grade 2 and 3 pancreatitis, interrupt alpelisib dose until improvement to Grade ≤1 and resume at next lower dose level. Only one dose reduction is permitted. If toxicity recurs, permanently discontinue alpelisib treatment.

For Grade 2 total bilirubin elevation, interrupt alpelisib dose until recovery to Grade ≤1 and resume at the same dose if resolved in ≤14 days or resume at the next lower dose level if resolved in >14 days.

6.6.3. Combination with Everolimus

The recommended dose of everolimus is 10 mg tablet taken orally QD. There is a potential for PK DDI between elacestrant and everolimus mediated by:

- P-gp, as elacestrant is an inhibitor of P-gp and everolimus is a P-gp substrate, leading to an increase in everolimus exposure. However, the increase in everolimus exposure due to P-gp inhibition by elacestrant is expected to be low (an increase of 27% in C_{max} and 13% in AUC was observed for digoxin [P-gp substrate] when coadministered with 345 mg elacestrant in a DDI clinical study)
- CYP3A4, as everolimus is a CYP3A4 inhibitor and elacestrant is a CYP3A4 substrate, leading to an increase in elacestrant exposure. However, the increase in elacestrant exposure due to CYP3A4 inhibition by everolimus is expected to be low (coadministration of everolimus with midazolam [CYP3A substrate] in a clinical study resulted in 25% increase in midazolam C_{max} and 30% increase in midazolam AUC).

Consequently, the starting dose of everolimus was selected to be 5 mg, while elacestrant starting dose was decreased to 258 mg (300 mg elacestrant dihydrochloride).

Everolimus will be administered QD for 28 consecutive days (1 cycle).

The use of prophylactic steroid mouthwash is strongly encouraged to help for prevention of mucositis.

The dose levels that will be evaluated during Phase 1b and reduction guidelines are presented in Table 11.

Table 11: Dose Levels and Reduction Guidelines for the Combination of Elacestrant and Everolimus

Drug in Combination	Starting Dose/mg	Dose –1/mg	Dose –2/mg
Everolimus (mg)	5 mg QD	2.5 QD	2.5 every other day
	7.5 QD	5.0 QD	2.5 QD
	10 QD	7.5 QD	5.0 QD
Elacestrant (mg)	300 QD	200 QD	100 QD
	400 QD	300 QD	200 QD

QD=once daily.

Recommendations for dose modification are presented in Table 12.

Table 12: Everolimus Combination - Recommendation for Dose Modifications

CTCAE Grade	Everolimus	Elacestrant
Neutropenia		
Grade 1 (ANC <2.0 - 1.5 x 10^9 /L) or Grade 2 (ANC <1.5 – 1.0 x 10^9 /L)	No dose adjustment is required.	No dose adjustment is required.
Grade 3 (ANC <1.0 - 0.5 x 10^9 /L)	Withhold until improvement to Grade 0, 1, or 2. Resume at same dose.	No dose adjustment is required.
Grade 4 (ANC <0.5 x 10^9 /L)	Withhold until improvement to Grade 0, 1, or 2. Resume at 50% of previous dose; change to every other day dosing if the reduced dose is lower than the lowest available strength.	No dose adjustment is required.
Febrile neutropenia (Grade 3: ANC <1.0 x 10^9 /L with a single temperature of >38.3°C [101°F] or a sustained temperature of ≥38°C [100.4°F] for more than 1 hour)	Withhold until improvement to Grade ≤2 and no fever. Resume at 50% of previous dose; change to every other day dosing if the reduced dose is lower than the lowest available strength.	No dose adjustment is required.
Febrile neutropenia (Grade 4: Life-threatening consequences; urgent intervention indicated)	Permanently discontinue.	Interrupt until improvement to Grade ≤2 and no fever. Resume at same dose level.
Thrombocytopenia		
Grade 1 (platelet count <100.0 - 75.0 x 10^9 /L)	No dose adjustment is required.	No dose adjustment is required.
Grade 2 (platelet count <75.0 - 50.0 x 10^9 /L)	Withhold until improvement to Grade 0 or 1. Resume at same dose.	No dose adjustment is required.
Grade 3 or 4 (platelet count <50.0 x 10^9 /L)	Withhold until improvement to Grade ≤1. Resume at 50% of previous dose; change to every other day dosing if the reduced dose is lower than the lowest available strength.	No dose adjustment is required.
Non-infectious pneumonitis		
Grade 1	No dose adjustment is required.	No dose adjustment is required.
Grade 2	Withhold until improvement to Grade ≤1. Resume at 50% of previous dose; change to every other day dosing if the reduced dose is lower than the lowest available strength. Permanently discontinue if toxicity does not	No dose adjustment is required.

CTCAE Grade	Everolimus	Elacestrant
	resolve or improve to Grade 1 within 4 weeks.	
Grade 3	Withhold until improvement to Grade ≤1. Resume at 50% of previous dose; change to every other day dosing if the reduced dose is lower than the lowest available strength. If toxicity recurs at Grade 3, permanently discontinue.	No dose adjustment is required.
Grade 4	Permanently discontinue.	No dose adjustment is required.
Stomatitis		
Grade 2	Withhold until improvement to Grade ≤1. Resume at same dose. If recurs at Grade 2, withhold until improvement to Grade 0 or 1. Resume at 50% of previous dose; change to every other day dosing if the reduced dose is lower than the lowest available strength.	No dose adjustment is required.
Grade 3	Withhold until improvement to Grade ≤1. Resume at 50% of previous dose; change to every other day dosing if the reduced dose is lower than the lowest available strength.	No dose adjustment is required.
Grade 4	Permanently discontinue.	No dose adjustment is required.
Hyperglycemia[a]		
FPG >ULN – 160 mg/dL or > ULN – 8.9 mmol/L (CTCAE v4.03 Grade 1)	No dose adjustment is required.	No dose adjustment is required.
FPG >160 – 250 mg/dL or >8.9 – 13.9 mmol/L (CTCAE v4.03 Grade 2)	No dose adjustment is required.	No dose adjustment is required.
FPG >250 – 500 mg/dL or >13.9 - 27.8 mmol/L (CTCAE v4.03 Grade 3)	Withhold until improvement to Grade ≤2. Resume at 50% of previous dose; change to every other day dosing if the reduced dose is lower than the lowest available strength.	No dose adjustment is required.
FPG >500 mg/dL or >27.8 mmol/L (CTCAE v4.03 Grade 4)	Permanently discontinue.	No dose adjustment is required.
Metabolic events (e.g., dyslipidemia)		
Grade 1 or Grade 2	No dose adjustment is required.	No dose adjustment is required.

CTCAE Grade	Everolimus	Elacestrant
Grade 3	Withhold until improvement to Grade ≤2. Resume at 50% of previous dose; change to every other day dosing if the reduced dose is lower than the lowest available strength.	No dose adjustment is required.
Grade 4	Permanently discontinue.	No dose adjustment is required
Other non-hematologic toxicities		
Grade 1	No dose adjustment is required.	No dose adjustment is required.
Grade 2	If toxicity becomes intolerable, withhold until improvement to Grade ≤1. Resume at same dose. If toxicity recurs at Grade 2, withhold until improvement to Grade ≤1. Resume at 50% of previous dose; change to every other day dosing if the reduced dose is lower than the lowest available strength.	If toxicity becomes intolerable, withhold until improvement to Grade ≤1. Resume at same dose.
Grade 3	Withhold until improvement to Grade 0 or 1. Consider resuming at 50% of previous dose; change to every other day dosing if the reduced dose is lower than the lowest available strength. If recurs at Grade 3, permanently discontinue.	Withhold if clinically indicated. When toxicity resolves to Grade 1 or baseline, resume at a lower dose level. If the Grade 3 toxicity recurs, withhold treatment. When toxicity resolves to Grade 1 or baseline, resume at another lower dose level.
Grade 4	Permanently discontinue.	Withhold treatment. When toxicity resolves to Grade 1 or baseline, resume at a lower dose level. If the Grade 4 toxicity recurs, permanently discontinue.

Laboratory values that meet criteria listed in this table should be repeated and confirmed within 24 hours of the initial results being reported.
ANC=absolute neutrophil count; CTCAE=Common Terminology Criteria for Adverse Events, version 5.0; FPG=fasting plasma glucose; ULN=upper limit of normal.
a. Hyperglycemia grading follows the CTCAE criteria version 4.03.

6.6.4. Combination with Palbociclib

The recommended dose of palbociclib is 125 mg capsule taken orally QD with food. There is a potential for PK DDI between elacestrant and palbociclib mediated by CYP3A4, as palbociclib is

a weak CYP3A4 inhibitor and elacestrant is a CYP3A4 substrate. This may lead to an increase in elacestrant exposure. Consequently, the starting dose of palbociclib was selected to be 100 mg, and the starting dose of elacestrant was selected to be 258 mg (i.e., 300 mg elacestrant dihydrochloride).

Palbociclib will be administered QD for 21 consecutive days followed by 7 days off treatment to comprise a complete cycle of 28 days. Elacestrant will be administered QD continuously during the treatment cycle.

Dose levels to be evaluated during Phase 1b and reduction guidelines are presented in Table 13.

Table 13: Dose Levels and Reduction Guidelines for the Combination of Elacestrant and Palbociclib

Drug in Combination	Starting Dose	Dose -1	Dose -2
Palbociclib (mg)	100 QD	75 QD	75 every other day
	125 QD	100 QD	75 QD
Elacestrant (mg)	300 QD	200 QD	100 QD
	400 QD	300 QD	200 QD

QD=once daily.

Recommendations for dose modification are presented in Table 14.

Table 14: Palbociclib Combination - Recommendation for Dose Modifications

CTCAE Grade	Palbociclib	Elacestrant
Hematologic toxicity[a]		
Grade 1 or 2	No dose adjustment is required.	No dose adjustment is required.
Grade 3	Day 1 of cycle: Withhold palbociclib, repeat complete blood count monitoring within 1 week. When recovered to Grade ≤2, start the next cycle at the same dose. Day 15 of first 2 cycles: If Grade 3 on Day 15, continue palbociclib at current dose to complete cycle and repeat complete blood count on Day 22. If Grade 4 on Day 22, see Grade 4 dose modification guidelines below. Consider dose reduction in cases of prolonged (>1 week) recovery from Grade 3 neutropenia or recurrent Grade 3 neutropenia on Day 1 of subsequent cycles.	No dose adjustment is required.

CTCAE Grade	Palbociclib	Elacestrant
Febrile neutropenia Grade 3 (ANC <1.0 × 10^9/L) with a single temperature of >38.3°C (101°F) or a sustained temperature of ≥38.0°C (100.4°F) for more than 1 hour) neutropenia with fever ≥38.5 ºC and/or infection	At any time: Withhold palbociclib until recovery to Grade ≤2. Resume at the next lower dose.	No dose adjustment is required.
Grade 4	At any time: Withhold palbociclib until recovery to Grade ≤2. Resume at the next lower dose.	Interrupt until improvement to Grade ≤2, and no fever. Resume at same dose level.
Non-hematologic toxicity		
Grade 1 or 2	No dose adjustment is required.	No dose adjustment is required.
Grade ≥3 (if persisting despite optimal medical treatment)	Withhold until symptoms resolve to Grade ≤1 or Grade ≤2 (if not considered a safety risk for the patient). Resume at the next lower dose.	No dose adjustment is required.

Laboratory values that meet criteria listed in this table should be repeated and confirmed within 24 hours of the initial results being reported.
ANC=absolute neutrophil count; CTCAE=Common Terminology Criteria for Adverse Events, version 5.0
a. Guidance applies to all hematologic adverse reactions except lymphopenia (unless associated with clinical events, e.g., opportunistic infections).

6.6.5. Combination with Abemaciclib

The recommended Phase 2 dose for the combination of elacestrant and abemaciclib is evaluated in the ongoing ELECTRA trial (ClinicalTrials.gov Identifier: NCT04791384).

The recommended dose of abemaciclib in combination with fulvestrant, tamoxifen, or an AI is 150 mg taken orally twice daily, with or without food.

Abemaciclib will be administered twice daily for 28 consecutive days (1 cycle).

Drug in Combination	Starting Dose	Dose -1	Dose -2
Abemaciclib (mg)	100 mg BID	50 mg BID	50 mg QD
	150 mg BID	100 mg BID	50 mg BID
Elacestrant (mg)	300 QD	200 QD	100 QD
	400 QD	300 QD	200 QD

Recommendations for dose modification are presented in Table 15.

Table 15: Abemaciclib Combination - Recommendation for Dose Modifications

CTCAE Grade	Abemaciclib	Elacestrant
Hematologic toxicity		
Grade 1 or 2	No dose adjustment is required.	No dose adjustment is required.
Grade 3	Suspend dose until toxicity resolves to Grade ≤2. Dose reduction is not required.	No dose adjustment is required.
Grade 3, recurrent	Suspend dose until toxicity resolves to Grade ≤2. Resume at next lower dose.	No dose adjustment is required.
Grade 4	Suspend dose until toxicity resolves to Grade ≤2. Resume at next lower dose.	Suspend dose until toxicity resolves to ≤Grade 2. Resume at next lower dose.
Diarrhoea		
Grade 1	No dose adjustment is required. Instruct patients to start antidiarrheal therapy such as loperamide at the first sign of loose stools, increase oral fluids, and notify their healthcare provider for further instructions and appropriate follow up.	No dose adjustment is required.
Grade 2	If toxicity does not resolve within 24 hours to Grade ≤1, suspend dose until resolution. No dose reduction is required.	No dose adjustment is required.
Grade 2 that persists or recurs after resuming the same dose despite maximal supportive measures	Suspend dose until toxicity resolves to Grade ≤1. Resume at next lower dose.	No dose adjustment is required.
Grade 3 or 4 or requires hospitalization	Suspend dose until toxicity resolves to Grade ≤1. Resume at next lower dose.	Suspend dose until toxicity resolves to Grade ≤2. Resume at next lower dose.
Increased aminotransferases[a]		
Grade 1 or Grade 2, WITHOUT increase in total bilirubin above 2 × ULN	No dose modification is required.	No dose modification is required.
Persistent or recurrent Grade 2, or Grade 3, WITHOUT increase in total bilirubin above 2 × ULN	Suspend dose until toxicity resolves to baseline or Grade 1. Resume at next lower dose.	Suspend dose until toxicity resolves to baseline or Grade 1. Resume at next lower dose.
Elevation in AST and/or ALT >3 × ULN	Discontinue abemaciclib.	Suspend dose until toxicity resolves to

CTCAE Grade	Abemaciclib	Elacestrant
WITH total bilirubin >2 × ULN, in the absence of cholestasis		baseline or Grade 1. Resume at next lower dose.
Grade 4	Discontinue abemaciclib.	Suspend dose until toxicity resolves to baseline or Grade 1. Resume at next lower dose.
ILD/pneumonitis		
Grade 1 or 2	No dose adjustment is required.	No dose adjustment is required.
Persistent or recurrent Grade 2 toxicity that does not resolve with maximal supportive measures within 7 days to baseline or Grade 1	Suspend dose until toxicity resolves to baseline or Grade 1. Resume at next lower dose.	No dose modification is required.
Grade 3 or 4	Discontinue abemaciclib.	Suspend dose until toxicity resolves to baseline or Grade 1. Resume at next lower dose.
Venous Thromboembolic Events		
Grade 1 or 2	No dose modification is required.	No dose modification is required.
Grade 3 or 4	Suspend dose and treat as clinically indicated. Resume dosing when patient is clinically stable.	Suspend dose and treat as clinically indicated. Resume dosing when patient is clinically stable.
Other toxicities		
Grade 1 or 2	No dose adjustment is required.	No dose adjustment is required
Persistent or recurrent Grade 2 toxicity that does not resolve with maximal supportive measures within 7 days to baseline or Grade 1	Suspend dose until toxicity resolves to baseline or ≤Grade 1. Resume at next lower dose.	Suspend dose until toxicity resolves to baseline or Grade ≤1. Resume at next lower dose.
Grade 3 or 4	Suspend dose until toxicity resolves to baseline or Grade ≤1. Resume at next lower dose.	Suspend dose until toxicity resolves to baseline or Grade ≤1. Resume at next lower dose.

Laboratory values that meet criteria listed in this table should be repeated and confirmed within 24 hours of the initial results being reported.
ALT=alanine aminotransferase increased; AST=aspartate aminotransferase increased; ILD=interstitial lung disease;

ULN=upper limit of normal.
a. Grade 1: >ULN - 3.0 × ULN if baseline was normal; 1.5 - 3.0 × baseline if baseline was abnormal, Grade 2: >3.0 - 5.0 × ULN if baseline was normal; >3.0 - 5.0 × baseline if baseline was abnormal, Grade 3: >5.0 - 20.0 × ULN if baseline was normal; >5.0 - 20.0 × baseline if baseline was abnormal, Grade 4: >20.0 × ULN if baseline was normal; >20.0 × baseline if baseline was abnormal.

6.6.6. Combination with Ribociclib

The recommended starting dose of ribociclib in combination with fulvestrant or an AI is 600 mg (3 × 200 mg tablets) taken orally QD, with or without food.

Coadministration of the sensitive CYP3A4 substrate midazolam with multiple doses of ribociclib at 400 mg increased the midazolam exposure by 280% (3.80-fold) in healthy subjects, compared with administration of midazolam alone. Simulations using physiologically-based pharmacokinetic models suggested that ribociclib given at the clinically relevant dose of 600 mg is expected to increase the midazolam AUC by 5.2-fold (Kisqali® SmPC). These results indicate that ribociclib is a moderate to strong CYP3A4 inhibitor at the doses of 400 mg and 600 mg, respectively. Therefore, as elacestrant is a sensitive CYP3A4 substrate, a moderate to high risk for a PK DDI is expected when coadministered with ribociclib 400 mg and 600 mg, respectively, leading to potential increase in elacestrant exposure. Consequently, the starting dose of ribociclib was selected to be 400 mg and the starting dose of elacestrant was selected to be 86 mg (i.e. 100 mg elacestrant dihydrochloride).

Ribociclib will be administered QD for 21 consecutive days followed by 7 days off treatment to comprise a complete cycle of 28 days. Elacestrant will be administered QD continuously during the treatment cycle.

Dose levels to be evaluated in Phase 1b are presented in Table 16.

Table 16: Dose Levels and Reduction Guidelines for the Combination of Elacestrant and Ribociclib

Drug in Combination	Starting Dose	Dose -1	Dose -2
Ribociclib (mg)	400 QD	200 QD	200 every other day
	600 QD	400 QD	200 QD
Elacestrant (mg)	100 QD	100 every other day	Discontinue elacestrant
	200 QD	100 QD	100 every other day
	300 QD	200 QD	100 QD
	400 QD	300 QD	200 QD

QD=once daily.

Recommendations for dose modification are presented in Table 17.

Table 17: Ribociclib Combination - Recommendation for Dose Modifications

CTCAE Grade	Ribociclib[a]	Elacestrant
Neutropenia		
Grade 1 or 2	No dose adjustment is required.	No dose adjustment is required.
Grade 3	Dose interruption until recovery to Grade ≤2. Resume ribociclib at the same dose level. If toxicity recurs at Grade 3, dose interruption until recovery, then resume ribociclib at the next lower dose level.	No dose adjustment is required.
Grade 3, febrile neutropenia	Dose interruption until recovery to Grade ≤2. Resume ribociclib at the next lower dose level.	No dose adjustment is required.
Grade 4	Dose interruption until recovery to Grade ≤2. Resume ribociclib at the next lower dose level.	Suspend dose until toxicity resolves to Grade ≤2. Resume at next lower dose level.
AST and/or ALT elevations from baseline[b], WITHOUT increase in total bilirubin above 2 x ULN		
Grade 1	No dose adjustment is required.	No dose adjustment is required.
Grade 2	Baseline at Grade <2: Dose interruption until recovery to ≤ baseline grade, then resume ribociclib at same dose level. If Grade 2 recurs, resume ribociclib at next lower dose level. Baseline at Grade 2: No dose interruption.	No dose adjustment is required.
Grade 3	Dose interruption until recovery to ≤baseline[b] grade, then resume at next lower dose level. If Grade 3 recurs, discontinue ribociclib.	Suspend dose until toxicity resolves to Grade ≤2 Resume at the same dose level.
Grade 4	Discontinue ribociclib	Suspend dose until toxicity resolves to Grade ≤2 Resume at next lower dose
Combined elevations in AST and/or ALT WITH total bilirubin increase, in the absence of cholestasis	If patients develop ALT and/or AST > 3 × ULN along with total bilirubin > 2 × ULN irrespective of baseline grade, discontinue ribociclib.	Suspend dose until toxicity resolves to Grade ≤2 Resume at next lower dose.
QT prolongation		
ECGs with QTcF >480 ms	Interrupt ribociclib	No dose adjustment is required.

CTCAE Grade	Ribociclib[a]	Elacestrant
	o If QTcF prolongation resolves to <481 ms, resume ribociclib at the next lower dose level; o If QTcF ≥481 ms recurs, interrupt dose until QTcF resolves to <481 ms; then resume ribociclib at next lower dose level.	
ECGs with QTcF >500 ms	Interrupt ribociclib o If QTcF prolongation resolves to <481 ms, resume ribociclib at the next lower dose level. o Permanently discontinue ribociclib if QTcF interval prolongation is either >500 ms or greater than 60 ms change from baseline AND associated with any of the following: Torsades de Pointes, polymorphic ventricular tachycardia, unexplained syncope, or signs/symptoms of serious arrhythmia.	Suspend dose until QTcF prolongation resolves to <481 ms. Resume at same dose level.
ILD/pneumonitis		
Grade 1 (asymptomatic)	No dose adjustment is required. Initiate appropriate medical therapy and monitor as clinically indicated.	No dose modification is required.
Grade 2 (symptomatic)	Dose interruption until recovery to Grade ≤1, then resume ribociclib at the next lower dose level.	No dose modification is required.
Grade 3 or 4 (severe)	Discontinue ribociclib.	Suspend dose until toxicity resolves to baseline or Grade 1. Resume at next lower dose.
Other toxicities		
Grade 1 or 2	No dose adjustment is required. Initiate appropriate medical therapy and monitor as clinically indicated.	No dose adjustment is required.
Grade 3	Dose interruption until recovery to Grade ≤1, then resume ribociclib at same dose level. If Grade 3 recurs, resume ribociclib at the next lower dose level.	Suspend dose until toxicity resolves to Grade ≤2. Resume at next lower dose.
Grade 4	Discontinue ribociclib.	Discontinue elacestrant.

Laboratory values that meet criteria listed in this table should be repeated and confirmed within 24 hours of the initial results being reported.
ALT=alanine aminotransferase increased; ANC=absolute neutrophil count; AST=aspartate aminotransferase increased; CTCAE=Common Terminology Criteria for Adverse Events, version 5.0; ILD=interstitial lung disease; QTcF=QT interval corrected by Fridericia's formula; ULN=upper limit of normal.
a. An individualized benefit-risk assessment should be performed when considering resuming ribociclib.
b. Baseline=prior to treatment initiation

6.6.7. Combination with Capivasertib

The recommended starting dose of capivasertib in combination with fulvestrant is 400 mg (2 × 200 mg tablets) orally taken twice daily (BID), with or without food, for 4 days followed by 3 days off.

There is a low potential for PK DDIs between elacestrant and capivasertib mediated by CYP3A4 enzyme as capivasertib is a weak inhibitor of CYP3A4 while elacestrant is a CYP3A4 substrate. Consequently, the starting dose of capivasertib was selected to be 320 mg and the starting dose of elacestrant dihydrochloride was selected to be 300 mg.

Capivasertib will be administered BID for 4 consecutive days followed by 3 days off treatment for 4 weeks during the treatment cycle. Elacestrant will be administered QD continuously during the treatment cycle.

Dose levels to be evaluated in Phase 1b are presented in Table 18.

Table 18 Dose Levels and Reduction Guidelines for the Combination of Elacestrant and Capivasertib

Drug in Combination	Starting Dose	Dose -1	Dose -2
Capivasertib	200 BID 4 days on/3 off	-	-
	320 BID 4 days on/3 off	200 BID 4 days on/3 off	-
	400 BID 4 days on/3 off	320 BID 4 days on/3 off	200 BID 4 days on/3 off
Elacestrant (mg)	300 QD	200 QD	100 QD
	400 QD	300 QD	200 QD

BID=twice daily; QD=once daily.

Recommendations for dose modification are presented in Table 19.

Table 19 Capivasertib Combination - Recommendation for Dose Modifications

CTCAE Grade	Capivasertib	Elacestrant
Hyperglycemia		
FPG >ULN – 160 mg/dL or > ULN – 8.9 mmol/L (CTCAE v4.03 Grade 1) or HbA1C >7%	No dose adjustment required. Consider initiation or intensification of oral antidiabetic treatment.	No dose adjustment required.
FPG >160 – 250 mg/dL or >8.9 – 13.9 mmol/L (CTCAE v4.03 Grade 2)	Withhold capivasertib until FG decrease ≤ 160 mg/dL (or ≤ 8.9 mmol/L). If recovery occurs in ≤ 28 days, resume capivasertib at same dose level.	No dose adjustment required.

CTCAE Grade	Capivasertib	Elacestrant
	If recovery occurs in > 28 days, resume capivasertib at one lower dose level.	
FPG >250 –500 mg/dL or >13.9 – 27.8 mmol/L (CTCAE v4.03 Grade 3)	Withhold capivasertib until FG decrease ≤ 160 mg/dL (or ≤ 8.9 mmol/L). If recovery occurs in ≤ 28 days, resume capivasertib at one lower dose level. If recovery occurs in > 28 days, permanently discontinue capivasertib.	No dose adjustment required.
FPG >500 mg/dL or >27.8 mmol/L (CTCAE v4.03 Grade 4) or life-threatening sequelae of hyperglycemia at any FPG level	For life-threatening sequelae of hyperglycemia or if FG persists at ≥ 500 mg/dL after 24 hours, permanently discontinue capivasertib. If FG ≤ 500 mg/dL (or ≤ 27.8 mmol/L) within 24 hours, then follow the guidance in the table for the relevant grade.	No dose adjustment required.
Cutaneous Adverse Reactions (i.e. rash)[a]		
Grade 2	Withhold capivasertib until recovery to ≤ Grade 1. Resume capivasertib at the same dose level. Persistent or recurrent: reduce capivasertib by one lower dose level.	No dose adjustment required.
Grade 3	Withhold capivasertib until recovery to ≤ Grade 1. If recovery occurs in ≤ 28 days, resume capivasertib at same dose level. If recovery occurs in > 28 days, resume capivasertib at one lower dose level. For recurrent Grade 3, permanently discontinue capivasertib.	No dose adjustment required.
Grade 4	Permanently discontinue capivasertib.	No dose adjustment required.

CTCAE Grade	Capivasertib	Elacestrant
Diarrhea		
Grade 2	Withhold capivasertib until recovery to ≤ Grade 1. If recovery occurs in ≤ 28 days, resume capivasertib at same dose or one lower dose as clinically indicated. If recovery occurs in > 28 days, resume at one lower dose as clinically indicated. For recurrence, reduce capivasertib by one lower dose level.	Interrupt elacestrant dose until recovery to Grade ≤1, then resume at same dose level.
Grade 3	Withhold capivasertib until recovery to ≤ Grade 1. If recovery occurs in ≤ 28 days, resume capivasertib at same dose or one lower dose level as clinically indicated. If recovery occurs in > 28 days, permanently discontinue capivasertib.	Interrupt elacestrant until recovery to Grade ≤1, resume at the next lower dose level.
Grade 4	Permanently discontinue capivasertib.	Interrupt elacestrant until recovery to Grade ≤1, resume at the next lower dose level.
Other Toxicity		
Grade 2	Withhold capivasertib until recovery to ≤ Grade 1. Resume capivasertib at the same dose level.	Grade 1: No dose adjustment is required. Grade 2: If toxicity becomes intolerable, withhold until improvement to Grade ≤1. Resume at same dose level.
Grade 3	Withhold capivasertib until recovery to ≤ Grade 1. If recovery occurs in ≤ 28 days, resume capivasertib at same dose. If recovery occurs in > 28 days, resume capivasertib at one lower dose level.	Withhold if clinically indicated. When toxicity resolves to Grade 1 or baseline, resume at a lower dose level. If the Grade 3 toxicity recurs, withhold treatment. When toxicity resolves to Grade 1 or baseline, resume at another lower dose level.
Grade 4	Permanently discontinue capivasertib.	Withhold treatment. When toxicity resolves to Grade 1 or baseline, resume at a lower dose level.

CTCAE Grade	Capivasertib	Elacestrant
		If the Grade 4 toxicity recurs, permanently discontinue.

a. Per TRUQAP™ (capivasertib) package insert, cutaneous adverse reactions may include butterfly rash, dermatitis, allergic dermatitis, dry skin, eczema, erythema multiforme, hand dermatitis, palmar-plantar erythrodysesthesia syndrome, pruritus, rash, erythematous rash, maculo-papular rash, popular rash, skin decolorization, skin fissures, skin reaction, skin ulcer, urticaria, purpura, erythema, and drug eruption.

6.7. Study Steering Committee and Dose Escalation Committee

6.7.1. Membership

The SSC is a group consisting of individuals who collectively are highly experienced clinicians with experience in breast cancer and in conducting clinical studies. The membership and responsibilities of the SSC are described in the SSC Charter.

In Phase 1b, the study will implement a DEC that will report to the SSC.

6.7.2. Responsibilities

The SSC is responsible for ensuring the trial is being conducted with the highest scientific and ethical standards. In particular, the SSC is responsible for:

- Providing input to the Clinical Study Protocol and potential amendments to the Clinical Study Protocol
- Reviewing reports relevant to trial conduct
- Reviewing and endorsing authorship qualification criteria and publication strategy for study related publications (abstract, manuscripts)

In Phase 1b, subject safety will be monitored throughout the study by the DEC established by the Sponsor. The DEC membership will consist of participating investigators, Sponsor representatives, and additional Sponsor staff as appropriate. This team will monitor treatment-emergent data on an ongoing basis throughout study conduct for the purpose of ensuring the continued safety of subjects enrolled in this study. The DEC will discuss the subjects enrolled at each dose level and review subject data, including but not limited to demographics, PK results (when available), dosing, concomitant medications, hematology and serum chemistry, and AEs. The team will meet at regular frequency while the study is being conducted. Additionally, if unexpected safety findings are identified between scheduled DEC meetings, an ad hoc meeting will be convened. The DEC will review and confirm all cases of potential DLTs. The DEC will be responsible for determining and documenting agreement that dose escalation and/or expanding an existing dose level is/are considered appropriately safe. The DEC will also use all available information to recommend the RP2D at the completion of dose escalation for each combination. Decisions with the potential to impact subject safety (e.g., unfavorable change in benefit/risk assessment) will be promptly communicated to investigators and regulatory authorities as appropriate. Documentation of meeting outcomes will be maintained by the Sponsor and/or Sponsor's designee. The recommendations made by the DEC will be reported to the SSC for final review and decision. The decision to declare RP2D for a specific combination

will be made by the SSC following evaluation of all available safety and PK data and review and recommendation from the DEC. Once the RP2D has been determined based on the totality of safety, tolerability, and PK data, the Phase 2 arm may be opened for enrollment.

In Phase 2, available safety, efficacy, and PK data will be periodically reviewed by the SSC (at least every 6 months).

The SSC may also meet in ad hoc meetings at its discretion, as needed in response to events occurring in the study.

6.8. Participant Diversity

The strategy to achieve a more diverse patient population in the study includes the selection of high-performing sites based on the population that they serve to support the increased enrollment of underrepresented populations.

6.9. Criteria for Study Termination

The Sponsor reserves the right to discontinue the study for medical reasons or any other reason at any time. If a study is prematurely terminated or suspended, the Sponsor will promptly inform the investigators/institutions and regulatory authorities the reasons for the termination or suspension. The IRB/IEC will also be informed promptly by the Sponsor or by the investigator/institution, as specified by the applicable regulatory requirements.

The Investigator reserves the right to discontinue the study at his/her institution, as per their clinical judgment. If the Investigator terminates or suspends a study without prior agreement of the Sponsor, the Investigator should inform the institution where applicable, and the Investigator/institution should promptly inform the Sponsor and the IRB/IEC and provide a detailed written explanation of the termination or suspension.

6.9.1. End of Clinical Trial and Archiving

The end of the study is defined as the last visit of the last patient, which is expected 24 months after the date of the global last patient in the study.

For clinical investigational sites located in the EU, a declaration of the end of the clinical study will be made according to the procedure outlined in EU Regulation No. 536/2014 of 16 April 2014; for other countries, local regulations will be followed.

The study activities will end with the collection and analysis of study data and the issue of the clinical study report. Patient records must be archived according to national law requirements. Additional details are provided in Section 15.3.4.

The patients will be identified by the PI or sub-investigators trained and delegated to conduct the study at sites. The sites and the PIs will use their networks within their near geographical domain to recruit these patients. In case other sources will be evaluated, in line with the local regulations, to recruit patients, these resources will be assessed and approved by the EC/IRB as required.

All access to the personal data of the potential patients for the study will be done in accordance with all applicable regulations.

The PI or delegated site staff will obtain informed consent from the patients. If the PI or delegee is the treating specialist of the patient, the Investigator is allowed to notify their patient about the study. If the patient is interested in participating in the study, the patient will be referred to one of the other study team members to explain the study in detail and obtain informed consent.

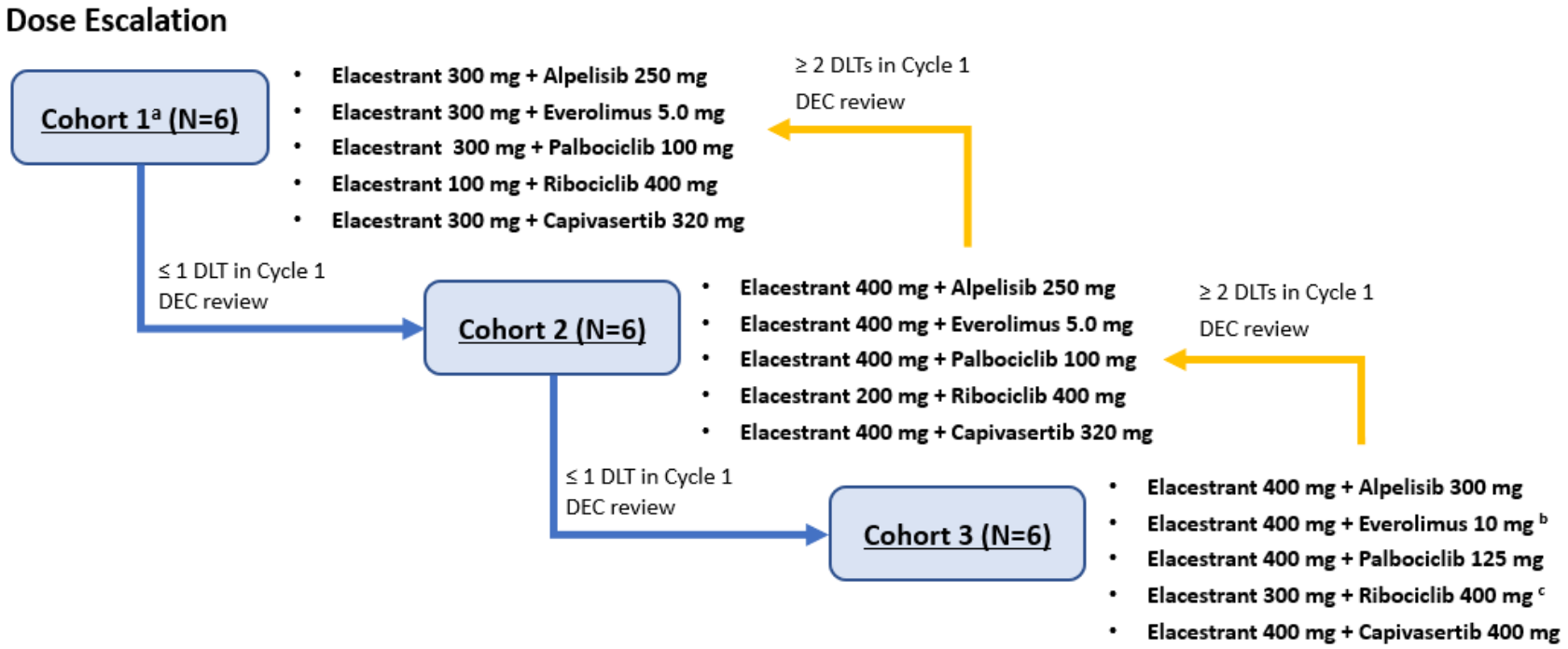
There is no strict time limit to this, but at a minimum patients will be given 24 hours and hence overnight to read and consider information in the patient information sheet. Patients will be advised that they may share information provided in the informed consent with family members for their opinions if they wish. The Investigator or delegated site staff will fully inform the patient of all pertinent aspects of the trial including the written information approved/favorably assessed by the EC. Furthermore, the potential patients will get the chance to ask questions during the informed consent process.

Given the need for close safety monitoring and ability to communicate efficiently with clinical staff regarding any AEs, patients who are not able to understand the ICF or who demonstrate inability to communicate with clinical staff will not be included in the study. As alternative approved informed consent in the patient's language is used to allow the patient study evaluation and eventually his/her participation. All the national and global guidance/regulations will be followed in the procedure of recruitment and informed consent process.

In case a patient is unable to write, and if local regulations allow, consent may be given and recorded through appropriate alternative means in the presence of at least one impartial witness identified by the Investigator or site staff personnel involved in the informed consent process. The witness is required to sign and date the ICF.

6.10. Schematic Study Treatments and Visit Schedule

Figure 3: Phase 1b Study Design

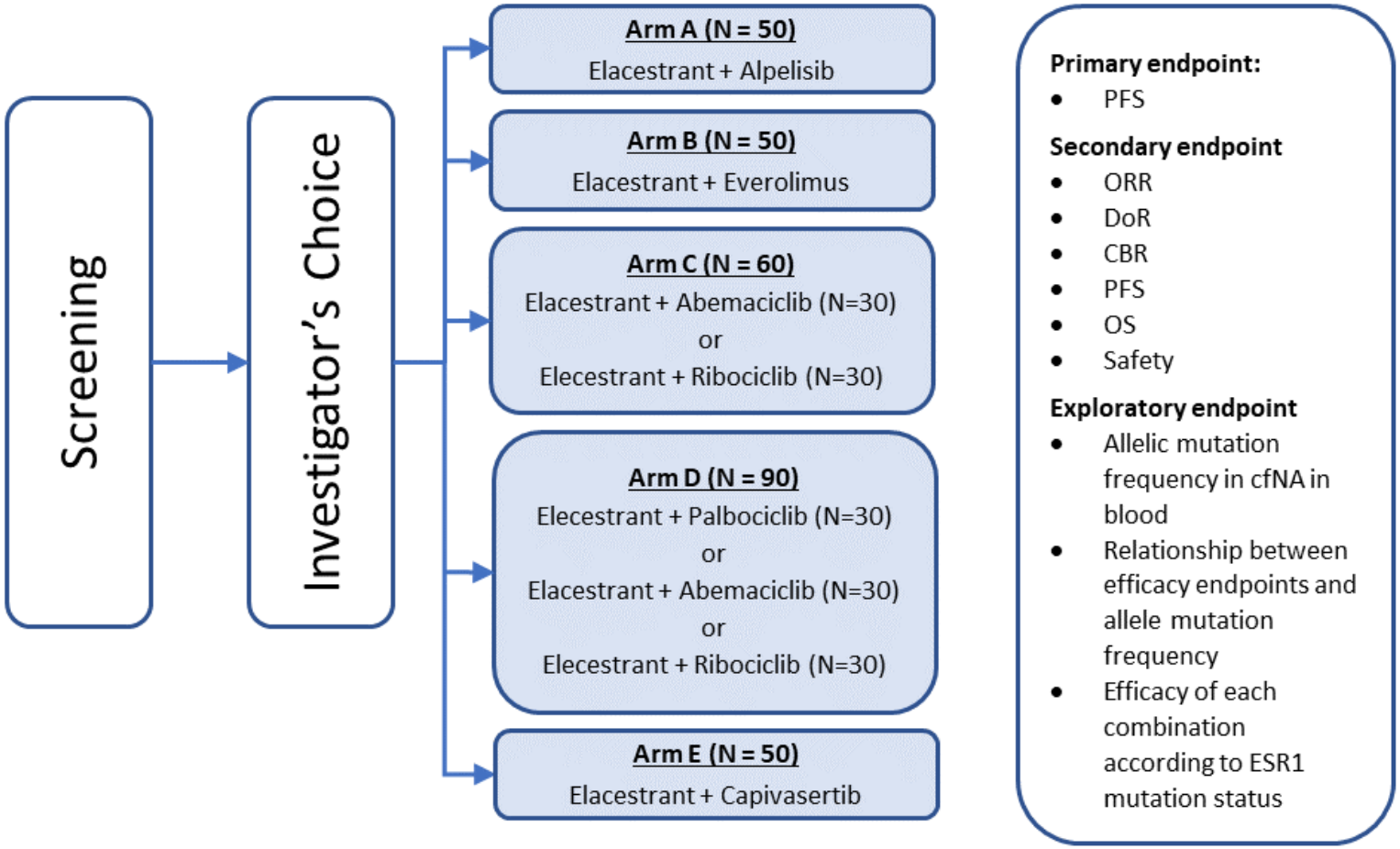


[a] If ≥ 2 DLTs in Cohort 1, Cohort -1 or Cohort 0 may be opened with one or more reduced doses

[b] Based on data from Cohorts 1 and 2, everolimus dose of 10 mg may be used in Cohort 3

[c] Based on data from Cohort 1 and Cohort 2, elacestrant dose of 400 mg and ribociclib dose of 600 mg may be used in Cohort 3
DEC=Dose Escalation Committee; DLT=dose-limiting toxicity (during Cycle 1); RP2D=recommended dose for Phase 2.

Figure 4: Phase 2 Study Design



CBR=clinical benefit rate; cfNA=cell-free nucleic acid; DoR=duration of response; ORR=overall response rate; OS=overall survival; PFS=progression-free survival.

A Schedule of Assessments is presented in Table 20 and Table 21.

Table 20: Schedule of Assessments, Combinations with Alpelisib, Everolimus, Palbociclib, Ribociclib, and Abemaciclib

Period	**Screening**	**Cycle 1**			**Cycle 2**		**Cycle 3+**	**EOT[a]**	**Safety Follow-up[b]**	**Survival[c] Follow-up**
Day	**D-28 to D1**	**D1[d]** (±1)	**D8** (±1)	**D15** (±2)	**D1** (±2)	**D15** (±1)	**D1** (±2)	**--**	**28 days** (+7)	**--**
Informed consent[r]	×									
Eligibility criteria	×									
Medical history	×									
Concomitant medications	×	×	×	×	×	×	×	×	×	
AEs	×	×	×	×	×	×	×	×	×	
Physical examination[e]	×	×	×	×	×	×	×	×	×	
ECOG PS	×	×			×		×	×	×	
Triplicate 12-lead ECG[f]	×	×		×	×		×	×	×	
Pregnancy testing[g]	×	×								
CBC[h]	×	×	×	×	×	×	×	×	×	
CMP[i]	×	×		×	×	×	×	×	×	
Lipid Panel	× [j]	×			× [j]		× [j]	×		
HBV, HCV[k]	×									
Study treatments dispensed/returned; accountability[l]		×	×	×	×	×	×	×		
PK samples[m] (plasma, except for everolimus/whole blood)				×						
cfNA whole blood[n]		×		×			×	×		
Whole-body bone scan[o]	×									
CT/MRI scans[p]	×						×	×		
Brain CT or MRI[q]	×									
Survival follow-up										×

AE=adverse event; ALT=alanine aminotransferase; AST=aspartate aminotransferase; BOR=best overall response; BP=blood pressure; BUN=blood urea nitrogen; C1/D15=Cycle 1, Day 15; CBC=complete blood cell count; cfNA=circulating free nucleic acid; CMP=comprehensive metabolic panel; CO_2=carbon

dioxide; CrCl=creatinine clearance; CT=computed tomography; D=day; ECG=electrocardiogram; ECOG=Eastern Cooperative Oncology Group; eGFR=estimated glomerular filtration rate; EOT=end of treatment; HDL=high-density lipoproteins; Hgb=hemoglobin; HBV=hepatitis B virus; Hct=hematocrit; HCV=hepatitis C virus; HR=heart rate; MRI=magnetic resonance imaging; PK=pharmacokinetics; PS=performance status; RR=response rate; SAE=serious adverse event; SOC=standard of care; T_{max}=time of the maximum observed plasma (whole blood for everolimus) concentration.

Any protocol assessments, including laboratory tests, performed within 3 days prior to D1 of the next cycle (including Cycle 1, which has a 7-day window) do not need to be repeated on Day 1 of the new cycle.

a. EOT visit to be performed within 14 days of the date of last treatment administration. If a patient discontinues trial therapy for any reason, EOT assessments will be conducted, unless the EOT visit falls within 7 days of another scheduled visit in which case assessments do not need to be repeated.

b. All patients will be followed for AEs for 28 days after the last treatment administration or until resolution or stabilization of all treatment-related AEs to either Grade ≤2 or baseline, whichever is longer, or until the patient is lost to follow-up. At any time after 28 days from the last dose of study treatment, the Investigator may report any SAE that he/she believes is related to study treatment. Patients who discontinue elacestrant will be discontinued from the study, except for patients receiving abemaciclib as per Investigator's judgment.

c. All patients will be followed for survival approximately every 3 months for up to 24 months after enrollment of the last patient. Survival follow-up may be done by telephone call at the Investigator's discretion. The following information will be collected for the first anticancer systemic therapy used after discontinuation of study therapy: regimen of individual treatments, start date, end date, and BOR.

d. Scheduled ECGs, vital signs, and laboratory tests should be performed pre-dose. Assessments performed during the 7 days prior to Cycle 1 Day 1 do not need to be repeated, unless otherwise indicated.

e. Physical examinations will include measurements of weight and vital signs (resting HR, BP, RR, and body temperature). There is no need to repeat physical examinations on C1/D1, if the baseline physical examination was performed ≤3 days before. Height will be recorded at the screening visit only. Physical examinations after screening or C1D1 can be symptom-driven.

f. At all indicated visits, a standard 12-lead ECG will be performed in triplicate pre-dose after the patient has been resting for 5 minutes; these 3 should be done approximately 2 minutes apart per local practice. An additional ECG will be performed at the T_{max} of each drug for all patients enrolled in Phase 1b at C1/D15, in time, as close as possible to the time of scheduled PK sampling.
   - alpelisib: 3 hours (±30 minutes),
   - everolimus: 1.5 hours (±15 minutes),
   - palbociclib: 9 hours (±1 hour),
   - abemaciclib: 8 hours (±60 minutes), and
   - ribociclib: 2.5 hours (±30 minutes)

g. This is only required for pre- and peri-menopausal women. Pregnancy test during the screening period should be serum (there is no need to repeat on C1/D1 if this is within 72 hours). If a urine test is positive, serum testing should be performed for confirmation.

h. CBC count consisting of Hgb, Hct, WBC with 5-part differential, and platelet count CBCs will be collected before treatment at every visit indicated and at the Safety Follow-up visit. Note: Laboratory assessments may be repeated during Screening after supplementation or transfusion (red blood cells transfusion is allowed once ≤2 weeks before study enrollment).

i. CMP includes BUN, Cr, CrCl, eGFR, sodium, calcium (corrected for serum albumin), potassium, chloride, phosphate, and magnesium, CO_2, glucose, lipase, albumin, total and direct bilirubin, ALP, AST, and ALT. Note: Laboratory assessments may be repeated during the Screening Phase after supplementation or transfusion (red blood cells transfusion is allowed once ≤2 weeks before study enrollment). If hyperglycemia is suspected, repeat glucose testing after fasting per local practice.

j. Monitor lipids (total cholesterol, HDL, LDL, and triglycerides) at Screening, C1D1 (pre-dose as baseline), C2D1, C3D1, C4D1, C8D1, C12D1 + every 6 months + EOT, and as clinically indicated.

k. Testing only required in patients with known or suspected infection. See Appendix B for HBV and HCV testing criteria.

l. One cycle is 4 weeks (28 days). Study therapy should be dispensed, and patient's drug administration diary (Patient Diary) should be reviewed at the beginning of each cycle. Note that palbociclib and ribociclib should be administered daily for 21 days, followed by 7 days off treatment, per cycle; all other treatment will be administered daily (or twice daily (BID) for abemaciclib) per treatment.

m. PK (Phase 1b and Phase 2 subset of patients) :PK (plasma except for everolimus combination, whole blood) will be taken on C1D15 and C1D16 for Phase 1b patients, and sparce sampling for a subset of Phase 2 patients. Please refer to Appendix G for details.

n. Whole blood for cfNA will be collected only on C1/D1 (or within 3 days prior to C1/D1), C1/D15, C3/D1 and at EOT (maximum total of 4 samples on study). Samples should be collected before dosing. Two aliquots of 10 mL whole blood will be collected for each timepoint.

o. A whole-body bone scan using ^{99}m-technetium will be performed during screening to establish a baseline (a bone scan performed per local SOC within 6 weeks before C1/D1 is acceptable), approximately every 24 weeks (in conjunction with a scheduled tumor assessment visit), and as clinically indicated. Lesions identified on bone scans should be followed with cross-sectional imaging (CT or MRI).

p. Tumor assessments will be performed at baseline and every 8 weeks ±7 days after C1/D1 until radiological disease progression, and will include:

- Chest: CT with contrast,
- Abdomen, pelvis, and other known or suspected sites of the of disease: CT (with contrast) or MRI (with contrast)
    - If imaging with contrast is contraindicated or not possible for other reasons, imaging without contrast should be performed and the reason for the contraindication should be captured in source documents.

Tumor assessments will be performed using the same imaging modality and under the same operating conditions as the baseline scans.

Note: Standard-of-care scans obtained outside the screening window may be considered for baseline if approved by Sponsor.

Note: Patients who have discontinued study treatment for other reasons without progression will be followed for disease assessments every 8 weeks from the date of the last tumor assessment until radiological disease progression is documented, death, or another anticancer therapy is initiated, whichever occurs first, unless the study is terminated.

q. Brain CT with contrast or MRI (pre- and postcontrast): Brain assessment at baseline is only required if the patient has history of brain metastasis or symptoms suggestive of brain metastasis. Brain assessment may also be required during the trial if clinically indicated.

r. Informed consent and authorization for release of medical records may be obtained before the 28-day screening window. No study specific assessments can be performed prior to signing informed consent. In certain cases where repeating a standard of care assessment would cause the patient additional and unnecessary discomfort or grave inconvenience, a standard of care assessments done prior to signature of ICF and within the 28-day screening window may be used ONLY if the site documents prior approval from Sponsor or Sponsor designee (CRO medical monitor).

Table 21 Schedule of Assessments, Combination With Capivasertib

Period	Screening	Cycle 1				Cycle 2		Cycle 3+	EOT[a]	Safety Follow-up[b]	Survival[c] Follow-up
Day	D-28 to D1	D1[d] (±1)	D8 (±1)	D11 (-1)	D21 (±2)	D1 (±2)	D15 (±1)	D1 (±2)	--	28 days (+7)	--
Informed consent[r]	×										
Eligibility criteria	×										
Medical history	×										
Concomitant medications	×	×	×		×	×	×	×	×	×	
AEs	×	×	×		×	×	×	×	×	×	
Physical examination[e]	×	×	×		×	×	×	×	×	×	
ECOG PS	×	×				×		×	×	×	
Triplicate 12-lead ECG[f]	×	×		×[f]		×		×	×	×	
Pregnancy testing[g]	×	×									
CBC[h]	×	×	×		×	×	×	×	×	×	
CMP[i]	×	×			×	×	×	×	×	×	
Lipid Panel	×[j]	×				×[j]		×[j]	×		
HBV, HCV[k]	×										
Study treatments dispensed/returned; accountability[l]		×	×		×	×	×	×	×		
PK samples[m] (plasma)				×							
cfNA whole blood[n]		×			×			×	×		
Whole-body bone scan[o]	×										
CT/MRI scans[p]	×							×	×		
Brain CT or MRI[q]	×										
Survival follow-up											×

AE=adverse event; ALT=alanine aminotransferase; AST=aspartate aminotransferase; BOR=best overall response; BP=blood pressure; BUN=blood urea nitrogen; C1/D15=Cycle 1, Day 15; CBC=complete blood cell count; cfNA=circulating free nucleic acid; CMP=comprehensive metabolic panel; CO_2=carbon dioxide; CrCl=creatinine clearance; CT=computed tomography; D=day; ECG=electrocardiogram; ECOG=Eastern Cooperative Oncology Group;

eGFR=estimated glomerular filtration rate; EOT=end of treatment; HDL=high-density lipoproteins; Hgb=hemoglobin; HBV=hepatitis B virus; Hct=hematocrit; HCV=hepatitis C virus; HR=heart rate; MRI=magnetic resonance imaging; PK=pharmacokinetics; PS=performance status; RR=response rate; SAE=serious adverse event; SOC=standard of care; T_{max}=time of the maximum observed plasma (whole blood for everolimus) concentration.

Any protocol assessments, including laboratory tests, performed within 3 days prior to D1 of the next cycle (including Cycle 1, which has a 7-day window) do not need to be repeated on Day 1 of the new cycle.

a. EOT visit to be performed within 14 days of the date of last treatment administration. If a patient discontinues trial therapy for any reason, EOT assessments will be conducted, unless the EOT visit falls within 7 days of another scheduled visit in which case assessments do not need to be repeated.

b. All patients will be followed for AEs for 28 days after the last treatment administration or until resolution or stabilization of all treatment-related AEs to either Grade ≤2 or baseline, whichever is longer, or until the patient is lost to follow-up. At any time after 28 days from the last dose of study treatment, the Investigator may report any SAE that he/she believes is related to study treatment. Patients who discontinue elacestrant will be discontinued from the study, except for patients receiving abemaciclib as per Investigator's judgment.

c. All patients will be followed for survival approximately every 3 months for up to 24 months after enrollment of the last patient. Survival follow-up may be done by telephone call at the Investigator's discretion. The following information will be collected for the first anticancer systemic therapy used after discontinuation of study therapy: regimen of individual treatments, start date, end date, and BOR.

d. Scheduled ECGs, vital signs, and laboratory tests should be performed pre-dose. Assessments performed during the 7 days prior to Cycle 1 Day 1 do not need to be repeated, unless otherwise indicated.

e. Physical examinations will include measurements of weight and vital signs (resting HR, BP, RR, and body temperature). There is no need to repeat physical examinations on C1/D1, if the baseline physical examination was performed ≤3 days before. Height will be recorded at the screening visit only. Physical examinations after screening or C1D1 can be symptom driven.

f. At all indicated visits, a standard 12-lead ECG will be performed in triplicate pre-dose after the patient has been resting for 5 minutes; these 3 should be done approximately 2 minutes apart per local practice. An additional ECG will be performed at the T_{max} of each drug for all patients enrolled in Phase 1b at C1/D11, in time, as close as possible to the time of scheduled PK sampling.
   - capivasertib: 2 hours (±30 minutes),

g. This is only required for pre- and peri-menopausal women. Pregnancy test during the screening period should be serum (there is no need to repeat on C1/D1 if this is within 72 hours). If a urine test is positive, serum testing should be performed for confirmation.

h. CBC count consisting of Hgb, Hct, WBC with 5-part differential, and platelet count CBCs will be collected before treatment at every visit indicated and at the Safety Follow-up visit. Note: Laboratory assessments may be repeated during Screening after supplementation or transfusion (red blood cells transfusion is allowed once ≤2 weeks before study enrollment).

i. CMP includes BUN, Cr, CrCl, eGFR, sodium, calcium (corrected for serum albumin), potassium, chloride, phosphate, and magnesium, CO_2, glucose, lipase, albumin, total and direct bilirubin, ALP, AST, and ALT. Note: Laboratory assessments may be repeated during the Screening Phase after supplementation or transfusion (red blood cells transfusion is allowed once ≤2 weeks before study enrollment). If hyperglycemia is suspected, repeat glucose testing after fasting per local practice.

j. Monitor lipids (total cholesterol, HDL, LDL, and triglycerides) at Screening, C1D1 (pre-dose as baseline), C2D1, C3D1, C4D1, C8D1, C12D1 + every 6 months + EOT, and as clinically indicated.

k. Testing only required in patients with known or suspected infection. See Appendix B for HBV and HCV testing criteria.

l. One cycle is 4 weeks (28 days). Study therapy should be dispensed, and patient's drug administration diary (Patient Diary) should be reviewed at the beginning of each cycle. Capivasertib is taken for 4 days followed by 3 days off, for the 4 weeks in each cycle; elacestrant will be taken daily for each day of the study cycle.

m. Plasma PK (Phase 1b and Phase 2 subset of patients) will be taken on C1D11 (-1 day) and C1D12 (-1 day) for Phase 1b patients, and sampling for a subset of Phase 2 patients. Please refer to Appendix G for details.

n. Whole blood for cfNA will be collected only on C1/D1 (or within 3 days prior to C1/D1), C1/D21, C3/D1 and at EOT (maximum total of 4 samples on study). Samples should be collected before dosing. Two aliquots of 10 mL whole blood will be collected for each timepoint.

o. A whole-body bone scan using ^{99}m-technetium will be performed during screening to establish a baseline (a bone scan performed per local SOC within 6 weeks before C1/D1 is acceptable), approximately every 24 weeks (in conjunction with a scheduled tumor assessment visit), and as clinically indicated. Lesions identified on bone scans should be followed with cross-sectional imaging (CT or MRI).

p. Tumor assessments will be performed at baseline and every 8 weeks ±7 days after C1/D1 until radiological disease progression, and will include:

- Chest: CT with contrast,
- Abdomen, pelvis, and other known or suspected sites of the of disease: CT (with contrast) or MRI (with contrast)
  - If imaging with contrast is contraindicated or not possible for other reasons, imaging without contrast should be performed and the reason for the contraindication should be captured in source documents.

Tumor assessments will be performed using the same imaging modality and under the same operating conditions as the baseline scans.

Note: Standard-of-care scans obtained outside the screening window may be considered for baseline if approved by Sponsor.

Note: Patients who have discontinued study treatment for other reasons without progression will be followed for disease assessments every 8 weeks from the date of the last tumor assessment until radiological disease progression is documented, death, or another anticancer therapy is initiated, whichever occurs first, unless the study is terminated.

q. Brain CT with contrast or MRI (pre- and postcontrast): Brain assessment at baseline is only required if the patient has history of brain metastasis or symptoms suggestive of brain metastasis. Brain assessment may also be required during the trial if clinically indicated.

r. Informed consent and authorization for release of medical records may be obtained before the 28-day screening window. No study specific assessments can be performed prior to signing informed consent. In certain cases where repeating a standard of care assessment would cause the patient additional and unnecessary discomfort or grave inconvenience, a standard of care assessments done prior to signature of ICF and within the 28-day screening window may be used ONLY if the site documents prior approval from Sponsor or Sponsor designee (CRO medical monitor).

7. SELECTION AND WITHDRAWAL OF PATIENTS

This umbrella trial will evaluate the safety and efficacy of elacestrant in combination with alpelisib, everolimus, palbociclib, abemaciclib, ribociclib and capivasertib in patients satisfying all the following eligibility criteria. Additional eligibility criteria per combination arm are applied.

7.1. Patient Inclusion Criteria Common to Phase 1b and 2 for All Treatment Arms

1. Patient has signed the ICF before all study-specific activities are conducted.
2. Women or men aged ≥18 years (or the minimum age of consent in accordance with the local law), at the time of informed consent signature. Female patients may of any menopausal status.
   - Postmenopausal status is defined by:
     a) Age ≥60 years
     b) Age <60 years and amenorrhea for 12 or more months (without an alternative cause) and FSH value and an estradiol level within the postmenopausal range per local laboratory reference
     c) Documentation of bilateral oophorectomy, at least 1 month before first dose of trial therapy.
   - Premenopausal and perimenopausal women (who do not fit postmenopausal criteria) and men must be receiving a LHRH agonist and must be initiated at least 3 weeks before the start of trial therapy and are planning to continue LHRH agonist treatment during the study treatment.
3. Histopathological or cytological confirmed ER+/HER2– breast cancer, per local laboratory, as per the ASCO/CAP guidelines (Allison et al, 2020; Wolff et al, 2018). *Note*: In the context of this trial, ER status will be considered positive if ≥10% of tumor cells demonstrate positive nuclear staining by immunohistochemistry, with or without PGR positivity.
4. At least 1 not previously irradiated measurable lesion as per RECIST version 1.1 and/or at least 1 lytic or mixed (lytic +sclerotic) bone lesion with identifiable soft tissue components meeting the definition of measurability by RECIST version 1.1 that can be evaluated by CT or MRI; patients with sclerotic/osteoblastic bone lesions only in the absence of measurable disease are not eligible.
5. ECOG performance status of 0 or 1.
6. Patient has adequate bone marrow and organ function, as defined by the following laboratory values:
   a. ANC ≥1.5 × 10^9/L
   b. Platelets ≥100 × 10^9/L
   c. Hemoglobin ≥9.0 g/dL

d. Potassium, sodium, calcium (corrected for serum albumin) and magnesium CTCAE v5.0 Grade ≤1
e. Creatinine is ≤ 1.5 x ULN or if Creatinine is > 1.5 x ULN, then Creatinine clearance must be ≥50 mL/min based on the Cockcroft-Gault formula. *Note:* C-G formula:
    - CrCl (male) = ([140-age in years] × weight in kg)/ ([serum creatinine in mg/dL] × 72)
    - CrCl (female) = (0.85 × [140-age in years] × weight in kg)/ ([serum creatinine in mg/dL] × 72)
f. Serum albumin ≥3.0 g/dL (≥30 g/L)
g. In absence of liver metastases, ALT and AST ≤3.0 × ULN. If the patient has liver metastases, ALT, and AST ≤5 × ULN
h. Total serum bilirubin <1.5 × ULN except for patients with Gilbert's syndrome who may be included if the total serum bilirubin is ≤3.0 × ULN or direct bilirubin ≤1.5 × ULN.

Note: Laboratory assessments may be repeated during the Screening Phase after supplementation or transfusion (red blood cells transfusion is allowed once ≤2 weeks before study enrollment).

7.2. Patient Exclusion Criteria Common to Phase 1b and 2 for All Treatment Arms

1. Active or newly diagnosed CNS metastases, including meningeal carcinomatosis.

    Note: Patients with stable brain or subdural metastases are allowed if the patient has completed local therapy and was on a stable or decreasing dose of corticosteroids at baseline for management of brain metastasis for at least 4 weeks before starting treatment in this study. The dose must be ≤2.0 mg/day of dexamethasone or equivalent. Any signs (e.g., radiologic) or symptoms of brain metastases must be stable for at least 4 weeks before starting study treatment.

2. Patients with advanced, symptomatic visceral spread, that are at risk of life-threatening complications in the short term, including massive uncontrolled effusions (peritoneal, pleural, pericardial), pulmonary lymphangitis, or liver involvement >50%.
3. Prior chemotherapy in the advanced/metastatic setting.
4. Patients with known germline BRCA mutation without prior treatment with a PARP inhibitor before study entry.
5. Prior therapy with elacestrant or other investigational selective estrogen receptor degraders (SERDs), or investigational alike agents such as selective estrogen receptor modulators (SERM), selective estrogen receptor covalent antagonists (SERCANs), complete estrogen receptor antagonists (CERANs), and proteolysis-targeting chimeras (PROTACs), in the metastatic setting. Prior treatment with fulvestrant is not exclusionary as it is an approved medication.
6. Patient has a concurrent malignancy or history of invasive malignancy within 3 years of enrollment, except basal or squamous cell skin cancer, superficial bladder cancer, or

carcinoma in situ of the cervix that has completed curative therapy. Other malignancies with low risk of recurrence may be considered eligible with Sponsor approval.

7. Uncontrolled significant active infections.
   - Patients with HBV and/or HCV infection must have undetectable viral load during screening (See Appendix B).
   - Patients known to be HIV+ are allowed if they have undetectable viral load at baseline.
8. Documented pneumonitis/ILD prior to Cycle 1 Day 1.
9. Major surgery within 28 days before starting trial therapy.
10. Inability to take oral medications, refractory or chronic nausea, gastrointestinal conditions (including significant gastric or bowel resection), history of malabsorption syndrome, or any other uncontrolled gastrointestinal condition that impact the absorption of the study drug.
11. Known intolerance to elacestrant or any of its excipients (see Table 22).
12. Pregnant and breast-feeding women are excluded from the study. In addition, women of childbearing potential are excluded who:
    - Within 28 days before starting trial therapy, did not use a highly effective method of contraception.
    - Do not agree to use a highly effective method of contraception (Appendix F) or abstain from heterosexual intercourse throughout the entire study period and for 120 days after trial therapy discontinuation.
13. Men or women who do not agree to abstain from donating sperm or ova, or who do not use a highly effective method of contraception (Appendix F), during the treatment period and for 120 days after the last dose of study treatment.
14. Patient is currently receiving or received any of the following medications prior to first dose of trial therapy:
    - Anti-cancer therapy within 14 days (28 days for anticancer antibody-based treatment) or 5 half-lives, whichever is shorter.

      Please note: Toxicity from prior therapy must be resolved to NCI CTCAE version 5.0 Grade ≤1, except alopecia and peripheral sensory neuropathy (Grade ≤2).
    - Known strong or moderate inducers or inhibitors (including foods and herbal preparations) of CYP3A4 within 14 days or 5 half-lives, whichever is shorter (refer to http://medicine.iupui.edu/clinpharm/ddis/ or https://www.fda.gov/drugs/drug-interactions-labeling/drug-development-and-drug-interactions-table-substrates-inhibitors-and-inducers).
    - Herbal preparations/medications within 7 days. These include, but are not limited to, St. John's wort, kava, ephedra (ma huang), gingko biloba, DHEA, yohimbe, saw palmetto, and ginseng.

- Vaccination, including but not limited to vaccination against COVID-19, during the 7 days prior to randomization.

15. Evidence of ongoing alcohol or drug abuse as assessed by the Investigator.
16. Any severe medical or psychiatric condition that, in the Investigator's opinion, would preclude the patient's participation in a clinical study.

7.3. Additional Eligibility Criteria by Combination

In general, the SmPC of the respective combination drug should be consulted for instructions/restrictions with respect to interactions with concomitant medications.

7.3.1. Alpelisib Combination (Phase 1b and Arm A)

Inclusion:

1. PIK3CA mutation by local laboratory assessment.
2. One or up to two prior hormonal therapies in the advanced or metastatic setting, one of which was in combination with a CDK4/6 inhibitor.

Exclusion:

1. Prior therapy with alpelisib or any other PI3K inhibitor.
2. Type 1 diabetes or uncontrolled type 2 diabetes (fasting plasma glucose level of >140 mg/dL [7.7 mmol/L], or HbA1c level of >6.4%).
3. Known intolerance to alpelisib or any of its excipients.
4. Patient is currently receiving or received drugs known to be a BCRP inhibitor (e.g., curcumin, cyclosporine A, darolutamide, eltrombopag, febuxostat, fostamatinib, rolapitant, teriflunomide) within 14 days or 5 half-lives, whichever is shorter, prior to first dose of trial therapy (refer to Table 5.2 of https://www.fda.gov/drugs/drug-interactions-labeling/drug-development-and-drug-interactions-table-substrates-inhibitors-and-inducers).
5. Patient has ongoing osteonecrosis of the jaw from previous or concurrent treatment with bisphosphonates or denosumab.

7.3.2. Everolimus Combination (Phase 1b and Arm B)

Inclusion:

1. One or up to two prior hormonal therapies in the advanced or metastatic setting, one of which was in combination with a CDK4/6 inhibitor.

Exclusion:

1. Prior therapy with everolimus.
2. Known intolerance to everolimus or any of its excipients.

7.3.3. Abemaciclib Combination (Arm C)

Inclusion:

1. One or up to two prior hormonal therapies in the advanced or metastatic setting, one of which was in combination with a CDK4/6 inhibitor.

Exclusion:

1. Prior therapy with abemaciclib in the advanced or metastatic setting. Adjuvant therapy with abemaciclib is exclusionary if the patient relapsed within the past 12 months.
2. Known intolerance to abemaciclib or any of its excipients.

7.3.4. Ribociclib Combination (Phase 1b and Arm C)

Inclusion:

1. One or up to two prior hormonal therapies in the advanced or metastatic setting, one of which was in combination with a CDK4/6 inhibitor.

Exclusion:

1. Prior therapy with ribociclib in the metastatic setting. Prior adjuvant therapy with ribociclib is also exclusionary if the patient relapsed within the past 12 months.
2. Known intolerance to ribociclib or any of its excipients.
3. QTcF values ≥450 msec.
4. Patients who already have or who are at significant risk of developing QTc prolongation, including patients with:
   - Long QT syndrome
   - Uncontrolled or significant cardiac disease including recent (6 months) myocardial infarction, congestive heart failure, unstable angina, and brady-arrhythmias
   - Electrolyte abnormalities
5. Patient is currently receiving or received drugs known to prolong QT interval within 14 days or 5 half-lives, whichever is shorter, before the first dose of trial therapy. See Appendix E.

7.3.5. Palbociclib Combination (Phase 1b)

Inclusion:

1. One or up to two prior hormonal therapies in the advanced or metastatic setting, one of which was in combination with a CDK4/6 inhibitor.

Exclusion:

1. Prior therapy with palbociclib in the metastatic setting.
2. Known intolerance to palbociclib or any of its excipients.

7.3.6. Palbociclib Combination (Arm D)

Inclusion:

1. One or up to two prior hormonal therapies in the advanced or metastatic setting.

Exclusion:

1. Prior therapy with a CDK4/6i in the metastatic setting.
2. Known intolerance to palbociclib or any of its excipients.

7.3.7. Abemaciclib Combination (Arm D)

Inclusion:

1. One or up to two prior hormonal therapies in the advanced or metastatic setting.

Exclusion:

1. Prior therapy with any CDK4/6i in the metastatic setting.
2. Known intolerance to abemaciclib or any of its excipients.

7.3.8. Ribociclib Combination (Arm D)

Inclusion:

1. One or up to two prior hormonal therapies in the advanced or metastatic setting.

Exclusion:

1. Prior therapy with a CDK4/6i in the metastatic setting.
2. Known intolerance to ribociclib or any of its excipients.
3. QTcF values ≥450 msec.
4. Patients who already have or who are at significant risk of developing QTc prolongation, including patients with:
   - Long QT syndrome
   - Uncontrolled or significant cardiac disease including recent (6 months) myocardial infarction, congestive heart failure, unstable angina, and brady-arrhythmias
   - Electrolyte abnormalities
5. Patient is currently receiving or received drugs known to prolong QT interval within 14 days or 5 half-lives, whichever is shorter, before the first dose of trial therapy. See Appendix E.

7.3.9 Capivasertib Combination (Phase 1b and Arm E)

Inclusion:

1. PIK3CA/AKT1/PTEN-alteration as detected by an FDA and/or locally approved test (local result).
2. One or up to two prior hormonal therapies in the advanced or metastatic setting OR patients who have radiological evidence of breast cancer recurrence or progression within 12 months from the end of adjuvant treatment with ET, as these patients are considered as first line relapsed patients. Prior CDKi treatment is allowed but not required.

Exclusion:

1. Prior treatment with any of the following: AKT, PI3K and mTOR inhibitors
2. Clinically significant abnormalities of glucose metabolism as defined by any of the following: Patients with diabetes mellitus type 1; patients with diabetes mellitus type 2 requiring insulin treatment or patients with HbA1c level of >8.0% (63.9 mmol/mol).

7.4. Patient Withdrawal Criteria

7.4.1. Treatment Discontinuation

Patients will continue to receive study treatment until any of the following occur:

1. Disease progression
2. Death
3. Development of unacceptable toxicity considered related to any of the study drugs
4. Patient requests to discontinue treatment
5. Withdrawal of consent
6. Occurrence of pregnancy
7. Investigator's decision
8. Lost to follow-up
9. Inability to comply with protocol requirements

The reason for treatment discontinuation will be documented. If a patient discontinues study treatment, the patient will enter the Follow-Up Phase and complete protocol-specified off-treatment visits, procedures, and survival follow-up unless the patient withdraws consent. The Investigator should confirm whether a patient will discontinue trial therapy but agree to continue protocol-specified, off-treatment study visits, procedures, and survival follow-up, or whether the patient will withdraw consent.

If a patient withdraws consent, the date will be documented in the source documents. In such a case, data collection for visits or assessments occurring after the date of consent withdrawal will not be allowed unless the patient agrees to be followed for survival information. Patients who withdraw consent will always be asked about the reason(s). Data already collected will be used and analyzed for the purpose of the study. With regard to biological samples already collected, the patient will be asked if samples already obtained but not yet analyzed shall be destroyed or analyzed.

During the Follow-Up Phase, patients who have discontinued study treatment for other reasons without progression will be followed for disease assessments every 8 weeks from the date of the last tumor assessment until radiological disease progression is documented or another anticancer therapy is initiated. Patients will be followed for AEs for 28 days after the last treatment administration or until resolution or stabilization of all treatment-related AEs to either Grade ≤2 or baseline, whichever is longer, or until the patient is lost to follow-up. At any time after 28 days from the last dose of study treatment, the Investigator may report any SAE that he/she believes is related to study treatment.

Patients who discontinue elacestrant will be discontinued from the study, except for patients receiving abemaciclib as per Investigator's judgment.

All patients will be followed for survival until death approximately every 3 months for up to 24 months after enrollment of the last patient, except when a patient withdraws consent for this procedure.

Average treatment duration is expected to be approximately 6 months for Arms A, B, and C, and 9 months for Arm D.

7.5. Duration of Treatment

Treatment will continue until disease progression, death, development of unacceptable toxicity, withdrawal of consent, or other reasons.

Average treatment duration is estimated to be 6 months in all cohorts except Arm D, when the treatment duration is estimated to be 9 months.

7.6. Duration of Study

The study duration for each patient is estimated to be:

- **Screening Phase**: Up to 28 days prior to C1/D1.
- **Treatment Phase:** From C1/D1 until the date of radiologically documented progression, or treatment discontinuation due to other reasons.
- **Survival Follow-Up Phase:** All patients will be followed for survival approximately every 3 months, up to 24 months after enrollment of the last patient in each arm.

Patients will be followed for AEs for 28 days after the last treatment administration, whether it is elacestrant or the combination drug.

7.7. Study Stopping Rules

The following stopping rules will apply to both phases of the trial with each combination evaluated separately. Events believed to be related to elacestrant could be evaluated as an aggregate regarding the following:

- Any fatal adverse event considered related to any of the study drugs,
- Four or more patients have Grade 4 (NCI CTCAE v5.0) non-hematological adverse events leading to individual permanent drug discontinuation and considered related to the study drugs

In case Study Stopping Rules are met, recruitment in the corresponding treatment arm will be put on hold (temporary stopping), and the DEC will promptly meet to review the safety data and make a recommendation to the SSC. The SSC will be consulted for a final decision on if the specific combination or the entire study should be stopped.

7.7.1. End of Study

Arms A, B, C, and E: The cut-off date for the main analysis will be 6 months after enrollment of the last patient.

Arm D: The cut-off date for the main analysis will be 12 months after enrollment of the last patient.

The cut-off date for an updated analysis for all arms, including OS will be 24 months after enrollment of the last patient in each arm.

A patient is considered to have completed the study if the patient: has completed all periods of the study, including survival follow-up, within 24 months after enrollment of the last patient in his/her arm of the study; has died; has withdrawn consent for survival follow-up; or was lost to follow-up, whichever occurs first.

The end of the study is defined as the last visit of the last patient that is expected 24 months after the date of the global last patient in the study.

8. TREATMENT OF PATIENTS

8.1. Description of Study Drug

Table 22: Investigational Product

Product Name:	**Elacestrant**	**Alpelisib**
Dosage Form:	Film-coated tablet	Film-coated tablet
Unit Dose	100 or 400 mg of elacestrant dihydrochloride per tablet	50, 150, 200 mg
Route of Administration	Oral	Oral
Physical Description	Blue, immediate-release Opadry II film-coated tablet, containing 100 or 400 mg of elacestrant dihydrochloride drug substance per tablet (corresponding to 86 and 345 mg of elacestrant, respectively) and microcrystalline cellulose, silicified microcrystalline cellulose, crospovidone, colloidal silicon dioxide and magnesium stearate (nonbovine)	Film-coated tablet; for detailed information. cf. SmPC Alpelisib for Piqray® https://www.ema.europa.eu/en/documents/product-information/piqray-epar-product-information_en.pdf cf. USPI Alpelisib for Piqray® https://www.accessdata.fda.gov/drugsatfda_docs/label/2019/212526s000lbl.pdf
Packaging and Labeling	The secondary packaging and labeling of elacestrant will be performed under the responsibility of the Department of Pharmaceutical Development of A.MRBS GmbH. Elacestrant is packed in HDPE bottles (primary packaging) by Catalent Greenville, Inc. (formerly Mayne Pharma Inc dba Metrics Contract Services). The bottles will be labeled as patient kits according to the study design. Elacestrant will be labeled in compliance with current valid international and corresponding national requirements.	Authorized market preparations will be used as combination medication. Alpelisib should be obtained commercially through a local pharmacy or licensed distributor. It can also be provided under the responsibility of the Department of Pharmaceutical Development of A.MRBS with the following modification: – The authorized market preparations will not be modified except for re-packaging (secondary packaging and labelling). No additional substances or materials apart from secondary packaging will be added to the product. The label will comply with the valid international and corresponding national requirements.

Product Name:	**Everolimus**[a]	**Palbociclib**
Dosage Form:	Tablet	Film-coated tablet
Unit Dose	2.5, 5, 7.5, 10 mg	75, 100, 125 mg
Route of Administration	Oral	Oral
Physical Description	Tablet, for detailed information cf. SmPC Everolimus for Votubia®. https://www.ema.europa.eu/en/documents/product-information/votubia-epar-product-information_en.pdf cf. SmPC Everolimus for Affinitor®. https://www.ema.europa.eu/en/documents/product-information/afinitor-epar-product-information_en.pdf cf. USPI Everolimus for Afinitor® https://www.accessdata.fda.gov/drugsatfda_docs/label/2010/022334s6lbl.pdf	Film -coated tablet, for detailed information. cf. SmPC Palbociclib for Ibrance® https://www.ema.europa.eu/en/documents/product-information/ibrance-epar-product-information_en.pdf cf. USPI Palbociclib for Ibrance® https://www.accessdata.fda.gov/drugsatfda_docs/label/2019/212436lbl.pdf
Packaging and Labeling	Authorized market preparations will be used as combination medication. Everolimus should be obtained commercially through a local pharmacy or licensed distributor. It can also be provided under the responsibility of the Department of Pharmaceutical Development of A.MRBS with the following modification: – The authorized market preparations will not be modified except for repackaging (secondary packaging and labelling). No additional substances or materials apart from secondary packaging will be added to the product. The label will comply with the valid international and corresponding national requirements.	Authorized market preparations will be used as combination medication. Palbociclib should be obtained commercially through a local pharmacy or licensed distributor. It can also be provided under the responsibility of the Department of Pharmaceutical Development of A.MRBS with the following modification: – The authorized market preparations will not be modified except for repackaging (secondary packaging and labelling). No additional substances or materials apart from secondary packaging will be added to the product. The label will comply with the valid international and corresponding national requirements.

[a] Use of generics that are locally approved is allowed if they are also approved in either the US or in the European Union.

Product Name	**Abemaciclib**	**Ribociclib**
Dosage Form	Tablet or film-coated tablet	Tablet or film-coated tablet
Unit Dose	50, 100, and 150 mg tablet	200 mg
Route of Administration	Oral	Oral
Physical Description	Tablets for oral use. cf. USPI for Abemaciclib: Verzenio® https://www.accessdata.fda.gov/spl/data/977140a1-3e11-4020-b744-4edad117cf97/977140a1-3e11-4020-b744-4edad117cf97.xml. See Section 9 of EU SmPC for abemaciclib: Verzenios® https://www.ema.europa.eu/en/documents/product-information/verzenios-epar-product-information_en.pdf	Film-coated tablet; for detailed information. cf. SmPC ribociclib for Kisqali® https://www.ema.europa.eu/en/documents/product-information/kisqali-epar-product-information_en.pdf cf. USPI ribociclib for Kisqali® https://www.accessdata.fda.gov/drugsatfda_docs/label/2017/209092s000lbl.pdf Film-coated tablet; for detailed information, see IMPD.
Packaging and labeling	Authorized market preparations will be used as combination medication. Abemaciclib should be obtained commercially through a local pharmacy or licensed distributor. It can also be provided under the responsibility of the Department of Pharmaceutical Development of A.MRBS with the following modification: – The authorized market preparations will not be modified except for re-packaging (secondary packaging and labelling). No additional substances or materials apart from secondary packaging will be added to the product. The label will comply with the valid international and corresponding national requirements.	Authorized market preparations will be used as combination medication. Ribociclib should be obtained commercially through a local pharmacy or licensed distributor. It can also be provided under the responsibility of the Department of Pharmaceutical Development of A.MRBS with the following modification: – The authorized market preparations will not be modified except for re-packaging (secondary packaging and labelling). No additional substances or materials apart from secondary packaging will be added to the product. The label will comply with the valid international and corresponding national requirements.

Product Name	**Capivasertib**
Dosage Form	Film-coated tablet
Unit Dose	160 and 200 mg
Route of Administration	Oral
Physical Description	Film-coated tablets for oral use. cf. USPI for Capivasertib: TRUQAP® https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/218197s000lbl.pdf
Packaging and labeling	Authorized market preparations will be used as combination medication. Capivasertib should be obtained commercially through a local pharmacy or licensed distributor. It can also be provided under the responsibility of the Department of Pharmaceutical Development of A.MRBS with the following modification: – The authorized market preparations will not be modified except for re-packaging (secondary packaging and labelling). No additional substances or materials apart from secondary packaging will be added to the product. The label will comply with the valid international and corresponding national requirements.

8.2. Study Drug Storage

Storage conditions for each of the drugs in this study may be found in the Pharmacy Manual.

8.3. Administration

Patients should take elacestrant with the combination drug (alpelisib, everolimus, palbociclib or ribociclib) at the same time in the morning every day (except for off days for palbociclib, ribociclib or capivasertib). Both drugs should be taken with a low-fat meal. In the case of the elacestrant and abemaciclib combination, the evening dose of abemaciclib should also be taken with a low-fat meal. For the capivasertib combination on dosing days, the morning dose should be taken with elacestrant with a low-fat meal and the evening dose of capivasertib only can be taken with or without food.

Elacestrant and the combination drug should be swallowed whole. Tablets or capsules should not be ingested if they are broken or cracked. Dosing should not be repeated if a patient vomits. If the elacestrant or the combination drug doses are missed by more than 6 hours, they should be skipped, and the next dose should be taken at the usual time. A diary will be given to the patient to record the intake of the study drugs.

8.4. Study Drug Accountability

Upon receipt of all study treatment, study site personnel or the designated pharmacist will open the shipment package, verify the contents as stated on the enclosed Delivery Note and confirm the receipt in the IRT system.

The IRT will record study treatment delivery to study sites and patients and inventory at the sites, including dates, quantities, expiry dates, and batch/serial number.

The Investigator will be responsible for documenting dispensing of study treatment to patient by entering the unique medication number in the source documents, which is reported in the eCRF to allow for drug accountability.

Patients will be instructed to return used or unused study drugs at each visit to allow for drug accountability.

In addition, sites will maintain drug accountability forms to document dispensed and administered study treatment for each patient.

8.5. Study Drug Handling and Disposal

Throughout and at the end of the study, all remaining study treatment will be reconciled under the responsibility of the Investigator at the study site.

No later than the Site Closeout Visit, the used and unused elacestrant and the above listed study drugs shall be returned to A.MRBS/A.MRBS' depot for destruction, provided this is not in conflict with any national export legislation. In case local destruction is required due to national legislation a certificate of destruction, indicating the batch number and the box number, needs to be provided.

Any remains of used commercially obtained study drugs provided by local pharmacy or licensed distributor shall be destroyed locally.

8.6. Treatment Compliance

Records of treatment compliance for each patient will be kept during the study. A diary will be provided to patients on Day 1 of each cycle to document the date and time of each study drug dose for all treatment cycles, as well as the incidence of vomiting following administration. The Investigator and/or delegated site staff will review and document treatment compliance for each patient at each visit. Using this diary, the site personnel will enter the dosing information of study drug in the eCRF (at the required timepoints (Cycle 1 and Cycle 2), and a summary for all cycles. Clinical research associates will review treatment compliance during site visits and at the completion of the study. In the exceptional situation in which the drug's Patient Diary retrieval is unattainable, the Investigator and/or designee will be responsible for documenting treatment compliance per his/her knowledge.

8.7. Concomitant Medications

Patients will be instructed not to take any additional medications during the study without prior consultation with the treating investigator.

In general, the SmPC of the respective combination drug should be consulted for instructions/restrictions with respect to interactions with concomitant medications.

8.7.1. Permitted Concomitant Medications

All medications should be administered according to local guidelines and at the discretion of the treating investigator. The following medications, commonly used in the target patient population, may be administered:

- Erythropoietic and granulocyte growth factors and transfusions. Prophylactic use of growth factors during Cycle 1 of Phase 1b is not allowed. If required, suspend dosing of the combination drug for at least 48 hours after the last administration of blood cell growth factor and until toxicity resolves to Grade ≤2. Resume at the next lower dose unless already performed for the toxicity that led to the use of the growth factor. The dose of elacestrant does not need to be suspended or reduced in case of cytopenia.
- Bone-modifying agents, including bisphosphonates and receptor activator of nuclear factor kappa-B ligand inhibitors, if the patient has been on a stable dose for at least 1 month prior to initiation of trial therapy.
- Low molecular weight heparin, dabigatran, apixaban, rivaroxaban, or edoxaban.
- Antacids such as PPIs and histamine H2-receptor antagonists. The use of antiemetics for the management of nausea and vomiting is also allowed.
- For abemaciclib arms the use of antidiarrheal is recommended; see USPI.
- Other medications considered necessary for the patient's management if they are not included in the list of prohibited concomitant medications below.
- Short-course palliative radiotherapy for symptomatic isolated bone metastases, if these were clearly present at the time of study treatment start, and do not represent signs of progression. In absence of data about the interaction of elacestrant concurrently with radiotherapy, it is recommended to interrupt elacestrant 2 days before (48 hours) and resume when clinically indicated, as per Investigator's judgment.
- The use of non-drowsy antihistamines, such as loratadine, is strongly encouraged for prophylaxis in patients taking alpelisib.
- The use of prophylactic steroid mouthwash is strongly encouraged to help for prevention of mucositis in patients taking everolimus.
- Instruct patients to start antidiarrheal therapy (such as loperamide) at the first sign of loose stools, increase oral fluids, and notify their healthcare provider for further instructions and appropriate follow up.

8.7.2. Medications to be Administered with Caution

- Elacestrant coadministration with digoxin results in increased exposure of digoxin. Thus, patients receiving digoxin should be monitored for ADRs.
- Elacestrant coadministration with rosuvastatin results in increased exposure of rosuvastatin. Thus, patients receiving rosuvastatin should be monitored for ADRs.

- Ribociclib, everolimus and palbociclib may increase the exposure of sensitive CYP3A4 substrates with a narrow therapeutic index (e.g., alfentanil, aripiprazole, astemizole, cisapride, cyclosporine, dihydroergotamine, ergotamine, fentanyl, pimozide, quinidine, sirolimus, tacrolimus and terfenadine) when coadministered. Therefore, in ribociclib, everolimus and palbociclib combination arms, patients receiving sensitive CYP3A4 substrates with a narrow therapeutic index should be monitored for ADRs and their dose may need to be reduced according to their respective SmPCs.
- Caution should be exercised in case of coadministration with other P-gp (e.g. dabigatran etexilate, fexofenadine) and BCRP (e.g. atorvastatin, cimetidine, fluvastatin, glyburide, prazosin, simvastatin, sulfasalazine) substrates. Reduce the dosage of P-gp substrates or BCRP substrates per their prescribing information when minimal concentration changes may lead to serious or life-threatening adverse reactions. The use of alternatives to P-gp and/or BCRP substrates is recommended whenever possible.
- Caution is advised for the use of medicinal cannabis (marijuana), where legally allowed, as there is no formal assessment for potential interactions based on the previous EMERALD study. The strain, dose and the administration schedule should be properly documented in the eCRF.

8.7.3. Prohibited Concomitant Medications

The following treatments are prohibited while on this study:

- Other investigational drugs.
- Other anticancer drugs or any radiotherapy not specifically authorized in Section 8.7.1.
- Hormone replacement therapy (not including thyroid treatment, which is allowed on study), topical estrogens (intravaginal preparations), megestrol acetate and selective estrogen modulators (e.g., basedoxifene, ospemifene, raloxifene, tamoxifen) are prohibited during the active treatment phase.
- Strong CYP3A4 inhibitor drugs, food and herbal products, including but not limited to*: ceritinib, clarithromycin, cobicistat, danoprevir, dasabuvir, elvitegravir, idelalisib, itraconazole, ketoconazole, lopinavir, nefadozone, nelfinavir, ombitasvir, paritaprevir, posaconazole, ritonavir, saquinavir, telithromycin, tipranavir, troleandomycin, voriconazole.
- Moderate CYP3A4 inhibitor drugs, food and herbal products, including but not limited to*: amprenavir, aprepitant, atazanavir, casopitant, cimetidine, ciprofloxacin, conivaptan, crizotinib, cyclosporine, darunavir, darunavir/ritonavir, diltiazem, dronedarone, erythromycin, faldaprevir, fluconazole, fluvoxamine, fosamprenavir, imatinib, isavuconazole, letermovir, netupitant, nilotinib, octreotide, tofisopam, verapamil.

- Strong CYP3A4 inducer drugs, food and herbal products, including but not limited to*: carbamazepine, enzalutamide, lumacaftor, mitotane, phenobarbital, phenytoin, rifampin, rifabutin.
- Moderate CYP3A4 inducer drugs, food and herbal products, including but not limited to*: bosentan, dabrafenib, efavirenz, etravirine, lesinurad, lopinavir, modafinil, nafcillin, thioridazine, tipranavir/ritonavir, telotristat ethyl
- In the ribociclib arm in both phases avoid concomitant use of drugs known to prolong QT interval including, but not limited to: amiodarone, arsenic trioxide, astemizole, azithromycin, bepridil, chloroquine, chlorpromazine, ciprofloxacin, cisapride, citalopram, clarithromycin, disopyramide, dofetilide, domperidone, droperidol, erythromycin, flecainide, halofantrine, haloperidol, ibutilide, levofloxacin, levomethadyl acetate, mesoridazine, methadone, moxifloxacin, ondansetron, pentamidine, pimozide, probucol, procainamide, quinidine, sotalol, sparfloxacin, sulpiride, terfenadine, thioridazine, vandetanib (https://crediblemeds.org)
- In the alpelisib arm in both phases, avoid concomitant use of drugs known to be a BCRP inhibitor including, but not limited to: curcumin, cyclosporine A, darolutamide, eltrombopag, febuxostat, fostamatinib, rolapitant, teriflunomide
- Herbal preparations, including, but are not limited to, St. John's wort, kava, ephedra (ma huang), gingko biloba, DHEA, yohimbe, saw palmetto, and ginseng.
- Grapefruit, grapefruit juice, and grapefruit-containing beverages.
- Systemic corticosteroids unless the patient was on a stable or decreasing dose of corticosteroids at baseline for management of brain metastasis. Single doses, topical applications, inhaled sprays, eye drops, local injections, and short-term courses of corticosteroids, including mouthwashes, are allowed. Systemic short-term steroids are also allowed for treatment of suspected alpelisib-related rash as appropriate.

* For the most updated information, refer to the following Web address, supported by Indiana University: Flockhart D (http://medicine.iupui.edu/clinpharm/ddis/) or FDA website (https://www.fda.gov/drugs/drug-interactions-labeling/drug-development-and-drug-interactions-table-substrates-inhibitors-and-inducers).

9. STUDY ASSESSMENTS

9.1. Assessment of Efficacy

All study images for tumor assessments obtained during both phases of the trial will be used as the basis of Investigator-determined response assessments and will be performed at each assessment time point and entered in the eCRF.

Tumor assessments will include:

- Chest: CT with contrast
- Abdomen, pelvis, and other known or suspected sites of the disease: CT (with contrast) or MRI (with contrast)
- Brain: CT with contrast, or MRI (pre- and postcontrast). Please note: Brain assessment is only required if the patient has a history of brain metastasis or symptoms suggestive of brain metastasis.

Tumor assessments will be performed during Screening/Baseline period (Day -28 to Cycle 1 Day 1, as close as possible to Cycle 1 Day 1), every 8 weeks (±7 days) after Cycle 1 Day 1, and as indicated if disease progression is suspected clinically.

Use of IV contrast CT scan:

The modality of contrast study should be considered by the Investigator, based on local practice and access to other available alternatives.

For severe hypersensitivity/contraindication to iodinated IV contrast, consider alternative options based on local practice (i.e., change in type of contrast, pre-medication etc.), see link below. If there is a severe shortage of IV contrast: imaging without contrast could be performed for the appropriate body section, see below. The reason for contraindication/unavailability should be captured in the source documents. Every effort should be made to ensure that the same imaging modality is used throughout the study.

- Chest: CT with contrast > CT without contrast. Magnetic resonance imaging of the chest is not recommended.
- CT with contrast of abdomen, pelvis, and other known or other suspected sites of the disease > MRI (with contrast) is recommended. Do not change back to CT once MRI has been instituted.
- Brain CT (with or without contrast) > MRI (with or without contrast)

For other recommendations, see details in the American College of Radiology IV contrast manual (https://www.acr.org/-/media/ACR/Files/Clinical-Resources/Contrast_Media.pdf).

A ^{99}m-Technetium based bone scan will be performed during screening (a historical bone scan performed within 6 weeks before Cycle 1 Day 1 is acceptable), approximately every 24 weeks (in conjunction with a scheduled tumor assessment visit), and as clinically indicated. Lesions identified on bone scans must be confirmed and followed with cross-sectional imaging (i.e., CT scan or MRI).

All responses must be confirmed no less than 28 days following the initial assessment of response. For stable disease to be considered the BOR, it must occur ≥6 weeks following the first dose of study drug.

Patients who discontinue study treatment without radiological disease progression will continue to have tumor assessments per the Schedule of Assessments until radiological disease progression, death, or initiation of another anticancer therapy, whichever occurs first, unless the study is terminated.

9.2. Assessment of Safety

Safety assessments will consist of monitoring and recording all adverse events, including all CTCAE v5.0 grades (for both increasing and decreasing severity); regular monitoring of hematology, blood chemistry, and complete metabolic panel; periodic measurement of vital signs and ECGs; and performance of physical examinations as detailed in Table 20 and Table 21. All safety analyses will be performed using SAS version 9.4 or a later version.

9.2.1. Demographic/Medical History

Demographic information such as sex, age, race, ethnicity, as well as weight, height, and breast cancer–specific history and prior therapy status and medical and surgical history and current medical conditions will be recorded at the Screening Visit. All other clinically significant medical and surgical history within 5 years must be noted in the Medical History, Previous and Concomitant Medication, and Procedures eCRF.

9.2.2. Physical Examination and Vital Signs

A complete physical examination (including also neurological examination) will be performed at Screening, and symptom-driven throughout the study at each study visit. It will include a general appearance observation and a complete examination of the following body systems/areas: HEENT/Neck, Lymph Nodes, Thyroid, Abdomen, Skin, Cardiovascular, Respiratory, Gastrointestinal, Neurological and Musculoskeletal/Extremities.

Vital signs will be recorded throughout the study at each study visit from Screening to the Safety Follow-up Visit. The following parameters will be measured:

- Heart rate (beats/minute)
- BP (systolic and diastolic, mmHg)
- Respiratory rate (breaths/minute)
- Body temperature (ºC)

Body weight (to the nearest 0.1 kg in indoor clothing, but without shoes) will be measured during the Screening Period and at Day 1 of subsequent cycles. Height will be recorded at the baseline visit only.

No need to repeat physical examinations on Cycle 1 Day 1 if the baseline physical examination was performed within ≤72 hours.

9.2.3. Electrocardiogram

Local 12-lead ECG machines will be utilized in this study. Electrocardiograms will be performed according to the Schedule of Assessments (Table 20 and Table 21). Each ECG, both at Screening and during the study, is to be performed in triplicate approximately 2 minutes apart as per local practice. For each of the ECG parameters (heart rate, QTcF interval, QTcB interval, PR interval, and QRS duration), values from each of the triplicate measurements will be entered in the eCRF.

9.2.4. Laboratory Assessments

All laboratory assessments will be performed as shown in the Schedule of Assessments (Table 20 and Table 21).

9.2.5. Hematology

Hematological parameters include the following laboratory tests: CBC consisting of hematocrit, hemoglobin, total WBC with 5-part differential, and platelet count. Hematology assessment does not need to be repeated on Cycle 1 Day 1 if it was performed within 7 days of baseline. The CBCs are collected and results assessed at every visit before treatment.

9.2.6. Comprehensive Metabolic Panel

The CMP includes the following: BUN (or urea per local practice), Cr, glomerular filtration (CrCl), sodium, calcium (corrected for serum albumin), potassium, chloride, phosphate, and magnesium, carbon dioxide, glucose, lipase, albumin, total and direct bilirubin, ALP, AST, and ALT. Additionally a lipid panel (total cholesterol, HDL, LDL, triglycerides) should be performed at baseline and periodically as per Table 20 and Table 21. Estradiol and FSH will be measured also at Screening for women under 60 years old (if these tests have been performed within 3 months prior to Screening Period in the context of the standard subject's management they can be reported in the eCRF, and these tests will not be repeated). If hyperglycemia is suspected, repeat glucose testing after fasting per local practice.

9.2.7. Virus Serology

Testing criteria for HBV and HCV are provided in Appendix B. Testing during screening is only required for patients with a history of HBV or HCV infection or suspected infections.

9.2.8. Pregnancy Screen

In accordance with the definition of WOCBP cited in clinical trial facilitation and coordination group (https://www.hma.eu/fileadmin/dateien/Human_Medicines/01-About_HMA/Working_Groups/CTFG/2020_09_HMA_CTFG_Contraception_guidance_Version_1.1_updated.pdf), a woman is considered of childbearing potential (WOCBP), i.e., fertile, following menarche and until becoming post-menopausal unless permanently sterile. Permanent sterilization methods include hysterectomy, bilateral salpingectomy, and bilateral oophorectomy. A postmenopausal state per the CTFG is defined as no menses for 12 months without an alternative medical cause. For this protocol, postmenopausal status for women is defined as:

a) Age ≥60 years
b) Age <60 years and amenorrhea for 12 or more months (without an alternative cause) **and** FSH value **and** an estradiol level within the postmenopausal range per local laboratory reference.

In addition, since pre-/peri- menopausal women on study are required to undergo chemical menopause with the addition of LHRH agonists at least 3-4 weeks prior to first dose of study drug and remain on LHRH agonists throughout the trial, no woman is expected to have child-bearing potential while on the study treatment.

A serum pregnancy test must be performed at Screening and Cycle 1 Day 1 for all women of childbearing potential (i.e., pre- and perimenopausal patients). Since all pre- and perimenopausal women are required to take LHRH agonists prior to study start and during study treatment, and thus rendered medically post-menopausal, pregnancy tests during study will only be done if deemed necessary by treating investigator, for instance in cases of suspected non-compliance with LHRH agonist therapy. Serum testing should be performed to confirm a positive urine test.

9.3. Pharmacokinetic Assessments

Pharmacokinetic assessments will be performed for all patients enrolled in Phase 1b according to Table 20, Table 21, and Appendix G. If samples were missed or not collected as described in Table 20, Table 21, and Appendix G, or in the case of serious SAEs where PK may be beneficial for assessment of causality, please contact Sponsor to determine if additional PK sampling is required.

9.3.1. Receipt of Pharmacokinetic Samples

The PK samples will be delivered by Cerba Research to the Clinical Laboratory "Menarini Ricerche S.p.A.", Via Tito Speri 10, 00071 Pomezia (Rome), Italy.

Upon receipt at Clinical Laboratory "Menarini Ricerche S.p.A." all samples will be checked by the Clinical Laboratory team against the Shipping Manifest provided by Cerba Research to determine if there are discrepancies or confirm that the samples have been received in an appropriate way by filling in and signing an acknowledgement of sample receipt and send it by email to Cerba Research.

The Clinical Laboratory team will inform the Sponsor by email in case of discrepancies of the sample arrival condition and if they will not be suitable for the processing and analysis.

In any case, the Sponsor ensures the quality of samples from their collection to their destruction by ensuring they are always stored at appropriate temperature during lab storage and transfer.

9.3.2. Pharmacokinetic Samples Bioanalysis

The PK sample bioanalysis will be performed according to previously validated chromatographic (LC-MS/MS) bioanalytical methods, in compliance with the GCP in a GLP-compliant facility.

The Sponsor will inform the Clinical Laboratory "Menarini Ricerche S.p.A." in case of informed consent withdrawn by a trial patient.

9.3.3. Retention of Pharmacokinetic Samples

After analysis, all leftover PK samples will be stored at the appropriate temperature and disposed of after the finalization of the bioanalytical report and never longer than 1 year after the end of the clinical trial.

9.4. Pharmacodynamic and Other Biomarkers Assessments

Blood samples for cfNA will be collected during Cycle 1 Day 1, Cycle 1 Day 21, Cycle 3 Day 1, and at the EOT visit (maximum total of 4 samples on study). Samples should be collected before dosing and immediately sent to the central laboratory for processing.

9.5. Adverse and Serious Adverse Events

Monitoring of AEs will be conducted throughout the study; SAEs that are assessed as related to treatment that occur >28 days after last dose of study treatment also are to be reported. All AEs should be monitored until 28 days after the last treatment administration or until resolution or stabilization of all treatment-related AEs to either Grade ≤2 or baseline, whichever is longer, or until the patient is lost to follow-up.

9.5.1. Adverse Event Reporting Period

Investigators will seek information on AEs at each patient contact or visit. All AEs, whether reported by the subjects or noted by the study personnel, will be recorded in the subject's medical record and on the AE page of the eCRF.

All AEs and SAEs, regardless of attribution, will be collected every visit after a patient signs the ICF through 28 days after last dose of study treatment or until study discontinuation or until initiation of subsequent anticancer therapy, whichever occurs first. Subjects will be assessed at the Safety Follow-Up visit to determine if any new AEs have occurred. After this period, Investigators should report only SAEs that are considered related to study treatment (elacestrant or any other study drug in the combination).

9.5.2. Definition of Adverse Events

9.5.2.1. Adverse Event

An AE is any untoward medical occurrence in a subject to whom a medicinal product is administered, and which does not necessarily have a causal relationship with this treatment.

An AE can be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom or disease temporally associated with the use of a medicinal product, whether or not considered related to the medicinal product (see Section 2A1 of ICH E2A23).

In addition, any unfavorable and unintended sign (including an abnormal laboratory finding), symptom or disease temporally associated with any intervention conducted due to the subject participation in the clinical trial, even if not associated to a medicinal product, should also be considered as an AE.

Clinically significant abnormal laboratory findings are considered AEs, however abnormal laboratory findings may not be considered as AEs if there is no change compared to baseline values (at randomization).

Disease progression will not be considered an AE (or an SAE) in this study, but will be collected as a reason for treatment discontinuation, as appropriate, and in the Tumor Assessment section of the eCRF. Adverse events (or SAEs) considered complications of disease progression should be reported. See Section 9.5.2.1.1 for more detailed instructions regarding the reporting of disease progression.

All AEs that occur after a patient signs the ICF, through 28 days following the cessation of treatment, whether or not considered to be related to the study, must be recorded on the eCRFs.

If the Investigator becomes aware of a SAE after the reporting period has ended, and he/she believes that there is a possibility that it is related to the study treatment, he/she should report it immediately.

9.5.2.1.1. Progression of Disease

Progression of metastatic breast cancer should not be reported as an AE if it is clearly consistent with the suspected progression of the underlying cancer as defined by [RECIST v1.1]. Clinical symptoms of progression may be reported as AEs if the symptoms led to a clinical diagnosis differentiated from the progression of the subject's underlying metastatic breast cancer or does not fit the expected pattern of progression for the disease (e.g., pulmonary embolism, pneumonia, sepsis, stroke).

9.5.2.1.2. Adverse Drug Reaction (ADR)

An Adverse Drug Reaction (ADR) is any untoward and unintended response to an Investigational Medicinal Product related to any dose administered.

The definition implies a reasonable possibility of a causal relationship between the event and the IMP. That means that there are facts (evidence) or arguments to suggest a causal relationship (see Section 9.6.).

9.5.2.2. Serious Adverse Event/Serious Adverse Reaction

A serious AE occurring during any study phase (i.e., baseline, treatment, or follow-up) and at any dose of the investigational product is considered as serious when it fulfills 1 or more of the following outcomes:

- Results in death
- Is immediately life-threatening
- Requires in-patient hospitalization or prolongation of existing hospitalization
- Results in persistent or significant disability or incapacity
- Results in a congenital abnormality or birth defect
- Is an important medical event that may jeopardize the patient or may require medical intervention to prevent one of the outcomes listed above.

A serious adverse reaction (SAR) is all noxious and unintended responses to an investigational medicinal product related to any dose administered meeting the seriousness criteria above.

The terms serious and severe are not synonymous. Severity refers to the intensity of an AE (mild, moderate, or severe, or according to NCI CTCAE 5.0 criteria); the event itself may be of relatively minor medical significance (such as severe headache without any further findings).

Notes:

1. "Life-threatening event" refers to an event in which the patient was at risk of death at the time of the event, it does not refer to an event which hypothetically might have caused death if it had been more severe.
2. "Persistent or significant disability or incapacity" means a permanent or significant and substantial disruption of a person's ability to carry out routine activities.
3. Hospitalization lasting less than 24 hours or pre-planned hospitalization for medical intervention, such as study treatment administration or PK sampling time-points, shall not qualify as SAE.

All SAEs that occur after a patient has signed the ICF through 28 days following the cessation of treatment, whether or not related to the study drugs, must be recorded on the eCRF.

Examples of pre-planned hospitalization scenario, which should not be considered as SAEs:

- Standard procedure for protocol therapy administration; however, hospitalization or prolonged hospitalization for a complication of therapy administration will be reported as an SAE.
- Administration of blood or platelet transfusion as routine treatment of studied indication; however, hospitalization or prolonged hospitalization for a complication of such transfusion remains reportable as an SAE.
- A procedure for a protocol/disease-related investigations (e.g., surgery, scans, endoscopy, sampling for laboratory test, bone marrow sampling); however, hospitalization or prolonged hospitalization for a complication of such procedures remains a reportable SAE.
- Hospitalization or prolongation of hospitalization for technical, practical, or social reasons, in the absence of an AE.
- Hospitalization for respite care.
- Routine treatment or monitoring of the studied indication not associated with any deterioration in condition.
- A procedure that is planned (e.g., planned prior to start of treatment on study); must be documented in the source document and the eCRF; hospitalization or prolonged hospitalizations for a complication remains a reportable SAE.
- An elective treatment or an elective procedure for a pre-existing condition, unrelated to the studied indication, that has not worsened from baseline.

9.5.2.3. Treatment-Emergent Adverse Events

Adverse events will be categorized as TEAE or non-TEAE. If an AE occurs for the first time or if it worsens in terms of seriousness or severity after the first study drug intake (elacestrant in combination with the other drugs), it will be classified as TEAE. Otherwise, it will be classified as non-TEAE.

9.5.2.4. Adverse Event/Adverse Drug Reaction Intensity (Severity)

Each event will be graded for severity using the classifications of NCI CTCAE v5.0. For events not addressed in the NCI CTCAE v5.0 classifications, the following grading will apply:

- Mild (Grade 1) – Mild; asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated.
- Moderate (Grade 2) – Moderate; minimal, local or non-invasive intervention indicated; limiting age-appropriate instrumental activity of daily living.
- Severe (Grade 3) – Severe or medically significant but not immediately life threatening; hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care activity of daily living.
- Life-threatening (Grade 4) – Life-threatening consequences; urgent intervention indicated.
- Death (Grade 5) – Related to adverse event.

9.5.2.5. Adverse Events of Special Interest

AESIs will be identified by the Drug Safety Physician and, if applicable, also by their Clinical Study Team Physician during the evaluation of safety data for the clinical study report. Adverse events of clinical importance, other than SAEs and AEs leading to discontinuation of the patient from the study, which may require further investigation for their characterization, once identified and properly assessed, may qualify as AESIs for the overall clinical development program of elacestrant. For each of these events, a narrative will be written and included in the clinical study report.

Currently, there are no AESIs identified for elacestrant.

9.6. Relationship to Study Drug

A related AE is any AE for which there is a reasonable possibility that the study treatment caused the AE.

Investigators should use their knowledge of the patient, the circumstances surrounding the event, and an evaluation of any potential alternative causes to determine whether an AE is considered related to any of study treatment, indicating on the eCRF that an AE is either related or unrelated.

The following guidance should be taken into consideration when determining the relationship of an AE to any of the study drugs or their combination. The following should be considered:

- Temporal relationship of event onset to initiation of study treatment
- Course of the event, considering especially the effects of dose reduction, discontinuation of study treatment, or reintroduction of study treatment (when applicable)
- Known association of the event with study treatment or with similar treatments
- Known association of the event with the disease under study
- Presence of risk factors in the patient or environment, or use of concomitant medications known to increase the occurrence of the event
- Presence of treatment-unrelated factors that are known to be associated with the occurrence of the event
- Whether there is a reasonable alternative explanation for the event.

The relationship of each AE to each of the study drugs in both phases and in each cohort will be recorded on the eCRF in response to the following question for each of the combination drugs:

Is there a reasonable possibility that the study drug caused the AE?

Yes (related)	A causal relationship between the study drug and the AE is a reasonable possibility (specifying the relationship with elacestrant, the combination therapy (alpelisib, everolimus, palbociclib, abemaciclib, ribociclib or capivasertib) or both).
No (not related)	A causal relationship between the study drug(s) and the AE is not a reasonable possibility.

9.7. Recording Adverse Events

Adverse events spontaneously reported by the patient and/or in response to an open question from the study personnel or revealed by observation will be recorded during the study at the investigational site.

The Investigator must record all available information concerning any non-serious AE (whether or not deemed related to any of the study treatment) in the corresponding section of the eCRF: AE pages, within 5 calendar days after the first knowledge of the occurrence of the event. When the site personnel enter a new AE in the eCRF, an automatic alert notification is generated and sent to the Sponsor.

During the clinical study, any laboratory test abnormality (newly occurring after the study drug administration or worsening of previously known abnormalities) considered as clinically significant per Investigator's judgment (such as values significantly above or under normal range or which require an intervention or diagnostic tests or may result in the discontinuation of the study treatment) must be reported as AEs.

Disease progression will not be considered an AE (or an SAE) and will be captured as an efficacy endpoint.

Any death (including those in which disease progression is the only known cause), observed while receiving trial therapy or within 28 days of last treatment administration, must be reported to the Sponsor on the Death Detail page of the eCRF within 24 hours of the first knowledge. In addition, any cause of death other than disease progression must be reported on the eCRF-AE page.

Information about AEs (regardless of seriousness) will be collected from the ICF signature until 28 days after the last dose of study drug. The AE term should be reported in standard medical terminology when possible, in English language and avoiding the use of abbreviations. For each AE, the Investigator will evaluate and report the onset (date and time), resolution (date and time), grade (per the CTCAE, version 5.0), causality, action taken, seriousness, and outcome (if applicable), and whether or not it caused the patient to discontinue any of the study drugs.

It is important to distinguish between serious and severe AEs. Severity is a measure of intensity whereas seriousness is defined by the criteria under Section 9.5.2.2.

9.8. Reporting Adverse Events

For patients who were considered as screen failures, AEs should be collected from ICF until the date when the screen failure was determined.

All SAEs (related and unrelated) will be recorded from the ICF signature until 28 days after the last dose of study drug. Any SAEs considered related to the investigational product and discovered by the Investigator at any time after the study end should be reported. All SAEs must be reported to the Sponsor within 24 hours of the first awareness of the event. The Investigator must complete, confirm the seriousness of the SAE in the eCRF, and verify the accuracy of the information recorded on the eCRF with the corresponding source documents. If the eCRF is no longer available, the paper CRF-SAE form must be used.

Additional follow-up information, if required or available, should all be notified to Stemline, Inc. within 1 day of receipt by updating the data in the eCRF (i.e., follow-up report).

9.9. Suspected Unexpected Serious Adverse Reaction

Any SAE judged by the Investigator or the Sponsor as drug-related (see Section 9.6) and considered as unexpected is qualified as a SUSAR.

SUSARs are subject to expedited reporting, as specified in Section 9.10.2, as having a "Reasonable Possibility" of relationship with any of the study treatments or their combination.

A treatment-related SAE is considered unexpected when its nature, intensity, or outcome is not consistent with the applicable product information provided in the Reference Safety Information (elacestrant IB, elacestrant IMPD- S&E and SmPC, alpelisib SmPC, everolimus SmPC, palbociclib SmPC, abemaciclib SmPC, ribociclib SmPC, and capivasertib USPI).

9.10. Management of Serious Adverse Events

9.10.1. Reporting Duties of the Investigator

All AEs (serious and non-serious) spontaneously reported by the subject and/or in response to an open-ended from study personnel or revealed by observation, physical examination, or other

diagnostic procedures will be recorded on the appropriate eCRF page. Any clinically relevant deterioration in laboratory assessments or other clinical findings is considered an AE and must be recorded on the appropriate eCRF page. When possible, a diagnosis should be provided in lieu of multiple signs and symptoms representing a single comprehensive medical diagnosis.

The Investigator (or designated representative/site staff) must record and save all the available information concerning any SAE (whether or not deemed related to any of the study treatments) in the corresponding section of the eCRF: eCRF-AE pages, no later than 24 hours after the first knowledge of the occurrence of the event. When the site personnel enter a new SAE in the eCRF, an automatic alert notification will be generated and sent to the SDSU.

The Investigator will be provided with the paper CRF-SAE form to be used only in case of breakdown of the eCRF system. In such a case, the Investigator will be responsible for sending the paper CRF-SAE form as described below **no later than 24 hours** after the first knowledge and subsequently entering the data in eCRF as soon as the system works again. Whenever the paper CRF-AE forms are used, they must be submitted by e-mail to the Sponsor's SDSU (Elevatesafety@menarinistemline.com). For the initial SAE, the Investigator should enter at least the following data:

- AE medical term.
- Seriousness criteria.
- Causality assessment.
- Study Code and Patient Identification (patient ID) [when the paper CRF-SAE form is used].

Reporter's name and contact details for clarification [when the paper CRF-SAE form is used]. If not already reported, the full description of the event and outcome must follow within 1 working day. The Sponsor's confirmation of reception of the SAE report must be kept in the patient's records. Any questions arising during the processing and medical review of the SAE will be managed by means of electronic queries (i.e., queries in the eCRF). In case of a breakdown of the eCRF system, queries will be sent by email.

Any information provided by the Investigator as a query reply or as a follow-up SAE report will be processed in the same way as the initial SAE report, within the required timeframe.

When relevant, the eCRF pages concerning medical history, concomitant medication and laboratory tests will also be retrieved.

Any further significant information and supporting documentation that become available (such as copies of laboratory reports, tests, procedures, autopsy evidence of the cause of death, etc.) shall be entered in the eCRF or provided by e-mail to the Sponsor (in case of eCRF unavailability), no later than 24 hours after they become known by the site.

The Investigator must also comply with the local applicable obligations on the reporting of ADRs and other safety cases and reports (such as SUSAR reporting to IRB in the US) to the central or local concerned IRBs/ECs if required, according to the specific country requirements.

9.10.2. Reporting Duties of the Sponsor

The Sponsor shall ensure that all relevant information about any SUSAR is expeditiously reported to the CAs and ECs/IRBs, following general and local rules and procedures. The deadlines to be complied with, starting on the date of first knowledge, intended as the day when the Sponsor or CRO receives the notification of the SUSAR, are the following:

- Fatal and life-threatening unexpected cases, no later than 7 days
- Other unexpected serious cases, no later than 15 days.

The Sponsor shall ensure that all relevant new information is also expeditiously reported as follow-up information within 15 days for all cases.

In accordance with the EU CTR 536/2014, SUSARs will be reported in EudraVigilance.

The following safety issues will be subjected to expedited management for the identification of possible necessary actions:

- SAEs associated with the study procedures
- Potential clinically significant findings emerging from non-clinical studies
- An anticipated end or suspension for safety reasons of another study with the same study treatment.

When appropriate and applicable per local and regional regulatory requirements, the Sponsor will arrange the adequate information also to be sent to the Investigators. The Sponsor will distribute the validated CIOMS I or MedWatch (for local cases in the US) form to the investigators.

For SUSARs that occurred in other ongoing studies with elacestrant, the Sponsor will comply with the expedited reporting to the ECs/IRBs and CAs involved in ELEVATE study.

The Sponsor will be responsible for expeditiously reporting SUSAR cases attributable to any of the investigational drugs, following the procedure described in the SMP, and informing the corresponding MAH (per elacestrant IB details) of the case and of the expedited reporting done.

9.11. Management of Pregnancy Exposure Cases

The Investigator is expected to record in a dedicated electronic “Pregnancy Exposure Report Form” any case of pregnancy occurring in a female patient or in a female partner of a male patient enrolled in the study, occurring during the treatment and/or follow-up periods. In case the eCRF is no longer available, it will be notified through the paper Pregnancy Exposure Form and sent to the Sponsor by email.

In case of pregnancy, the study treatment will be permanently discontinued. The Investigator is requested to follow each case of pregnancy exposure until the outcome (or longer, if required by local regulations), provided that the female patient or the female partner of a male patient enrolled in the study has signed the related pregnancy ICF.

The Investigator should fill out this electronic form within **24 hours** of becoming aware of the pregnancy and fully complete the form within **24 hours** after the outcome is known.

If the pregnancy results in an abnormal outcome, this will be recorded in the eCRF as an SAE and managed as described in Section 9.10. In case the eCRF is no longer available, this abnormal outcome recorded as SAE will be notified through the paper CRF-SAE form and sent to the Sponsor by email.

9.12. Management of Misuse and Overdose Cases

Study treatment overdose is the accidental or intentional overdose refers to the administration of a quantity of a medicinal product (either elacestrant and/or any of the combination drugs) given per administration or cumulatively, which is above the maximum dose being studied. An overdose or incorrect administration of study treatment is not an AE unless it results in an untoward medical effect. Observation and general supportive measures should be instituted in the setting of an accidental or intentional overdose.

A case of misuse describes situations where the medicinal product is intentionally and inappropriately used not in accordance with the terms of current protocol.

Although study drug misuse and overdose are not considered as AEs per se, both situations should be reported to the Sponsor within 24 hours, even if they may not result in an adverse outcome. Any study treatment overdose or incorrect administration of study treatment should be noted on the Patient Diary and the Adverse Event page on the eCRF.

If an AE (serious or non-serious) is associated with a misuse or overdose it will be recorded on the AE page in the eCRF, recording the overdose details. For SAEs resulting from misuse or overdose of any study treatments, should be reported within **24 hours** of awareness by the site personnel. Once the page is completed and saved by the staff involved in the study, an alert notification will be automatically sent to the Sponsor's SDSU. ONLY if the eCRF system does not work or if the eCRF is not available, the paper SAE report form shall be used and sent to the Sponsor by email.

If the pharmacy service discovers that an overdose has or may have been administered, they should immediately contact the Investigator and Sponsor (or their delegate) and let them know.

9.13. Follow-up of Subjects After Adverse Events

9.13.1. Sponsor Follow-up

For SAE, AEs of special interest, and pregnancies, the Sponsor or a designee may follow up with the Investigator by telephone, fax, electronic mail, and/or a monitoring visit to obtain additional case details and outcome information (e.g., from consultant reports, hospital discharge summaries, autopsy reports) to perform an independent medical assessment of the reported case.

9.13.2. Investigator Follow-up

The Investigator should follow each AE until the AE has resolved to baseline, the event is assessed as stable by the Investigator, the subject is lost to follow-up, or the subject withdraws consent. Every effort should be made to follow all SAEs considered related to study treatment or trial-related procedures until an outcome can be reported.

All pregnancies reported during the study should be followed until pregnancy outcome.

9.13.3. Post Study Follow-up

At treatment discontinuation visit, the Investigator should instruct each subject to report to the Investigator any subsequent AEs that the subject's investigator believes could be related to prior study treatment or study procedures. The Investigator should notify the Sponsor of any serious and related death, SAE, or other safety concern occurring at any time after a subject completed the study or study procedures. The Sponsor should also be notified if the Investigator becomes aware of the development of cancer or congenital anomaly/birth defect in a subsequently conceived offspring of a subject participating in the study.

9.14. Periodic Safety Reporting

9.14.1. Annual Safety Reporting

Given each of the compounds (alpelisib, everolimus, palbociclib, abemaciclib, ribociclib, and capivasertib) used in combination with elacestrant are approved and are being used in accordance with the approved indications, it is not anticipated that the benefit-risk profile of the individual agents or in combination with elacestrant would change. As such, a single consolidated annual safety report with a data cutoff based on the Development International Birth Date will be prepared yearly for all ongoing elacestrant trials using the relevant harmonized elacestrant DSUR, involving the different IMPs used in combination in these trials. The DSUR will serve as Annual Safety Report.

The Sponsor will submit to the concerned Health Authorities and Ethics Committees, as applicable, accounting for all new safety information received during the reporting period.

The annual safety report will be submitted to Health Authorities and Ethics Committees by the Sponsor within in accordance with all local and regional regulations following the DLP.

9.14.2. Periodic Line-listings Safety Reporting

All SUSARs occurring in the ELEVATE study and/or in any other ongoing trial using elacestrant as study drug and conducted by the Sponsor in any of the participating countries will be recorded in a line-listing prepared by the Sponsor every 6 months. Line-listings will serve the Sponsor in its continuous review of safety data to gain insight into increased frequency or signal of a particular AE.

The final line-listing of SUSARs will be sent to the applicable ECs and Investigators, in accordance with local and regional regulations. All Investigators are required to acknowledge their correct reception and reading of this information to the Sponsor by email. If no SUSARs occur, the line-listing will not be necessary.

9.15. Remote Visits

For patients who experience significant hardship coming to the study site, remote visits (Telehealth and home care visits), if allowed by local law, may be performed after sponsor approval instead of visits designated to be conducted at the research site. During these remote visits, the Health Care Provider will assess patient safety and collect relevant data, including but not limited to, drug compliance, compliance with contraception use as indicated, new AEs or SAEs, new concomitant medications, review of laboratory and tumor assessments.

10. STATISTICS

All statistical analyses will be performed by the sponsor or designee. Statistical analyses will be performed using SAS software or other validated statistical software as required. Details of the statistical analyses will be included in a separate SAP.

10.1. Definitions of Analysis Sets

Except for the DLT analysis set that will only apply to Phase 1b, all other analysis sets will be defined separately for Phase 1b and Phase 2 of each study arm.

- DLT Analysis Set: all patients who start trial therapy unless they receive less than 70% of their scheduled therapy during the first cycle for reasons other than toxicity. Any patient who experiences a DLT will also be considered evaluable, regardless of the percentage of scheduled therapy received during the first cycle. Patients enrolled in Phase 1b who are not DLT-evaluable will be replaced.
- Safety Analysis Set: all patients who receive at least 1 dose of any study drug. This will be the analysis set for all safety evaluations.
- The ITT Analysis Set: . all patients who receive at least 1 dose of any study drug. For this study, ITT population will be equal to the Safety population. This will be the analysis set for PFS and OS.
- RES: Includes patients in the SAS who have a baseline radiological tumor assessment and at least one post-baseline radiological tumor assessment. This will be the primary analysis set for efficacy assessments, except PFS and OS.
- PK Analysis Set: Includes patients in the SAS who have reliable drug assay data relevant for the PK analysis.
- PD Analysis Set: Includes patients in the SAS who have evaluable PD data (baseline and at least once after starting trial therapy).

10.2. Data Analysis

Descriptive statistics, including mean, median, standard deviations and ranges for all continuous measures will be tabulated and reported. Percentages and frequencies for all categorical measures will also be presented. Time-to-event endpoints will be reported using Kaplan-Meier estimates. Baseline demographics and other characteristics will be descriptively summarized.

In general, statistical analyses will be performed for the overall population and subgroups of population. Detailed presentation plan and definition of subgroups will be included in the SAP.

All analyses will be performed on the SAS version 9.4.

10.3. Subject Disposition

The number of subjects screened for participation and number enrolled overall and by study part will be tabulated along with the proportion in each analysis population. The proportion of subjects who discontinue study treatment will be tabulated, along with the primary reason for discontinuation.

10.4. Demographic and Other Baseline Characteristics

Demographic and baseline disease characteristic data will be summarized descriptively. Data to be tabulated will include at least demographics such as sex, age, and race, as well as weight and disease-specific status and medical history.

10.5. Prior and Concomitant Therapy

All investigator terms for medications recorded in the CRF will be coded to an 11-digit code using the current version of the WHO DD.

The number (percentage) of subjects who took prior and concomitant medications will be summarized on the SAS by study part, cohort and overall ATC class (i.e., anatomical class, therapeutic class, pharmacologic class, chemical class), as well as WHO Drug Dictionary preferred term.

- Prior medications will be defined as medications that stopped before the first dose of study drug.
- Concomitant medications will be defined as medications that started after the date of the first dose of study drug up to 28 days after the subject's last dose.

All medications will be presented in subject data listings.

10.6. Sample Size Phase 1b

Phase 1b aims at selecting the RP2D dose for each combination, defined as a dose that is associated with less than 33% of patients experiencing a DLT, that is, ≤1 patient experiencing a DLT out of 6 DLT-evaluable patients. DLT refers to toxicities experienced during Cycle 1 by patients enrolled in Phase 1b.

Patients who do not receive approximately 70% of the scheduled therapy of either drug during Cycle 1 of either drug, for reasons other than toxicity, will not be considered evaluable for DLT assessment and will be replaced if recommended by the study steering committee. Disease progression and AEs deemed related to disease progression will not be considered DLTs.

Phase 1b for each arm will have approximately between 6 and 18 DLT-evaluable patients. If additional cohorts or intermediate dose levels are needed to determine the RP2D, more patients may be treated if recommended by the study steering committee. The total number of patients in Phase 1b for all combinations will vary between approximately 30 to 90.

10.7. Sample Size Phase 2

10.7.1. Arm A

Alpelisib, in combination with fulvestrant, was associated with a PFS estimate at 6 months of 50.4% in patients with hormone receptor-positive, HER2–negative, advanced breast cancer with tumor PIK3CA mutation, following progression on or after previous therapy, including CDK4/6 inhibitors (Rugo et al, 2022).

A sample size of 50 patients receiving the combination of elacestrant and alpelisib, followed for at least 6 months, will provide a power higher than 80% against a hazard ratio of 0.55, relative to

the literature estimate (Rugo et al, 2022) with an alpha level of 0.05. The calculations against hazard ratios of 0.60, 0.65, and 0.70 give a power approximately higher than, 70%, 60%, and 45%, respectively.

10.7.2. Arm B

Everolimus, in combination with hormonal therapy, was associated with an average median PFS of 4 months (Dhakal et al, 2020; Nichetti et al, 2020; Rozenblit et al, 2021; Cook et al, 2021). Assuming exponential distribution, this corresponds to a PFS estimate at 6 months of 35.4%.

A sample size of 50 patients receiving the combination of elacestrant and everolimus, followed for at least 6 months, will provide a power higher than 85% against a hazard ratio of 0.55, relative to the literature estimate with an alpha level of 0.05. The calculations against hazard ratios of 0.60, 0.65, and 0.70 give a power approximately higher than 75%, 68%, and 55%, respectively.

10.7.3. Arm C

Abemaciclib, in combination with hormonal therapy, was associated with a median PFS of 5.3 months in patients who had disease progression on prior palbociclib (Seki et al, 2022). Similarly, Wander and colleagues reported a median PFS of 5.3 months in patients with ER+/HER2– mBC who received abemaciclib after disease progression on prior CDK4/6i (Wander et al, 2021). Currently, there is no published PFS estimate for the combination of ribociclib and hormonal therapy after disease progression on prior CDK4/6i. The power calculations for this arm assume that PFS estimates for the combination of elacestrant and ribociclib are similar to the estimates of elacestrant and abemaciclib in this group of patients. Assuming exponential distribution, a median PFS of 5.3 months corresponds to a PFS estimate at 6 months of 45.6%.

A sample size of 60 patients receiving the combination of elacestrant and abemaciclib or ribociclib (30 patients each) followed for at least 6 months, will provide a power higher than 85% against a hazard ratio of 0.55, relative to the literature estimate with an alpha level of 0.05. The calculations against hazard ratios of 0.60, 0.65, and 0.70 give a power approximately higher than 80%, 68%, and 55%, respectively.

10.7.4. Arm D

The literature provides the following estimates for median PFS for the combination of fulvestrant with CDK4/6 inhibitors after prior hormonal monotherapy:

- Palbociclib: 9.5 months, based on 347 patients (palbociclib USPI, 2022; PALOMA-3)
- Ribociclib: 14.6 months (data for second line subgroup of MONALEESA-3)
- Abemaciclib: 16.4 months, based on 446 patients (abemaciclib USPI, 2023; MONARCH-2)

An average of 13.4 months, weighted by the number of patients in each trial, corresponds to a 12-month PFS estimate of 53.7%.

A sample size of 90 patients receiving the combination of elacestrant with each of these 3 CDK4/6i (30 patients each) followed for at least 12 months, will provide a power higher than

90% against a hazard ratio of 0.55, relative to the literature estimate with an alpha level of 0.05. The calculations against hazard ratios of 0.60, 0.65, and 0.70 give a power approximately higher than 85%, 75%, and 60%, respectively.

10.7.5. Arm E

In the Phase 3 CAPItello-291 trial (Turner et al, 2023), patients with AI-resistant, HR+ HER2– advanced breast cancer the addition of capivasertib to fulvestrant provided a statistically significant improvement in the primary endpoints of PFS in the overall and AKT pathway-altered populations compared to fulvestrant alone. Overall, 708 patients underwent randomization; 289 patients (40.8%) had AKT pathway alterations, and 489 (69.1%) had received a CDK4/6 inhibitor previously for advanced breast cancer. In the overall population, the median progression-free survival was 7.2 months in the capivasertib–fulvestrant group, as compared with 3.6 months in the placebo–fulvestrant group (hazard ratio for progression or death, 0.60; 95% confidence interval [CI], 0.51 to 0.71; P<0.001). In the AKT pathway– altered population, the median progression-free survival was 7.3 months in the capivasertib–fulvestrant group, as compared with 3.1 months in the placebo–fulvestrant group (hazard ratio, 0.50; 95% CI, 0.38 to 0.65; P<0.001).

We assumed a reference of PFS rate 6 months of 56.1% corresponding to a median PFS of 7.2 months. A sample size of 60 patients receiving the combination of elacestrant and capivasertib, followed for at least 6 months, will provide a power higher than 82% against a hazard ratio of 0.55, relative to the literature estimate with an alpha level of 0.05. The calculations against hazard ratios of 0.60, 0.65, and 0.70 give a power approximately higher than 72%, 60% and 48% respectively.

10.8. Extent of Exposure

Treatment exposure will be summarized by combination and cohort (for Phase 1b) and by combination for each arm (for Phase 2) and as duration on treatment and extent of exposure to study drug. Duration on treatment will be calculated as the days from first dose until last dose and summarized quantitatively.

Measures of extent of exposure include the total number of doses per subject and compliance. Compliance will be calculated as the percentage of total number of doses relative to the expected number of doses and summarized quantitatively. The reasons for dose(s) withheld or delayed will also be summarized.

10.9. Efficacy Analysis

All efficacy parameters will be summarized descriptively for the Response-Evaluable Set, except PFS and OS, which will be calculated for the ITT population. ORR and CBR will be calculated with exact 95% confidence intervals (Clopper et al, 1934). DoR will be calculated for patients achieving a BOR of confirmed PR or CR. For DoR, PFS, and OS, medians will be calculated using Kaplan-Meier estimates. DoR, PFS, and OS will be reported in both summary tables and plotted with Kaplan-Meier curve. The consistency of the efficacy results across subgroups of potential prognostic value will be presented. Details will be provided in the SAP.

For Phase 2 PFS rate at 6 months (Arms A, B, C and E) and 12 months (Arm D) will be calculated together with its 95% CI. That will be the primary endpoint.

10.10. Safety Analysis

Safety data will be summarized using descriptive statistics (e.g., n, mean, standard deviation, median, minimum, maximum for continuous variables and n [%] for categorical variables). Safety variables include AEs, TEAEs, clinical laboratory parameters, vital signs, and 12-lead ECG results. Any TEAE that has a follow-up report will be counted as a single event with the highest CTCAE version 5.0 grade (CTCAE v5.0, 2017).

Safety will be assessed through the analysis of the reported incidence of AEs and TEAEs. The AEs will be coded using the MedDRA and summarized using SOC and PT. In addition, summaries of SAEs, and TEAEs leading to permanent treatment discontinuation, treatment interruption, and dose reduction; AEs by maximum NCI CTCAE v5.0 grade, and TEAEs related to trial therapy will also be presented by phase and treatment arm. Other safety endpoints, including laboratory results, vital signs, and ECG findings, will be summarized by treatment arm. All safety analyses will be performed using SAS version 9.4 or a later version.

More details will be provided in the SAP.

10.11. Pharmacokinetics Analysis

The assessment of potential drug interactions between elacestrant and the combination drugs will be carried out using the PK data for both elacestrant and the combination drug generated in Phase 1b prior to initiating Phase 2. The steady-state exposure for both drugs administered in combination in Phase 1 will be compared with historical PK data for both elacestrant and the combination drug single agents.

The PK analysis will be performed on the PK population.

The plasma concentrations of elacestrant (whole blood for everolimus) and the PK parameters for the combination drugs will be tabulated with the number of patients with data, mean, SD, median, minimum, maximum, geometric mean, and coefficient of variation (CV%, based on untransformed data).

PK parameters will be determined from the individual concentration time curves using non-compartmental methods with Phoenix™ WinNonlin® software, version 6.4 or higher (Pharsight Corp., Mountain View, California).

The PK parameters will include, as applicable, but will not be limited to C_{max}, T_{max} , $AUC_{0\text{-}tau}$, λ_z, $t_{1/2,z}$, CL/F, and V/F.

A population PK modeling approach might be used if deemed appropriate to compare with historical PK data.

10.12. Pharmacodynamics Analysis

The whole blood samples will be analyzed for evaluation of cfNA as a source of circulating biomarkers (i.e., DNA mutations). The samples may be used for exploratory analysis for evaluation of response-related outcomes.

10.12.1. Receipt of the Samples

Plasma/blood samples will be delivered from Cerba Research to the Clinical Laboratory for Analisi di Medicina Traslazionale (AMT), Via Tito Speri 10, 00071 Pomezia (Rome), Italy. Plasma/blood tubes will be inspected upon arrival at AMT Clinical Laboratory to verify their condition and their maintenance in dry ice before being stored at -80°C.

The AMT laboratory operator will inform the Sponsor by email in case of discrepancies of the sample arrival condition and if it will result not suitable for the processing and analysis.

In any case, the Sponsor ensures the quality of samples from their collection to their destruction by ensuring they are always stored at appropriate temperature during lab storage and transfer.

The AMT operator will also ensure traceability by cross-checking the information between the manifest document and the vial label and reporting the following information and steps in the relevant documentation: unique identification of sample, date of receipt, status of the sample at the time of the receipt (macro and main characteristic depending on the kind of sample e.g., “frozen” or “intact”), date of analysis, storage location details (if not reported in another document) and information about re-labelling (as applicable).

Moreover, the AMT laboratory will use a Laboratory Information Management System to provide information for the following processes:

- Property evidence management,
- Laboratory processing,
- Reagent lot numbers and usage,
- NAs analysis.

If there is a transfer to another facility, the traceability document will also feature the date of shipment/transfer, details of shipment including destination, as well as the above-mentioned measures to preserve the sample quality (e.g., dry ice, etc.), the status of the sample at the time of shipment/transfer (macro and main characteristics depending on the kind of sample, e.g., “frozen” or “intact”).

10.12.2. Sample Analysis

The presence and the frequency of mutations in cfNAs from patients’ plasma samples will be evaluated according to a validated Assay. and Genexus Integrated Sequencer workflow, as will be reported in detail in the Operative Instructions. After the processing of plasma samples, the data will be analyzed by a qualified and trained person.

The circulating free NAs extracted from plasma (cfNAs) and gDNA extracted from PBMCs will be assessed for quantity and purity using Quantus, Tapestation or equivalent methods. The cfDNA and gDNA quality will be assessed during the amplification step on Genexus instrument and reported as DNA QC on final report.

Stability of the samples for the analysis is:

- Plasma at -80°C no case no longer than 2 years
- cfDNA at -20/-80°C no case no longer than 5 years

Sponsor will be responsible for ensuring all biological sample collection and storage is secure and with adequate measures to protect confidentiality, per all applicable local regulations.

10.12.3. Retention of Samples

For the analysis, any unused biological samples, as well as secondary cfDNA samples derived from plasma processing, could be stored as residual material for possible technical repetitions if the quality of the samples is guaranteed as follows:

- Plasma at -80°C, in no case longer than 1 year after study completion for ctNAs extraction
- cfDNA at -20/-80°C, in no case longer than 1 year after study completion for analysis of biomarkers

The destruction of the samples will be authorized by the Sponsor.

If a patient prematurely terminates the study, the patient may choose if samples not yet analyzed shall be destroyed.

10.13. Interim Analyses

Phase 1b: After all patients in each cohort for each combination have completed the DLT observation period (Cycle 1, 28 days), all available safety and PK data will be reviewed by the DEC in consultation with the SSC and the decision to proceed to the next cohort or to the Phase 2 part of the trial, as applicable, will be taken at this meeting.

Phase 2: No formal interim analyses are planned. However, in addition to the standard pharmacovigilance activities, available safety, and efficacy will be reviewed periodically by the sponsors and the SSC.

11. DIRECT ACCESS TO SOURCE DATA/DOCUMENTS

11.1. Study Monitoring

This trial will be monitored in accordance with the ICH Note for Guidance on Good Clinical Practice. Monitoring will be carried out by Stemline under the responsibility of the CRO and will be described in detail in the Monitoring Plan. Monitoring will be performed through a combination of on-site visits and risk-based monitoring, particularly focusing on informed consents, adherence to inclusion/exclusion criteria, drug accountability, documentation of SAEs, and the proper recording of efficacy and safety measurements. All monitoring activities will be described in detail in the study-specific Monitoring Plan.

The site monitor will perform regular visits to the trial sites during study conduct. Facilities, study drug, storage area, eCRF, patient's source data, and all other study documentation will be inspected / reviewed by the site monitor for adherence to the protocol and GCP. At each site visit, the monitor will review the eCRFs for completion and accuracy. Accuracy will be checked by performing source data verification that is a direct comparison of the entries made onto the eCRFs against the appropriate source documentation. Any resulting discrepancies will be reviewed with the Investigator and his/her staff. The Investigator agrees to allow access to all

study related materials needed for the proper review of study conduct and to assist the monitor during the monitoring visits and during the data cleaning process.

This study will follow a quality risk management process which will be described in detail in the study-specific RMP. The process will include systematic implementation of overall risk assessment, critical processes and data identification, risk definition, evaluation and response planning, risk monitoring and control, and risk communication and reporting. The study -specific RMP will also contain risk-based monitoring strategy, which will provide guidance on blend of onsite, offsite, and central monitoring. Throughout the study, a Central Monitor will be responsible for central monitoring, performing periodic review of the risks, follow-up on the identified issues, and ensuring that the process of risk monitoring and review is appropriately documented. Central monitoring will be executed by a risk-based monitoring tool.

Before an investigational site can enter a patient into the study, a monitor from the CRO will visit the investigational study site to:

- Determine the adequacy of the facilities
- Discuss with the Investigator(s) and other personnel their responsibilities regarding protocol adherence, and the responsibilities of the Sponsor or its representatives including the CRO. This will be documented in a Clinical Study Agreement between Stemline and the Investigator.

During the study, a monitor from the CRO will have regular visits with the investigational site, for the following:

- Provide information and support to the investigators
- Confirm that facilities remain acceptable
- Confirm that the investigational team is adhering to the protocol, that data are being accurately recorded in the case report forms, and that investigational product accountability checks are being performed
- Perform source data verification. This includes a comparison of the data in the case report forms with the patient's medical records at the site or site, and other records relevant to the study. This will require direct access to all original records for each patient (e.g., clinic charts).
- Record and report any protocol deviations not previously sent to the Sponsor.
- Confirm AEs and SAEs have been properly documented on eCRFs and confirm any SAEs have been forwarded to the Sponsor and the specific reporting requirements of the applicable Ethical Committee/IRB and local authorities have been met.

The monitor will be available between visits if the Investigator(s) or other staff needs information or advice. More details of study monitoring will be outlined in the monitoring plan.

11.2. Audits and Inspections

Authorized representatives of the Sponsor, a regulatory authority, an Independent Ethics Committee, or an Institutional Review Board may visit the site to perform audits or inspections, including source data verification. The purpose of an audit or inspection by the Sponsor is to

systematically and independently examine all study-related activities and documents to determine whether these activities were conducted, and data were recorded, analyzed, and accurately reported according to the protocol, ICH-GCP Guidelines, and any applicable regulatory requirements. The Investigator should contact Stemline immediately if contacted by a regulatory agency about an inspection.

12. DATA QUALITY MANAGEMENT

12.1. Data Collection

Data collection activities will be carried out under the responsibility of the Sponsor. Patient data will be collected using an EDC system (see Section 12.1.1). Patients will be identified by the study identification number (patient ID), assigned during the Screening Period.

The patient ID will be a number composed of 8 digits CCCSSPPP:

- CCC is the international phone code of the country (with 1 or 2 leading zero for countries that have a 1- or 2-digit phone code).
- SS is the site number in the country. It will start from 01 for each country and will ascend.
- PPP is the patient number at the site. It will start from 001 and will ascend.

12.1.1. Electronic Case Report Form

Clinical data collected during the study at sites will be recorded in an eCRF using Medidata RAVE, which is a validated system. The Sponsor will be responsible to develop the eCRF based on this study protocol and to review and perform the user acceptance test of the eCRF to ensure protocol adherence.

The eCRF will be made available to the study personnel by means of the iMedidata interface which is a validated system. The accounts will be individual, and password protected.

The Investigator or designee will be responsible for entering study data into the eCRF in accordance with the eCRF completion guidelines provided by the Sponsor. To improve the quality of data collection and cleaning, data shall be entered into the eCRF as closely as possible to the time when they become available. The eCRF data will not be considered as source data (the definition of the source data can be found in Section 12.2).

The Principal Investigator will ensure the accuracy, completeness, and consistency of data entered, signing electronically the eCRF using the personal password.

An audit trail within the system will track all changes made to the data.

12.1.2. Interactive Response Technology

The IRT system, provided by Calyx, is a validated system used by the site personnel for the patient screening (including assignment of the patient number), kit assignment and patient status change. Site staff will be provided with a personal username and password to access the IRT portal. An IRT user manual will be prepared by Calyx and provided to the site. Details on the

IRT provider can be found in the specific manual. Some data such as patient numbers and KIT allocated through the IRT system will be automatically integrated in the eCRF.

12.1.3. Patient Diary

Patients are required to record their use of the drugs daily intake (including the time of each intake as well as the incidence of vomiting following administration) for at least the first 2 cycles of treatment duration in a Patient Diary that will be provided by the Sponsor. The patient must bring the diary to the site at each study visit, and the completed diary pages must be checked by the Investigator or designee. The Investigator or designee will be responsible for entering diary data into the eCRF. At the same time, a new diary will be dispensed to the patient for completion. In the exceptional situation in which the drug's Patient Diary retrieval is unattainable and for all cycles beyond cycle 3, the Investigator and/or designee will be responsible for documenting drug accountability per his/her knowledge.

12.1.4. Central Laboratory

Central laboratories will be used for the assessment of PK concentrations and for the analysis of biomarkers.

12.1.5. Data Capture Systems Versions and Validation Documentation

Versions of the data capture systems can change during the study. The Sponsor will maintain a list of the data capture system versions used and the validation documentation of each version. The validation documentation will be provided to the site on request (e.g., in case of audit or inspections).

12.1.6. Clinical Data Management

The eCRF data will be electronically verified via online and offline checks. Discrepancies in the data will be resolved by means of electronic queries. Data will be locked by the study data manager when all activities for the study, including medical revision of the data, are complete and no more entries are expected.

Data from sources other than the eCRF will be provided to the data manager on an agreed scheduled basis. The study data manager has the responsibility to reconcile data captured in the eCRF, with external data sources. Discrepancies found in the reconciliation of the data will be addressed by means of queries.

A clear overview of all clinical data management activities will be given in the data management plan.

12.2. Source Data

Source data are defined as all data in original records and certified copies of original records of clinical findings, observations or other activities in a clinical study that are necessary for the reconstruction and evaluation of the study.

Original documents and data records include but are not limited to site/patients' medical records, laboratory notes, ECG records, patients' diaries, patients' identification forms, and pharmacy

dispensing records. Study sites will also maintain paper drug accountability forms for the study treatment to document dispensed and returned study treatment, as applicable.

Source data should be held available for perusal by the Sponsor representatives for the study or to other authorized persons such as auditors and inspectors of Regulatory Authorities.

Direct access to source data is defined as the permission to examine, analyze, verify, and reproduce any records and reports that are important for evaluation of a clinical study (see Section 13).

Any party allowed direct access to study source data and documents should take all reasonable precautions within the constraints of the applicable regulatory requirements to maintain the confidentiality of patient identities and Sponsor proprietary information.

Data should be consistent with the source documents and discrepancies, if any, should be explained in writing. All the original documentation pertinent to the study procedures must be available for review in each patient's record.

13. QUALITY CONTROL AND QUALITY ASSURANCE

To ensure compliance with ICH-GCP Guidelines and all applicable regulatory requirements, the Sponsor may conduct a quality assurance audit. Please see Section 11.2 for more details regarding the audit process.

14. ETHICS

14.1. Ethics Review

Before starting the study in a study site, study protocol and relevant documentation must be submitted to and approved by the IEC(s) and the CA(s) of the participating country(ies). Before any study-related procedures may be performed, informed consent must be obtained from the patient by means of a signed declaration. The ICF must be approved in the corresponding local language and in accordance with local laws and regulations by the IEC prior to being submitted to the patient. Reasonable attempts will be made to enroll participants with characteristics of their local patient population that may vary in gender, race, and geographic location. Investigational product can only be supplied to an Investigator by Stemline or its authorized representative after documentation on all ethical and legal requirements for starting the study have been received by Stemline or its authorized representative.

The CA(s) and IEC(s) of the participating country(ies) will be informed about any changes in the study protocol, the end of the study, or the premature study termination as appropriate and within the requested time period.

The Investigator must also comply with the local applicable obligation(s) on the reporting of ADRs and other safety cases and reports (such as SUSAR reporting to IRB in the US) to the local concerned IRBs/ECs if required, according to the specific country requirements. The Sponsor will provide this information to the PI.

Progress reports and notifications of serious adverse drug reactions will be provided to the IRB or IEC according to local regulations and guidelines.

14.2. Ethical Conduct of the Study and General Aspects

This study will be conducted in compliance with the study protocol, the recommendations on biomedical research on human subjects of the *Declaration of Helsinki*, US FDA, GCP Guidelines, EU regulations in force (EU Directive 2005/28/EC and EU Regulation No 536/2014 of 16 April 2014), and national requirements of the participating countries.

The Sponsor has contracted a CRO (for details refer to the Study Operational Contact List) to perform some Sponsor trial-related duties and functions. The ultimate responsibility for the quality and the integrity of the study resides with the Sponsor. The study will be conducted in agreement with Sponsor's or CRO's SOP requirements as agreed.

14.3. Written Informed Consent

The PI(s) at each site will ensure that the patient is given full and adequate oral and written information about the nature, purpose, possible risk, and benefit of the study. Patients must also be notified that they are free to discontinue from the study at any time. The patient should be given the opportunity to ask questions and allowed time to consider the information provided.

The patient's signed and dated informed consent must be obtained before conducting any study procedures. Patients will also be informed about the measures taken to ensure their confidentiality according to the pertinent legislation.

The PI(s) must maintain the original, signed ICF. A copy of the signed ICF must be given to the patient. The process of obtaining the ICF must be documented in the source documents.

The first act of recruitment is the first patient signing first informed consent in line with first patient first visit and will be the study start date.

If a protocol amendment would affect the terms of the ICF, it will be revised to reflect the protocol change and submitted to IEC for approval. The Investigator will ensure that this new consent form is signed by all patients subsequently entered in the study and those currently in the study before the changes take effect on their participation in the trial. Patients who will not sign the new consent form must be terminated from the study participation.

14.4. Patient Insurance

The Sponsor has stipulated an insurance policy for patients participating in the study, in accordance with local regulatory requirements. Details on the insurance company, the insurance number and conditions will be made available to patients in the ICF and/or provided as a separate document, in accordance with national requirements. A copy of the insurance certificate will be provided to each Investigator and will be filed in the Investigator's File at the sites and in the eTMF of the study.

15. DATA HANDLING AND RECORDKEEPING

15.1. Inspection of Records

The Sponsor will be allowed to conduct site visits to the investigation facilities for the purpose of monitoring any aspect of the study. The Investigator agrees to allow the monitor to inspect the drug storage area, study drug stocks, drug accountability records, patient charts and study source documents, and other records relative to study conduct.

15.2. Retention of Records

As required by ICH-GCP guidelines, the Investigator will keep essential documents until at least 2 years after the last approval of a marketing application in an ICH region; until there are no pending or contemplated marketing applications in an ICH region; or at least 2 years have elapsed since the formal discontinuation of clinical development of the IMP or no more than 25 years from the end of the clinical trial (EU 536/2014), whichever occurs last. It is the responsibility of the Sponsor to inform the Investigator/institution as to when these documents no longer need to be retained. No study documents should be destroyed without prior written agreement between Sponsor and Investigator. Should the Investigator wish moving the study record to another location, he/she must notify the Sponsor in writing.

15.3. General Principles on Personal Data Privacy Compliance

All clinical trial information shall be recorded, processed, handled, and stored in such a way that it can be accurately reported, interpreted, and verified. At the same time, the confidentiality of records and of the personal data of the patients shall remain protected in accordance with the applicable law on personal data protection such as the EU GDPR 679/2016 and the EU Regulation on clinical trials on medicinal products for human use 536/2014 and HIPAA Privacy Rule, US Common Rule (45 CRF 46.116).

This section defines the appropriate technical and organizational measures that shall be implemented to protect information and personal data processed against unauthorized or unlawful access, disclosure, dissemination, alteration, or destruction or accidental loss as well as to assure the fulfillment of patients' privacy rights.

15.3.1. Acknowledgment

The Site, the PI, the Centralized Laboratory, the CRO as well as their appointed staff and service providers acknowledge that:

a. The performance of the will implies processing of sensitive personal data
b. Personal data processing is regulated by the applicable European (i.e., the EU GDPR 679/2016 and the EU Regulation on clinical trials on medicinal products for human use 536/2014) and local laws (i.e., the laws of the country where the study is conducted) as well as by the Sponsor's national legislation, where applicable
c. Strict compliance with the applicable data protection laws and this section of the protocol is deemed by the Sponsor as an essential condition of collaboration with the Site, the PI, the Centralized Laboratory, and the CRO.

15.3.2. Data Controllers and Data Processors

The Sponsor, the site, the PI, and the CRO acknowledge that for the purposes of the GDPR, Sponsor and site will act as independent data controllers while CRO and the PI will act as data processors, respectively, of the Sponsor and site. Before the beginning of the study, the Hospital will instruct in writing the Principal Investigator as its data processor. However, if specific local laws or regulations mandate a different definition of the privacy roles, the site, the Principal Investigator, and the CRO will implement the relevant legal instruments (e.g., if pursuant to the local laws, the Site is a data processor of the Sponsor, a Data Processing Agreement will be finalized. If pursuant to the local laws Sponsor and Site are joint controllers, a Joint Controllership Agreement will be finalized).

15.3.3. Duties of The Parties Involved in The Performance of The Study

Collection and use of patients' data, including their biological samples, will be carried out in full respect of the provisions of the information notices submitted to patients, as well as the privacy rights, the fundamental freedoms, and the dignity of patients. All the parties involved in this study undertake to adopt adequate measures to warrant that data will always be processed securely and in compliance with privacy laws.

The Site, the Principal Investigator, the Sponsor, the CRO, and the Centralized Laboratory, as well as their appointed staff and service providers, each in its respective remit and within the limits of their specific role in the study, shall implement the following safety measures (physical, logical, organizational, technical, electronic, IT, etc.) to ensure adequate protection of the personal data of the patients involved in the study. In particular:

i. DATA SAFETY. The Site and/or the Principal Investigator shall adopt all the necessary measures to prevent or minimize the risks of theft, fire, flooding, partial or total loss, accidental disclosure or illegal/unauthorized access to patient's data or Sponsor's proprietary confidential information; to this extent, before the beginning of the study, the Site and/or the Principal Investigator shall ensure that the actual measures they have implemented are fit-for-purpose and law-compliant, and in particular:

   - To minimize the risk of unauthorized access and theft, the hardware on which patients' personal data are stored shall be placed in a restricted-access area, accessible only to those individuals who need to retrieve the patients' personal data included in the database for professional purposes. The same safeguards shall be put in place for non-electronic databases.
   - Any electronic database containing the patients' personal data shall be password-protected by means of a strong password. Systems shall be set so that passwords must be updated at least every 2 months and feature at least 8 characters, with upper-case and lower-case recognition, containing "special" characters, such as upper-case letters [A-Z], lower-case letters [a-z], numbers [0-9], symbols [!, #, $, etc.], or other special characters [Á, ë, ö, etc.]. Passwords shall not include elements that may easily be associated with the assignee or information regarding him/her, such as name and year of birth (e.g. "johnbrown80") or easily predictable strings of characters (e.g. "qwerty", "12345", "admin", "user", etc.).

- Adequate cryptographic protection measures shall be put in place for data "at rest" and "in transit" (these include, for example, file system or database cryptography, or any other equivalent IT measure that renders data unintelligible to those who are not authorized to access them).
- High-level security measures shall be implemented also on the files or databases that contain the "key" to match patients' personal data (i.e., name, surname, etc.) with their respective "Patient IDs" (as defined at point [iv] below).
- Backup processes and other measures that ensure rapid restoration of business-critical systems shall be implemented.
- Updated antivirus and firewall programs shall be installed on the IT devices.

The Site shall regularly test and update the measures listed above.

The Site shall, upon request from the Sponsor and/or the CRO, provide detailed written information about the measures listed above.

The Site shall ensure that the selected sites for the study have implemented the above listed measures.

ii. TRANSMISSION OF DATA. All the parties that transfer data through internet and/or to the centralized database(s) used to process study data or to generate statistical analyses shall implement secure protocols based on cryptographic standards that make data unintelligible to unauthorized individuals.

Diaries shall not feature any identifying information of patients and shall only feature the patients' code. Patient diaries will be handed over to and by the patient to the study staff. Only in case of emergencies that prevent the collection of the diary (e.g., lockdowns due to the COVID-19 outbreak), a secure, fully tracked courier, that shall deliver the diary in the hands of the Principal Investigator or study staff may be engaged.

iii. SECURITY OF THE CENTRALIZED DATABASE. The centralized database held by the Sponsor shall have the following safeguards in place:

- Appropriate authentication methods, which differentiate between different users according to their respective roles to ensure that access to a specific set of patient's data is permitted exclusively to those for whom access to such data is essential in the context of their work for the study
- Appropriate measures to ensure that the authentication credentials are periodically updated (i.e., password change)

iv. PSEUDONYMIZATION. All personal data that may allow identification of the patients involved in the study shall be adequately dissociated from the other data pertaining to the study ("pseudonymization" process). The Principal Investigator shall adequately dissociate the identification data of patients from the data pertaining to the study by linking results to an alphanumeric code ["Patient ID"], whose format shall not make it possible to identify the patient directly or indirectly, to ensure that only anonymous data are transmitted to the Sponsor, the Centralized Laboratory, and/or the CRO. Site/Principal Investigator shall securely store a separate list (e.g., identification log) with

the identification code, together with all signed informed consents, in accordance with the security measures as defined above.

v. SAMPLE STORAGE. As outlined below, samples shall only be stored for as long as strictly necessary for the study's performance and in no case no longer than 10 years. Biological samples shall bear Patient ID, and in no case will they bear other information that may lead to the direct or indirect identification of the patient, especially when, in accordance with this protocol, samples shall be forwarded and shared outside the clinical Site (e.g., in case of centralized reading or local laboratory analysis).

vi. TRAINING. The parties shall ensure that any personnel involved in the study have received proper training on data protection issues.

All actions related to the implementation of the aforementioned measures shall be provided by the Sponsor, the Site and/or the CRO to the competent authorities (including data protection authorities) and Ethics Committees if and when requested. If such authorities or the Sponsor consider the implementation of the aforementioned measures insufficient to guarantee an adequate level of protection of the patients' personal data, the Site, the PI, the CRO and the Centralized Laboratory should adopt all the necessary activities to overcome such remarks to assure the full compliance with the data protection laws.

15.3.4. Archiving of The Clinical Trial Master File and Code Pairing List

As required by ICH-GCP guidelines, the Investigator will keep essential documents until at least 2 years after the last approval of a marketing application in an ICH region; until there are no pending or contemplated marketing applications in an ICH region; or at least 2 years have elapsed since the formal discontinuation of clinical development of the IMP or no more than 25 years from the end of the clinical trial (EU 536/2014), whichever occurs last. It is the responsibility of the Sponsor to inform the Investigator/institution as to when these documents no longer need to be retained. No study documents should be destroyed without prior written agreement between Sponsor and Investigator. Should the Investigator wish moving the study record to another location, he/she must notify the Sponsor in writing.

The patient code pairing list (in which the Patient ID is linked to the patients' identification data such as name and surname), shall be archived under the care of the Principal Investigator.

The content of the clinical trial master file shall be archived in a way that ensures that it is readily available and accessible, upon request, to the competent authorities. Any transfer of ownership of the content of the clinical trial master file shall be documented. The new owner shall undertake the responsibilities set out in this protocol.

The Sponsor appoints the Study Manager or delegates as responsible person/s for archives. Access to archives shall be restricted to those individuals.

The media used to archive the content of the clinical trial master file shall be such that the content remains complete and legible throughout the entire archive period. Any modification to the content of the clinical trial master file shall be traceable.

15.3.5. Data Breach

Data Breach is an incident regarding personal data security and leading to the accidental or unlawful destruction, loss, alteration, unauthorized disclosure of, or access to, personal data transmitted, stored, or otherwise processed. In particular: destruction of personal data is where the data no longer exists, or no longer exists in a form that is of any use to the Site, Sponsor, CRO, PI, etc.; data loss is when the data may still exist, but the Site Sponsor, CRO, PI, etc. has lost control or access to it, or no longer has it in its possession; damage is where personal data has been altered, corrupted, or is no longer complete; data unavailability is where, following a data incident (such as a network outage, a natural or human disaster, etc.), personal data become temporarily inaccessible to the Site Sponsor, CRO, PI.

Anomalous Event is an event that is not part of the standard operational scope of an infrastructure, network, or service and which affects, or is likely to affect, personal data. This may include theft or loss of IT devices and other physical events (e.g., an unauthorized access to a locked storage room containing paper files with personal data), and/or electronic/IT anomalies (e.g., cyber-attacks, default or hacking of cloud services), which may in any way entail loss, unavailability, alteration, theft, copy or dissemination of personal data.

Whoever becomes aware in any way of an anomalous event and/or of a data breach (see definitions above affecting the [patients' personal data and/or personal data collected in the context of the study shall, as appropriate, immediately (and in any case no later than 24 hours from the knowledge of an anomalous event and/or of a data breach) inform the Study Manager, the Sponsor's Data Protection Officer, who may be contacted at dpo@menarini.com, the Site and the CRO's Data Protection Officer and shall provide the following information:

1. Anomalous event/data breach type (e.g., data loss, unauthorized access, loss of company device, etc.)
2. Person or source that first reported the anomalous event/data breach
3. Date and Time when the person who first reported the anomalous event/data breach became aware of it
4. Anomalous event/data breach date and time (actual or presumed)
5. Place (specify if actual or alleged) where the anomalous event/data breach occurred
6. Anomalous event/data breach description
7. Indicate the source of the anomalous event/data breach (e.g., I.P. source) - (if relevant)
8. Indicate the affected infrastructure/system/application/cloud/software/hardware/database and their location
9. List or describe the processing/storage systems affected by the anomalous event/data breach (if relevant)
10. Number of data patients involved (if known)
11. Amount of allegedly breached data
12. Other relevant information.

Once all the above information has been provided, the Sponsor and/or the Site should have a reasonable degree of certainty that a security incident has occurred that has led to personal data being compromised.

Then, as appropriate, Sponsor and Site, each one in its respective remit, shall manage the Data Breach in accordance with the applicable data protection regulations.

For Data Breach affecting personal data of patients enrolled within the European Union, Sponsor and Site, autonomously or jointly depending on the circumstances and their privacy responsibilities as defined by the Regulation 2016/679 shall:

1. Collect the necessary evidence and information
2. Categorize the breach
3. Determine the risk probability and level to the rights and freedom of the concerned patients
4. Identify and put in place appropriate remedies to minimize the impact of the Data Breach
5. Determine the notification and communication duties vis à vis the competent supervisory authority and/or the concerned patients.

15.3.6. Information Notice on Personal Data Protection and Pseudonymization

Prior to patients' enrollment in the study, the Principal Investigator and/or the Site (including their personnel) shall provide each patient with adequate, law-compliant "information notices and consent forms to process personal data" as included in the ICF (or, as the case may be, through a separate, specific form) provided by the Sponsor or delegated CRO and shall collect his/her written consent to the processing of personal data according to the actual performance conditions in which the study is carried out. The Principal Investigator is responsible for archiving the signed ICF in accordance with the security measures described above.

Among other things, the ICF (or the separate form) shall inform patients about:

1. the applicable data protection legislation
2. the kind of data that shall be collected during the study listing them in detail or by category
3. the purpose of data processing and the legal basis
4. whether granting the consent(s) to process personal data is a necessary or an optional condition to take part in the study (if the processing relies on consent as a legal basis)
5. the use of data for future scientific research / secondary use of data (if any). In such a case the future scientific purposes / secondary use shall be included
6. the pseudonymization procedure and scope
7. who can access patients' data and under what circumstances, e.g., PI, Site, Sponsor, regulatory authorities etc.
8. the period of data retention/storage, including the storage of biological samples
9. to which entities/countries outside the EU patients' data will be transmitted

10. patients' data protection rights as defined by the EU GDPR 679/2016 and other relevant national regulations
11. Data Controllers/Data Processors and the relevant contact details
12. Sponsor's DPO contacts
13. in case of genetic data processing the possible findings, also regarding unexpected findings that might be disclosed on account of the processing of the genetic data.

15.3.7. Genetic Data

The collection of genetic data for performing genetic tests and screening shall be limited to the personal and family information that is indispensable for performing the study.

The source, nature, and mechanisms for sample taking and storage will be described in the Laboratory Manual.

Without prejudice to applicable laws and regulations, the protocol shall be subject to confidentiality obligations that will assure the secrecy of the data for at least 1 year after the conclusion of the study.

The measures to keep patients' identification data separated from biological materials and genetic information are:

- Access to the premises where genetic data are stored shall be controlled by security staff and/or electronic devices also based on biometrics. Any person admitted after closing time, on whatever grounds, shall have to be identified and their data recorded.
- Preservation, use, and transportation of biological samples shall be carried out in such a manner as to also ensure their quality, integrity, availability, and traceability.
- Genetic data shall be transmitted electronically by certified electronic mail after encrypting and digitally signing the information to be transmitted. Web application-based communication channels may be used if they rely on secure communication protocols, and they can guarantee the digital identity of the server providing the service as well as of the client station from which the data are accessed by means of digital certificates issued by a certification authority in pursuance of the law.
- Electronically processed genetic data may be accessed if authentication systems are based on tokens/devices.
- Genetic data and biological samples contained in lists, registers and/or databases shall be processed with encryption techniques and/or by means of identification codes and/or any other techniques that can make them temporarily unintelligible also to the persons authorized to access them.
- To minimize the risks of accidental disclosure and/or unlawful/unauthorized access, patients' identities will be disclosed only when strictly necessary (e.g., to prevent a physical prejudice).
- Genetic and medical data will be processed separately from any other personal data that can identify the patients directly.

- The ICF and/or separate form will detail the possible findings regarding genetic data, also regarding unexpected findings that might be disclosed as result of the test/elaboration of genetic data.
- The ICF and/or separate form will detail whether the patient is allowed to limit the scope of communication of his/her genetic data and the transfer of biological samples, including their possible use for additional purposes.
- The ICF and/or separate form will detail the retention period of genetic data and biological samples (if different from the general retention period of other data processed in the context of the study).

15.3.8. Transfer of Patients' Data Outside the European Union

The study performance entails transferring patients' personal data (coded data) outside the EU. To this extent, the Sponsor, the Site, the Principal Investigator, the central laboratories, and the CRO undertake to export such data in compliance with adequate safeguards/legal basis as required by the Regulation 679/2016 including the commission decisions, the standard contract clauses, the privacy shield, and patients' specific consent. Examples of non-EU countries/entities include but are not limited to the United States. The complete list will be available upon request.

15.3.9. Exercise of Patients' Data Privacy Rights

Each study patient has the right to contact the Sponsor, the Site, the PI, the Centralized Laboratory, the IRT provider and the CRO to exercise the rights afforded to the patient by the law, including those afforded under articles 15 to 22 of Regulation (EU) 2016/679, namely: knowing whether or not any data referring to his/her is being processed in the context of the study; access his/her data; verify the data's content, origin, exactness, location (including, where applicable, the non-EU countries where the data might be); obtain a copy of the data including their transmission to another entity indicated by the patient; ask that the data are supplemented, updated, amended; in the circumstances set forth by the law, ask that the processing of data is restricted, that data are anonymized or frozen; oppose to the processing of his/her data for legitimate reasons. Each patient has the right to lodge a complaint with his/her local supervisory authority and/or to notify the DPO any use of his/her personal data the patient regards as inappropriate.

Each study patient is free to withdraw at any time from the study. In such case, each study patient may ask the Sponsor, the Site, the PI, the Centralized Laboratory, the CRO to destroy/delete his/her personal data and his/her biological samples, unless they have been permanently anonymized), thus preventing any further processing or analysis of his/her data. However, data and results of tests that may have been used to determine the results of the study shall not be deleted, to avoid altering or impairing altogether the results of the study.

The genetic test may reveal information that may be of use to the patient for health or treatment. Such information will be communicated to the patient if he/she wishes to keep informed

If the Site, the PI, the Centralized Laboratory, the IRT provider, or the CRO receive a request for data privacy rights exercise, the concerned recipient shall immediately inform the Sponsor DPO by email at dpo@menarini.com.

The request shall be fulfilled within the term set forth by the applicable privacy laws (normally 30 days). The Sponsor, the Site, the PI, the Centralized Laboratory, the IRT provider, and the CRO shall implement adequate organizational measures to reply to patients within the above-mentioned deadline.

15.3.10. Future Research

With patients' optional and additional consent, the Sponsor and/or the Site may use the data collected during the study for further medical and scientific research purposes. These may include, for example: retrospective clinical studies; clinical studies pertaining to the patients' pathology/medical condition(s) or similar conditions; studies which compare the data of this Study with those from other sources to identify the factors involved in a disease; registration of new drugs. As required by regulations the specification of the future analysis not described in this document will be submitted for evaluation and/or approval to EC/IRB and CA and additional consent will be required to the patients if necessary.

In the context of these additional research activities, patients' data will be processed, pseudonymized and transferred abroad, and may be shared with future research partners. Details on processing and storage will be provided in the study manual, care will be taken to ensure compliance of GDPR and Privacy.

16. PUBLICATION POLICY

All manuscripts, abstracts, or other modes of presentation arising from the results of the study must be reviewed and approved in writing by the Sponsor in advance of submission. The review is aimed at protecting the Sponsor's proprietary information existing either at the date of the commencement of the study or generated during the study.

The detailed obligations regarding the publication of any data, material results, or other information generated or created in relation to the study shall be set out in the agreement between each Investigator and the Sponsor.

The contents of this protocol and any amendments and results obtained during the study should be kept confidential by the Investigator, the Investigator's staff, and the IRB/IEC and will not be disclosed in whole or in part to others or used for any purpose other than reviewing or performing the study, without the written consent of the Sponsor. No data collected as part of this study will be used in any written work, including publications, without the written consent of the Sponsor. These obligations of confidentiality and non-use shall in no way diminish such obligations as set forth in either the Confidentiality Agreement or Clinical Trial Agreement executed between the Sponsor and the institution/Investigator.

All persons assisting in the performance of this study must be bound by the obligations of confidentiality and non-use set forth in either the Confidentiality Agreement or Clinical Trial Agreement executed between the institution/Investigator and the Sponsor.

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18. APPENDICES

APPENDIX A. ECOG PERFORMANCE STATUS CRITERIA

Grade	Description
0	Normal activity. Fully active, able to carry on all pre-disease performance without restriction.
1	Symptoms, but ambulatory. Restricted in physically strenuous activity, but ambulatory and able to carry out work of a light or sedentary nature (e.g., light housework, office work).
2	In bed < 50% of the time. Ambulatory and capable of all self-care, but unable to carry out any work activities. Up and about more than 50% of waking hours.
3	In bed > 50% of the time. Capable of only limited self-care, confined to bed or chair more than 50% of waking hours.
4	100% bedridden. Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair.
5	Dead

ECOG=Eastern Cooperative Oncology Group; PS=performance status.

APPENDIX B. HBV/HCV TESTING CRITERIA

Eligibility Based on Serologic Markers for Hepatitis B and Hepatitis C

Test	Result				
HBsAg	+	-	-	-	-
HBcAb	Any	+	-	+	-
HBsAb	Any	-	+	+	-
HCV Ab	Any	Any	-	-	-
Eligibility	Not Eligible	Not Eligible	Eligible	Eligible	Eligible

HBcAb=hepatitis B core antibody; HBsAb=hepatitis B surface antibody; HBsAg=hepatitis B surface antigen; HBV=hepatitis B virus; HCV=hepatitis C virus
If indeterminate results are obtained, viral DNA (hepatitis B) or RNA (hepatitis C) should be measured to confirm negative viral status.

✘ **HCV Ab positive:** Indicates active infection and risk for reactivation. These patients are not eligible for this trial.

✘ **HBsAg positive:** Indicates active infection and risk for reactivation with fulminant hepatitis. These patients are not eligible for this trial.

✘ **HBcAb positive and HBsAb negative:** indicate active infection and risk for reactivation. These patients are not eligible for this trial.

✓ **HBsAb positive:** As a standalone marker, it indicates successful vaccination or previous infection that has been successfully resolved if it is the only positive finding. These patients are eligible for this trial.

✓ **HBsAg negative, HBcAb positive, HBsAb positive**: Resolved or latent infection. These patients are eligible for this trial.

✓ **All markers negative**: No prior exposure or vaccination to hepatitis B and no prior exposure to Hepatitis C. Patients are eligible for this trial

APPENDIX C. RESPONSE EVALUATION CRITERIA IN SOLID TUMORS (RECIST) VERSION 1.1

The RECIST version 1.1 criteria summarized below are adapted from Eisenhauer et al., 2009.

Target Lesions	Non-Target Lesions	New Lesions	Overall Response	Best Response for this Category also requires
CR	CR	No	CR	4-6 weeks Confirmation
CR	Non-CR/Non-PD	No	PR	4-6 weeks Confirmation
CR	Not evaluated	No	PR	4-6 weeks Confirmation
PR	Non-PD or not all evaluated	No	PR	4-6 weeks Confirmation
SD	Non-PD or not all evaluated	No	SD	Documented at least once >6 weeks from baseline
PD	Any	Yes or No	PD	—
Any	PD	Yes or No	PD	—
Any	Any	Yes	PD	—
Not all evaluated	Non-PD	No	Unevaluable	—

CR=complete response; PD=progressive disease; PR=partial response; SD=stable disease
Subjects with global deterioration of health status requiring discontinuation of treatment without objective evidence of disease progression at that time should be reported as "symptomatic deterioration". Every effort should be made to document the objective progression even after discontinuation of treatment.

APPENDIX D. COCKROFT-GAULT FORMULA

- MALES: CrCl=([140-age in years] × weight in kg)/([serum creatinine in mg/dL] × 72)
- FEMALES: CrCl=0.85 × [140-age in years] × weight in kg)/([serum creatinine in mg/dL] × 72)

APPENDIX E. DRUGS KNOWN TO PROLONG THE QT INTERVAL

See discussion of drug-induced QT prolongation and drugs concerned at

https://www.crediblemeds.org

APPENDIX F. CONTRACEPTION GUIDANCE

Contraception method used by women and men, should be consistent with local regulation and availability, please see the note.

1. For Female Participants:

   Postmenopausal women are defined as not having menses for 12 months without an alternative medical cause. For the purpose of this protocol, we define postmenopausal status as women as:

   a) Age ≥60 years

   b) Age <60 years and amenorrhea for 12 or more months (without an alternative cause) **and** FSH value **and** an estradiol level within the postmenopausal range per local laboratory reference.

   A high follicle stimulating hormone (FSH) level in the postmenopausal range may be used to confirm a post-menopausal state in women not using hormonal contraception or hormonal replacement therapy. However, in the absence of 12 months of amenorrhea, a single FSH measurement is insufficient.

   Women of childbearing potential (WOCBP) are defined as fertile, following menarche and until becoming post-menopausal unless permanently sterile. Permanent sterilization methods include hysterectomy, bilateral salpingectomy and bilateral oophorectomy.

   In the absence of the above, female participants (pre- or perimenopausal) must use the appropriate highly effective, non-hormonal contraception method within 28 days of the first dose of study treatment, until 120 days of the last dose of study treatment (or longer based on companion drug).

   Highly effective methods of contraception are those methods that can achieve a failure rate of less than 1% per year when used consistently and correctly are considered as highly effective birth control methods. Such methods include:

   - intrauterine device (IUD) [A]
   - bilateral tubal occlusion [A]
   - vasectomized partner [A, B]
   - sexual abstinence [C] (i.e., refraining from heterosexual intercourse during the study treatment and 120 days after the last dose of study treatment).

2. Male Participants

   A man is considered fertile after puberty unless permanently sterile by bilateral orchidectomy.

   Male participants must use the following methods of contraception within 28 days of the first dose of study treatment until 120 days of the last dose of study treatment (or longer based on companion drug):

- sexual abstinence
- male participants (including males with a vasectomy) with a pregnant or non-pregnant WOCBP should use male condom in order to avoid exposure of an existing embryo/fetus.

3. Female and Male Participants

    - Must agree to abstain from donating ova or sperm by using highly effective contraception within 28 days of the first dose of study treatment, until 120 days of the last dose of study treatment (or longer based on companion drug). Investigators or site personnel should have discussions with male participants with female partners of childbearing potential regarding appropriate highly effective contraception methods.

4. Unacceptable contraception methods

    - Periodic abstinence (calendar, symptothermal, post-ovulation methods), withdrawal (coitus interruptus), spermicides only, and lactational amenorrhoea method are not acceptable methods of contraception. Female condom and male condom should not be used together.

A. Contraception methods that in the context of this guidance are considered to have low user dependency.
B. Vasectomized partner is a highly effective birth control method provided that partner is the sole sexual partner of the WOCBP trial participant and that the vasectomized partner has received medical assessment of the surgical success.
C. In the context of this guidance sexual abstinence is considered a highly effective method only if defined as refraining from heterosexual intercourse during the entire period of risk associated with the study treatments. The reliability of sexual abstinence needs to be evaluated in relation to the duration of the clinical trial and the preferred and usual lifestyle of the subject.

Note: Please refer to "Recommendations related to contraception and pregnancy testing in clinical trials" for additional details (https://www.hma.eu/fileadmin/dateien/Human_Medicines/01-About_HMA/Working_Groups/CTFG/2020_09_HMA_CTFG_Contraception_guidance_Version_1.1_updated.pdf)

APPENDIX G. PK SAMPLING TIMES

Phase 1b- PK Sampling times alpelisib, palbociclib and ribociclib combinations

	Cycle 1, Day 15 (± 2 days)						Cycle 1 Day 16 (24 hours after PK day)
Time	**Pre-dose (-1 hr)**	**1 hr (±10 min)**	**2 hr (±20 min)**	**3 hr (±30 min)**	**6 hr (±30 min)**	**8 hr (±30 min)**	**Pre-dose (24 hrs post-C1D15)**
PK plasma	X	X	X	X	X	X	X

These will be collected pre-dose (within 1 hour before administration of study drug) and post-dose at 1 (±10 minutes), 2 (±20 minutes), 3 (±30 minutes), 6 (±30 minutes), 8 (±30 minutes), and 24 hours (±60 minutes); the 24-hour PK must be done before administration of that day's dose of study drugs.

Phase 1b- PK Sampling times for everolimus combinations

	Cycle 1, Day 15 (± 2 days)						Cycle 1 Day 16 (24 hours after PK day)
Time	**Pre-dose (-1 hr)**	**1 hr (±10 min)**	**2 hr (±20 min)**	**3 hr (±30 min)**	**6 hr (±30 min)**	**8 hr (±30 min)**	**Pre-dose (24 hrs post-C1D15)**
PK whole blood	X	X	X	X	X	X	X

These will be collected pre-dose (within 1 hour before administration of study drug) and post-dose at 1 (±10 minutes), 2 (±20 minutes), 3 (±30 minutes), 6 (±30 minutes), 8 (±30 minutes), and 24 hours (±60 minutes); the 24-hour PK must be done before administration of that day's dose of study drugs.

Phase 1b- PK Sampling times capivasertib combination

	Cycle 1, Day 11 (- 1 day allowed)							Cycle 1 Day 12 (24 hours after PK day)
Time	**Pre-dose (-1 hr)**	**1 hr (±10 min)**	**2 hr (±20 min)**	**3 hr (±30 min)**	**6 hr (±30 min)**	**8 hr (±30 min)**	**12 hr (±30 min)**	**Pre-dose (24 hrs post-C1D15)**
PK plasma	X	X	X	X	X	X	X	X

On C1/D11, plasma samples will be taken to determine plasma (blood for everolimus) concentrations of study drugs and their metabolites. These will be collected pre-dose (within 1 hour before administration of study drug) and post-

dose at 1 (±10 minutes), 2 (±20 minutes), 3 (±30 minutes), 6 (±30 minutes), 8 (±30 minutes), 12 (±60 minutes, before the second capivasertib dose) and 24 hours (±60 minutes); the 24-hour PK must be done before administration of that day's dose of study drugs.

Phase 2 PK sampling will be performed in approximately 30 patients/combination/arm on Cycle 1 Day 15 and every 8 weeks starting from Cycle 3 Day 1 in conjunction with CBC and CT/MRI scan. Samples will be collected for up to 12 months. The objective of these PK samples is to explore the exposure-response relationship for safety and efficacy of elacestrant and in each combination drug.

Phase 2 PK:

Day	**Cycle 1 Day 15 (± 2 days)**	**Cycle 3 Day 1 (± 2 days)**	**Cycle 5 Day 1 (± 2 days)**	**Cycle 7 Day 1 (± 2 days)**	**Cycle 9 Day 1 (± 2 days)**	**Cycle 11 Day 1 (± 2 days)**	**Cycle 13 Day 1 (± 2 days)**
Time	**Pre-dose (-1 hr)**	**Pre-dose (-1 hr)**	**Pre-dose (-1 hr)**	**Pre-dose (-1 hr)**	**Pre-dose (-1 hr)**	**Pre-dose (-1 hr)**	**Pre-dose (-1 hr)**
PK (plasma except for everolimus combination- whole blood)	X	X	X	X	X	X	X

Samples are collected as pre-dose (within 1 hour before administration of study drug) while on study treatment.